BECOMING A DOCTOR OF

INTERNATIONAL ENERGY AND NATURAL RESOURCES LAW

DOCTORAL TEXT AND ADVANCED PROFESSIONAL REFERENCE



TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

BECOMING A DOCTOR OF INTERNATIONAL ENERGY AND NATURAL RESOURCES LAW

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International Energy and Natural Resources Law

A Doctoral Text and Advanced Professional Reference on Resource Sovereignty, Energy Transactions, Investment Protection, Environmental and Climate Governance, and International Dispute Resolution

TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

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Becoming a Doctor of International Energy and Natural Resources Law

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DEDICATION

To the scholars, regulators, negotiators, advocates, and arbitrators whose rigorous work turns the world's energy and natural resources into lawful, accountable, and equitable development —

and to the communities whose lands, waters, and futures that work must always serve.

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PART I

Jurisprudential and Institutional Foundations

The conceptual architecture of the discipline: its definition and history, its sources and principles, the doctrine of permanent sovereignty over natural resources, and the institutions through which global energy governance is organised.

1 CHAPTER

The Nature, Scope, and Evolution of International Energy and Natural Resources Law

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Define international energy and natural resources law as a field of legal study and distinguish it from adjacent fields, including public international law, international economic law, and international environmental law.
2. Trace the historical development of resource governance from the early concession era through decolonisation, the New International Economic Order, liberalisation, and the contemporary energy transition.
3. Explain how the internationalisation of energy markets has transformed the regulation of exploration, production, transportation, and trade.
4. Analyse the interaction between public law, private law, and transnational legal relations in energy and resource projects.
5. Evaluate the significance of legal pluralism and competing regulatory authorities in global resource governance.
6. Identify the principal foundational debates that structure doctoral research in the field.

Introduction

Energy and natural resources sit at the junction of almost every question that modern legal systems are asked to answer. Who owns what lies beneath the soil and the seabed? Under what conditions may a sovereign state grant rights to explore for and extract those resources, and on what terms may it later change its mind? How should the economic rent produced by a non-renewable asset be shared between the state, the investor, the host community, and future generations? What obligations attach to the transport of oil and gas across national frontiers, to the generation and trade of electricity, or to the emissions released

when fossil fuels are burned? How should disputes between powerful states and powerful corporations be resolved, and by whom? Each of these questions is, at bottom, a legal question, and the body of law that has developed to answer them is the subject of this book.

International energy and natural resources law is a young discipline in name but an old one in substance. The legal instruments through which states have granted, regulated, taxed, and expropriated mineral and petroleum rights have a history that stretches back more than a century, and the doctrines that frame those transactions — sovereignty, contract, property, state responsibility, and the protection of aliens — are older still. What is new is the recognition that these instruments and doctrines constitute a coherent field of study, with its own sources, principles, institutions, and methods, and with a distinctive set of intellectual problems that cannot be adequately addressed from within any single traditional legal category.

This opening chapter lays the foundation for the remainder of the book. It begins by defining the field and identifying its boundaries. It then traces the historical development of resource governance, explains the internationalisation of energy markets and regulation, and examines the interplay between public law, private law, and transnational legal relations. It considers the relationship between energy law and international economic law, the phenomenon of legal pluralism, and the contemporary challenges confronting global resource governance. The chapter closes by setting out the foundational debates that animate doctoral research and that recur throughout the chapters that follow.

The approach throughout is analytical rather than descriptive. A doctoral candidate in this field must be able to do more than recite the content of treaties and statutes; he or she must be able to explain why the law takes the form that it does, to identify the tensions and contradictions within it, to evaluate competing interpretations, and to propose reasoned reforms. The chapter therefore emphasises the conceptual architecture of the field — the ideas that hold it together and the fault lines along which it is contested.

1.1 Defining Energy and Natural Resources Law

The Problem of Definition

Any attempt to define a field of law must confront a familiar difficulty: legal categories are not natural kinds. They are constructed by scholars, practitioners, courts, and legislators for particular purposes, and their boundaries shift as those purposes change. “Energy law” and “natural resources law” are no exception. Neither phrase appears as a term of art in any multilateral treaty of general application, and neither corresponds to a single body of legislation in most domestic legal systems. Instead, each describes a cluster of rules, principles, institutions, and

practices that share a common subject matter — the ownership, development, transformation, trade, and use of energy sources and natural resources — and that are united by a common set of policy concerns.

A useful working definition, adopted for the purposes of this book, is the following:

> International energy and natural resources law is the body of rules, principles, institutions, and practices — derived from public international law, international economic law, comparative domestic law, and transnational commercial practice — that governs the allocation of rights over energy sources and natural resources, the conduct of activities involving their exploration, development, transformation, transport, trade, and use, and the distribution of the benefits and burdens arising from those activities, in situations that involve more than one legal order.

Several features of this definition deserve comment. First, it is *functional* rather than formal: it defines the field by reference to its subject matter and the problems it addresses, not by reference to a particular source of law. Second, it is *pluralist*: it expressly acknowledges that the relevant rules derive from multiple sources, including domestic legislation and private contracts as well as treaties and custom. Third, it is *distributive*: it identifies the allocation of benefits and burdens as a central concern, reflecting the fact that energy and resource law is fundamentally about who gets what, when, and how. Fourth, it is *transnational*: the field is concerned with situations that involve more than one legal order, whether because a resource straddles a boundary, because an investor is foreign, because a commodity is traded internationally, or because environmental effects cross frontiers.

Energy Law and Natural Resources Law: One Field or Two?

The conjunction "energy and natural resources" requires justification. Energy law, narrowly understood, is concerned with the production, conversion, transmission, distribution, and consumption of energy, whether derived from fossil fuels, nuclear fission, or renewable sources. Natural resources law, narrowly understood, is concerned with the ownership, management, and exploitation of resources found in nature, including minerals, petroleum, water, forests, fisheries, and biological diversity. The two fields overlap substantially — petroleum and natural gas are both natural resources and energy sources — but neither is wholly contained within the other. Renewable electricity generated from wind or solar radiation is an energy source but is not usually described as a natural resource in the extractive sense; copper and lithium are natural resources but are not themselves sources of energy, although they are indispensable to its transmission and storage.

There are strong reasons for treating the two fields together. Historically, the law governing petroleum developed out of the law governing hard-rock minerals, and many of the foundational instruments — concessions, licences, royalties, and the doctrine of permanent sovereignty — are common to both. Functionally, the energy transition is collapsing the distinction: the decarbonisation of energy systems depends on the extraction of critical minerals on an unprecedented scale, on access to land and water for renewable installations, and on the use of geological formations for the storage of carbon dioxide and hydrogen. Institutionally, the same ministries, regulators, and courts often have jurisdiction over both energy and minerals. Doctrinally, the same principles of international law — sovereignty over natural resources, the prohibition of transboundary harm, the protection of foreign investment, and the rights of indigenous peoples — govern both.

This book therefore treats energy law and natural resources law as a single integrated field, while recognising that particular chapters will emphasise one or the other. The integrated approach has the further advantage of exposing the connections between subjects that are too often studied in isolation: between petroleum fiscal regimes and renewable energy incentives, for example, or between mining law and climate law.

Relationship with Adjacent Fields

The field overlaps with several established branches of law, and a doctoral candidate must be able to locate it precisely in relation to each.

Public international law. The general rules of public international law — on sources, treaties, state responsibility, jurisdiction, immunity, and dispute settlement — supply the framework within which international energy and resource law operates. The doctrine of permanent sovereignty over natural resources, discussed in Chapter 2, is a principle of general international law. The law of the sea, international watercourse law, and international environmental law are specialised branches of public international law with direct application to resource activities.

International economic law. The law of international trade, investment, finance, and taxation governs the cross-border movement of energy commodities, capital, and revenues. The World Trade Organization (WTO) agreements apply to trade in energy goods and, to a more limited extent, services; bilateral and multilateral investment treaties protect foreign investors in the energy and mining sectors; and international tax rules shape the fiscal treatment of resource projects. The relationship between energy law and international economic law is examined in section 1.5 below.

International environmental and climate law. Resource activities are among the most significant sources of environmental harm, and international

environmental law imposes obligations of prevention, precaution, assessment, and cooperation that constrain the manner in which they may be carried out. Climate change law, in particular, is reshaping energy law by setting long-term objectives for the reduction of greenhouse gas emissions that are incompatible with the indefinite expansion of fossil fuel production.

Comparative and domestic law. Most of the rules that govern energy and resource activities on a day-to-day basis are domestic: constitutional provisions on resource ownership, petroleum and mining statutes, electricity regulation, environmental legislation, and tax codes. International energy law cannot be understood without close attention to these domestic regimes and to the patterns of convergence and divergence among them, which are explored in Chapter 24.

Transnational commercial law. Energy and resource projects are structured through complex networks of private contracts — joint operating agreements, sales and purchase agreements, engineering and construction contracts, financing agreements — that are frequently governed by English or New York law and subject to international commercial arbitration. The model contracts and standard terms produced by industry associations constitute a form of transnational private ordering that interacts with, and sometimes substitutes for, state regulation.

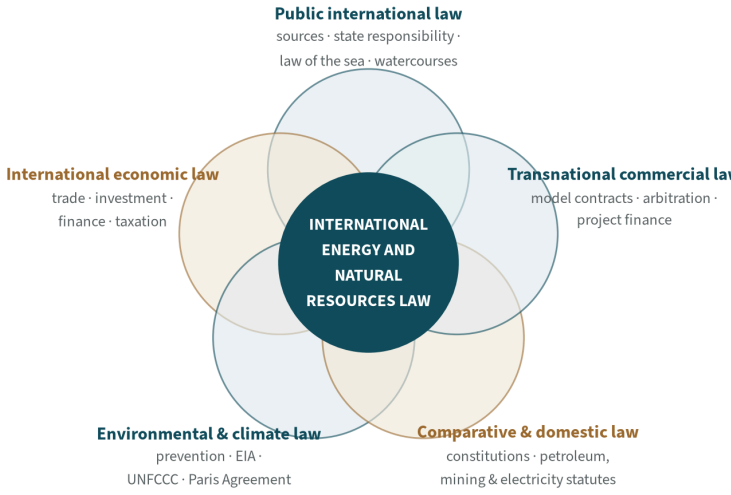


Figure 1.1 The constituent fields of international energy and natural resources law. The discipline is defined by its subject matter and draws on five overlapping bodies of law. *Source:* prepared for this edition.

The Scope of the Field

The scope of the field, as treated in this book, may be described along three axes.

The first axis is the *resource or energy source* concerned: petroleum (crude oil and natural gas), coal, hard-rock minerals including critical minerals, uranium and nuclear energy, renewable energy sources (solar, wind, hydropower, geothermal, biomass, marine), hydrogen and other energy carriers, electricity, water, and marine resources.

The second axis is the *stage of the value chain*: the allocation of rights to explore and develop resources (upstream), the transport and processing of those resources (midstream), and their sale, distribution, and consumption (downstream), together with the decommissioning of installations and the remediation of affected environments at the end of the project life cycle.

The third axis is the *legal function*: the allocation of property and access rights, the regulation of conduct, the protection of investment, the allocation of commercial risk by contract, the raising and distribution of revenue, the protection of the environment and human rights, and the resolution of disputes.

The chapters of this book are organised principally along the third axis, while particular chapters address specific resources (Chapters 5 to 12) and specific legal functions (Chapters 13 to 27).

Is There a "Lex Petrolea"?

One of the most contested questions in the definition of the field concerns the existence of a distinct transnational legal order governing the international petroleum industry — sometimes described as the *lex petrolea*. The idea has a long pedigree. In the arbitral award in *Kuwait v American Independent Oil Company (Aminoil)*, the tribunal referred to the existence of a body of practice in the oil industry that informed the interpretation of the concession, while stopping short of endorsing an autonomous legal system.[1] Proponents of the *lex petrolea* argue that the repeated use of standard contractual forms, the consistent practice of industry participants, the jurisprudence of arbitral tribunals, and the model contracts of bodies such as the Association of International Energy Negotiators (AIEN) have generated a body of transnational norms that operates partly independently of any national legal system.[2]

Critics respond that the concept is either trivially true — industry practice obviously influences the interpretation of industry contracts — or dangerously

[1] *Government of the State of Kuwait v American Independent Oil Company (Aminoil)*, Award of 24 March 1982, (1982) 21 ILM 976.

[2] See R. Doak Bishop, "International Arbitration of Petroleum Disputes: The Development of a *Lex Petrolea*" (1998) 23 *Yearbook of Commercial Arbitration* 1131; and Thomas C.C. Childs, "Update on *Lex Petrolea*: The Continuing Development of Customary Law Relating to International Oil and Gas Exploration and Production" (2011) 4 *Journal of World Energy Law and Business* 214.

overstated, in that it purports to detach contractual relations from the legal systems that ultimately give them force, and that it tends to privilege the interests of international oil companies over those of host states and affected communities. The debate is significant for doctoral researchers because it raises fundamental questions about the nature of law, the legitimacy of private ordering, and the relationship between state and non-state sources of normativity. It is revisited in Chapter 2 in the context of soft law and industry standards and in Chapter 26 in the context of arbitral practice.

1.2 Historical Development of Resource Governance

A historical perspective is indispensable to an understanding of contemporary energy and resource law. Many of the field's defining doctrines — the stabilisation clause, the doctrine of permanent sovereignty, the standard of compensation for expropriation, the treatment of production sharing contracts — were forged in specific historical controversies, and their meaning cannot be grasped without appreciating the circumstances in which they arose. This section traces five broad phases in the development of resource governance.

Phase One: The Classical Concession Era (c. 1900–1950)

The modern international petroleum industry was built on the classical concession. In 1901, William Knox D'Arcy obtained from the Shah of Persia a concession conferring the exclusive right to explore for, produce, and export petroleum over approximately three-quarters of the territory of Persia for a period of sixty years, in return for a modest initial payment and a share of annual net profits. The D'Arcy concession became the template for a generation of agreements across the Middle East, Latin America, and Asia.

The classical concession had several characteristic features. It covered vast areas, often a substantial proportion of the host state's territory. It endured for very long periods, frequently between sixty and ninety-nine years. It conferred upon the concessionaire not merely the right to extract petroleum but ownership of the petroleum once produced, together with broad managerial control over all aspects of operations. The host state's financial return was limited, typically to a fixed royalty per ton of production, and the state had little or no involvement in decisions concerning the rate of production, the pricing of crude oil, or the allocation of markets. The concessions were frequently negotiated between unequal parties, in circumstances in which the host state lacked the geological information, technical expertise, and bargaining power necessary to secure favourable terms.

The legal character of these instruments was ambiguous. From the perspective of the international oil companies and their home states, a concession was a contract that created vested rights protected by international law, and any unilateral modification or termination by the host state constituted a wrong for which

compensation was owed. From the perspective of host states, the concession was an act of sovereign authority — an administrative grant — that the state could revise in the public interest. This tension between contract and sovereignty is one of the enduring themes of the field.

During this period the industry was dominated by a small number of vertically integrated companies — later known as the "Seven Sisters" — which controlled exploration, production, refining, transportation, and marketing. Arrangements such as the Red Line Agreement of 1928, under which the partners in the Iraq Petroleum Company agreed not to compete with one another for concessions within a defined area of the former Ottoman Empire, and the Achnacarry or "As-Is" Agreement of the same year, which sought to stabilise market shares, illustrate the degree to which the international oil market was organised through private arrangements rather than public regulation.

Phase Two: Nationalisation and the Assertion of Sovereignty (1938–1960s)

The first major challenge to the concession system came from Latin America. In 1938, Mexico nationalised the holdings of foreign oil companies following a prolonged labour dispute and a confrontation with the companies over compliance with a decision of the Mexican Supreme Court. The Mexican nationalisation prompted a celebrated exchange of diplomatic notes in which the United States Secretary of State, Cordell Hull, asserted that international law required "prompt, adequate, and effective" compensation for the expropriation of foreign property — the formula that became known as the Hull standard. Mexico rejected the claim that international law imposed any such requirement, maintaining that compensation was a matter for domestic law. The dispute was eventually settled by agreement, but the doctrinal disagreement it exposed persisted for decades.

The second major challenge came in Iran. In 1951, the Iranian parliament enacted legislation nationalising the assets of the Anglo-Iranian Oil Company. The United Kingdom instituted proceedings before the International Court of Justice, but the Court held that it lacked jurisdiction because the relevant Iranian declaration accepting the Court's compulsory jurisdiction was limited to disputes concerning treaties, and the concession of 1933 was not a treaty between Iran and the United Kingdom.[3] The *Anglo-Iranian Oil Co* case is significant for its holding that a concession agreement between a state and a foreign company is not an international treaty, notwithstanding the involvement of the company's home state in its negotiation.

[3] *Anglo-Iranian Oil Co (United Kingdom v Iran)*, Preliminary Objection, Judgment, ICJ Reports 1952, p. 93.

The same period witnessed the gradual replacement of fixed royalty arrangements with profit-sharing formulas. Beginning with Venezuela in 1943 and spreading to Saudi Arabia in 1950, the so-called "fifty-fifty" principle — under which the host government received approximately half of the profits from petroleum operations through a combination of royalties and income taxes — became the industry norm. The fifty-fifty principle represented a significant shift in the distribution of resource rents, but it left the structure of the concession — and the managerial control of the companies — largely intact.

Phase Three: Decolonisation and the New International Economic Order (1960–1980)

The third phase was shaped by decolonisation and the emergence of a large number of newly independent states determined to assert control over their natural resources. Three developments were decisive.

First, in September 1960, Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela founded the Organization of the Petroleum Exporting Countries (OPEC). OPEC's initial objective was to resist reductions in posted prices unilaterally imposed by the companies, but it rapidly became a forum through which producing states coordinated their policies on pricing, taxation, and participation.

Second, the General Assembly of the United Nations adopted a series of resolutions articulating the principle of permanent sovereignty over natural resources. The most important of these, Resolution 1803 (XVII) of 14 December 1962, declared that "the right of peoples and nations to permanent sovereignty over their natural wealth and resources must be exercised in the interest of their national development and of the well-being of the people of the State concerned", while also providing that in cases of nationalisation "the owner shall be paid appropriate compensation, in accordance with the rules in force in the State taking such measures in the exercise of its sovereignty and in accordance with international law", and that foreign investment agreements "freely entered into by or between sovereign States shall be observed in good faith".[4] Resolution 1803 represented a carefully negotiated compromise between capital-exporting and capital-importing states, and it is widely regarded as reflecting customary international law in significant respects.

Third, in the early 1970s, producing states moved from profit-sharing to participation and outright nationalisation. Libya nationalised the interests of several foreign companies between 1971 and 1974; Algeria took a majority interest in French oil companies in 1971; Iraq nationalised the Iraq Petroleum Company in 1972; and the states of the Persian Gulf negotiated progressive participation

[4] UN General Assembly Resolution 1803 (XVII), "Permanent Sovereignty over Natural Resources", 14 December 1962, paras 1, 4, and 8.

agreements that culminated in full state ownership of their producing industries. The oil price increases of 1973–1974, associated with the Arab oil embargo and OPEC's assertion of control over pricing, transformed the economics of the industry and the balance of power between producing states and consuming states.

In 1974, the General Assembly adopted the Declaration on the Establishment of a New International Economic Order and the Charter of Economic Rights and Duties of States.[5] Article 2 of the Charter asserted the right of every state "to nationalize, expropriate or transfer ownership of foreign property, in which case appropriate compensation should be paid by the State adopting such measures, taking into account its relevant laws and regulations and all circumstances that the State considers pertinent", and provided that any controversy over compensation "shall be settled under the domestic law of the nationalizing State and by its tribunals", unless otherwise agreed. Unlike Resolution 1803, the Charter omitted any reference to international law as the governing standard for compensation and was adopted over the opposition or abstention of most capital-exporting states. Its status as a statement of customary international law has accordingly been doubted.

The legal consequences of the nationalisations of this period were addressed in a series of landmark arbitral awards, often referred to collectively as the "Libyan oil arbitrations" — *BP v Libya* (1973), *Texaco v Libya* (1977), and *LIAMCO v Libya* (1977) — together with the *Aminoil* award of 1982. These awards, which are examined in detail in Chapters 3 and 14, grappled with the legal status of concession contracts, the effect of stabilisation clauses, the applicable law, and the appropriate standard of compensation. Their reasoning remains foundational to the modern law of foreign investment.

[5] UN General Assembly Resolution 3201 (S-VI), "Declaration on the Establishment of a New International Economic Order", 1 May 1974; UN General Assembly Resolution 3281 (XXIX), "Charter of Economic Rights and Duties of States", 12 December 1974.



Figure 1.2 Selected milestones in the development of international energy and natural resources law, 1901–2025. *Source:* prepared for this edition.

Phase Four: Liberalisation, Investment Protection, and Market Integration (1980–2005)

The fourth phase was characterised by a reversal of the statist tendencies of the preceding period. The collapse of oil prices in the mid-1980s, the debt crisis in the developing world, the end of the Cold War, and the ascendancy of market-oriented economic policies led many states to reopen their resource sectors to foreign investment. Production sharing contracts — first used in Indonesia in the 1960s — became the dominant contractual form in many developing producer states, allowing governments to retain formal ownership of petroleum while engaging foreign companies as contractors entitled to a share of production. Mining codes across Africa and Latin America were rewritten, often with the support of international financial institutions, to attract foreign capital through security of tenure, fiscal stability, and liberal repatriation of profits.

At the same time, a dense network of bilateral investment treaties (BITs) emerged. From a few hundred in the 1980s, the number of BITs grew to more than two thousand by the early 2000s. These treaties conferred upon foreign investors substantive protections — fair and equitable treatment, full protection and security, protection against expropriation without compensation, national

and most-favoured-nation treatment — and, crucially, the right to bring claims directly against host states before international arbitral tribunals, typically under the auspices of the International Centre for Settlement of Investment Disputes (ICSID) established by the Washington Convention of 1965. The extension of investor–state arbitration transformed the enforcement of investor rights by removing the need for diplomatic protection by the investor's home state.

In the energy sector specifically, the Energy Charter Treaty (ECT), signed in Lisbon in December 1994 and in force since April 1998, established a multilateral framework for investment protection, trade, transit, and energy efficiency among the states of Europe and the former Soviet Union, together with a number of states in Asia. The ECT became the most frequently invoked investment treaty in the world, a fact that would later make it a central battleground in debates over the compatibility of investment protection with climate policy.

The liberalisation of domestic energy markets proceeded in parallel. In the electricity and gas sectors, the United Kingdom privatised its state-owned utilities in the late 1980s and 1990s; the United States introduced open access to transmission networks through Federal Energy Regulatory Commission (FERC) Order No. 888 of 1996; and the European Union adopted successive legislative packages, beginning with the first Electricity Directive of 1996, designed to create an internal energy market through the unbundling of vertically integrated utilities and the introduction of competition in generation and supply.

Phase Five: Resource Nationalism, Sustainability, and the Energy Transition (2005–present)

The fifth phase, which continues today, is marked by a complex and sometimes contradictory set of trends.

First, the commodity price boom of the 2000s prompted a revival of resource nationalism. Venezuela, Bolivia, Ecuador, and Russia, among others, increased state participation, renegotiated contracts, or expropriated foreign interests, generating a new wave of investment arbitrations, several of which resulted in awards of billions of United States dollars. Resource nationalism also took subtler forms, including increased royalties and windfall taxes, local content requirements, export restrictions on unprocessed minerals, and mandatory state participation in new projects.

Second, the international community increasingly emphasised transparency, good governance, and the equitable distribution of resource revenues. The Extractive Industries Transparency Initiative (EITI), launched in 2002 and now implemented in more than fifty countries, requires the disclosure of payments by companies to governments and of revenues received by governments from the extractive sector. Supply chain due diligence standards for conflict minerals, notably

the OECD *Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas*, sought to address the link between mineral extraction and armed conflict. The United Nations *Guiding Principles on Business and Human Rights*, endorsed by the Human Rights Council in 2011, articulated the responsibility of business enterprises, including extractive companies, to respect human rights.

Third, and most consequentially, climate change has moved from the margins to the centre of energy law. The Paris Agreement of 2015 set the objective of holding the increase in global average temperature to well below 2°C above pre-industrial levels and pursuing efforts to limit the increase to 1.5°C. At the twenty-eighth Conference of the Parties to the United Nations Framework Convention on Climate Change (COP28) in Dubai in 2023, the parties for the first time called in the outcome of the first global stocktake for "transitioning away from fossil fuels in energy systems, in a just, orderly and equitable manner". Climate litigation has proliferated in domestic courts, regional human rights bodies, and international tribunals, culminating in advisory opinions from the International Tribunal for the Law of the Sea (2024), the Inter-American Court of Human Rights (2025), and the International Court of Justice (2025) on the obligations of states in respect of climate change. These developments are examined in Chapter 18.

Fourth, the energy transition has created new legal frontiers: the regulation of renewable energy, hydrogen, energy storage, carbon capture and storage, and offshore wind; the governance of critical minerals essential to batteries, electric vehicles, and grid infrastructure; and the management of stranded assets and the decommissioning of fossil fuel infrastructure. Simultaneously, geopolitical events — notably the Russian invasion of Ukraine in February 2022 and the resulting disruption of European gas supply — have restored energy security to the forefront of policy and have demonstrated that the transition will be shaped as much by security considerations as by environmental ones.

KEY DOCTRINAL POINT

The history of resource governance is not a linear progression from one paradigm to another. Elements of each phase coexist in contemporary law. Classical concession-style licences remain in use in many jurisdictions; the doctrine of permanent sovereignty retains its normative force; investment treaties continue to protect investors even as they are reformed or terminated; and the imperatives of the energy transition are layered on top of these older structures rather than replacing them. A doctoral analysis must therefore be historically literate: it must recognise which layer of the law a given rule belongs to and how that rule has been reinterpreted over time.

1.3 Internationalisation of Energy Markets and Resource Regulation

The Global Character of Energy Markets

Energy markets are among the most internationalised of all economic sectors. Crude oil is a global commodity, traded on international exchanges and priced by reference to benchmark grades such as Brent and West Texas Intermediate. A substantial proportion of the world's oil production crosses at least one international border before it is consumed, often passing through maritime chokepoints such as the Strait of Hormuz and the Strait of Malacca. Natural gas markets, historically regional because of the constraints of pipeline transport, have become increasingly integrated through the growth of the liquefied natural gas (LNG) trade, which permits gas to be shipped across oceans and redirected in response to price signals. Coal is traded internationally in large volumes. Electricity, while still predominantly national, is increasingly traded across borders through interconnectors and regional power pools. The supply chains for renewable energy technologies — solar panels, wind turbines, batteries — are global, with manufacturing concentrated in a small number of states.

The internationalisation of markets has legal consequences. Domestic regulatory decisions in one state — on export restrictions, subsidies, environmental standards, or sanctions — can have profound effects on prices and supplies in others. Commercial disputes arising from cross-border transactions require rules on applicable law, jurisdiction, and the enforcement of judgments and awards. Infrastructure that crosses borders, such as pipelines and interconnectors, requires intergovernmental agreements to address transit, tariffs, and the allocation of regulatory authority.

Mechanisms of Internationalisation

The internationalisation of energy and resource regulation has proceeded through several mechanisms.

Treaty-making. States have concluded treaties addressing particular aspects of energy and resource activities: investment protection (BITs and the ECT), trade (the WTO agreements and regional trade agreements), transboundary pipelines (for example, the intergovernmental agreements underlying the Baku–Tbilisi–Ceyhan pipeline and the West African Gas Pipeline), nuclear safety and liability, the law of the sea, and environmental protection.

Institutional cooperation. International organisations such as the International Energy Agency (IEA), the International Atomic Energy Agency (IAEA), the International Renewable Energy Agency (IRENA), OPEC, and the Gas Exporting Countries Forum provide forums for the coordination of national policies, the exchange of information, and, in some cases, the adoption of binding

standards. Regional bodies — the Energy Community in South-East Europe, the West African Power Pool, the Southern African Power Pool — promote regional market integration.

Regulatory convergence. States often borrow regulatory models from one another, whether through voluntary emulation, the conditions attached to development finance, or the technical assistance of international organisations. The spread of independent regulatory agencies, of unbundling requirements in electricity, and of model petroleum agreements reflects this process of regulatory transplantation, which is examined in Chapter 24.

Private ordering. The standard contracts produced by industry associations — the AIEN Model Joint Operating Agreement, the International Federation of Consulting Engineers (FIDIC) suites of construction contracts, and the master agreements used in energy trading — create transnational norms that operate across jurisdictions.

Extraterritorial regulation. Some states regulate conduct outside their territory through legislation with extraterritorial effect. Examples include anti-corruption statutes such as the United States Foreign Corrupt Practices Act of 1977 and the United Kingdom Bribery Act 2010, sanctions regimes targeting the energy sectors of particular states, and supply chain due diligence obligations such as those contained in the European Union's Corporate Sustainability Due Diligence Directive and Batteries Regulation. These are discussed in Chapters 2 and 22.

The Limits of Internationalisation

Notwithstanding these developments, energy and resource regulation remains in important respects national. There is no comprehensive multilateral energy treaty of universal application; the ECT, the most ambitious attempt to create such an instrument, never attracted the participation of major producers and consumers such as the United States, China, Saudi Arabia, or Brazil, and its membership has contracted in recent years following withdrawals by the European Union and several of its member states. Petroleum, mining, and electricity remain subject primarily to domestic legislation, and the doctrine of permanent sovereignty ensures that the fundamental decisions about whether, when, and on what terms to develop natural resources are reserved to the state in which they are located. The internationalisation of energy law is thus partial, uneven, and contested — characteristics that make it a fertile subject for doctoral research.

1.4 Public Law, Private Law, and Transnational Legal Relations

The Public–Private Divide in Energy and Resource Law

One of the distinctive features of energy and resource law is the extent to which it straddles the conventional divide between public and private law. On the public side, the state acts as sovereign: it owns or controls subsurface resources, grants licences, sets fiscal terms, regulates conduct, protects the environment, and resolves disputes through its courts. On the private side, the state frequently acts as a commercial party: it enters into contracts with investors, participates in joint ventures through national oil or mining companies, sells resources on international markets, and borrows money to finance its share of project costs.

The dual character of the state — as sovereign regulator and as contracting party — generates persistent legal difficulties. When a state that has entered into a contract with a foreign investor subsequently changes its tax laws or environmental regulations in a manner that adversely affects the investor, is it acting as a sovereign exercising its regulatory powers, or as a contracting party in breach of its contractual obligations? The answer may determine whether the investor has a remedy, which tribunal has jurisdiction, and what law applies. The distinction between acts *jure imperii* and acts *jure gestionis*, familiar from the law of state immunity, and the distinction in investment law between a mere breach of contract and a breach of treaty, are both attempts to grapple with this problem.

The State Contract

The "state contract" — a contract between a state (or a state entity) and a foreign private party — is the central legal instrument of international resource development. Petroleum concessions, production sharing contracts, mining agreements, host government agreements for pipelines, and power purchase agreements with state-owned utilities are all species of state contract.

The legal status of state contracts has been debated for more than a century. Three principal positions may be distinguished.

The first position holds that state contracts are governed exclusively by the domestic law of the host state. On this view, the state retains its sovereign power to modify the contract by legislation, and the investor's protection lies only in such guarantees as domestic law provides. This position was associated with the Calvo doctrine in Latin America and with the claims of the New International Economic Order.

The second position holds that state contracts may be "internationalised" — that is, subjected to international law or to general principles of law — by the choice of the parties, by the inclusion of stabilisation and arbitration clauses, or by the nature of the contract as an economic development agreement. This

position found its fullest expression in the award of Sole Arbitrator René-Jean Dupuy in *Texaco v Libya*, which held that the concession contracts at issue had been internationalised and that Libya's nationalisation measures were a breach of its contractual obligations under international law.[6]

The third position, now widely accepted, holds that the contract is governed by the law chosen by the parties (often the host state's law, sometimes supplemented by international law or general principles), but that the investor may also enjoy independent protection under international law — most importantly through investment treaties — which may be invoked even where the contract itself is governed by domestic law. The distinction between contract claims and treaty claims, articulated by the *ad hoc* Committee in *Compañía de Aguas del Aconquija SA and Vivendi Universal v Argentina* (*Vivendi I*), is now a cornerstone of investment arbitration: "whether there has been a breach of the BIT and whether there has been a breach of contract are different questions".[7]

Transnational Legal Relations

Beyond the public–private divide lies a further category of relations that is neither purely public nor purely private, neither purely national nor purely international. Philip Jessup, writing in 1956, coined the term "transnational law" to include "all law which regulates actions or events that transcend national frontiers", embracing both public and private international law as well as "other rules which do not wholly fit into such standard categories".[8] Energy and resource activities exemplify transnational legal relations. A single offshore gas project may involve a host government agreement with the state, a production sharing contract with the national oil company, a joint operating agreement among the co-venturers, engineering contracts with international contractors, LNG sales agreements with buyers in several countries, financing agreements with commercial banks and export credit agencies, and an investment treaty between the host state and the home states of the investors. Each relationship is governed by a different combination of national and international rules, and disputes may be resolved in different forums.

The transnational character of these relations has three important implications. First, it means that no single legal order governs the project as a whole; the applicable law must be determined relationship by relationship. Second, it creates the risk of conflicting obligations, as when an environmental obligation imposed by domestic law conflicts with a stabilisation clause in a contract, or a

[6] *Texaco Overseas Petroleum Company and California Asiatic Oil Company v Government of the Libyan Arab Republic*, Award on the Merits of 19 January 1977, (1978) 17 ILM 1; (1979) 53 ILR 389.

[7] *Compañía de Aguas del Aconquija SA and Vivendi Universal v Argentine Republic*, ICSID Case No ARB/97/3, Decision on Annulment, 3 July 2002, para 96.

[8] Philip C. Jessup, *Transnational Law* (Yale University Press 1956), p. 2.

sanctions regime imposed by one state conflicts with a contractual obligation to supply. Third, it confers significant influence upon private actors — companies, banks, industry associations, arbitrators — in the creation and interpretation of norms, raising questions of legitimacy and accountability that recur throughout this book.

1.5 The Relationship Between Energy Law and International Economic Law

Energy as an Exception in Trade Law

The relationship between energy law and international economic law is close but historically uneasy. When the General Agreement on Tariffs and Trade (GATT) was negotiated in 1947, energy products were not excluded from its scope, but in practice trade in oil and gas was largely left outside the multilateral trading system. Many major producers were not parties to the GATT; the trade in crude oil was conducted largely within vertically integrated companies; and energy policy was regarded as so closely bound up with national security that states were unwilling to subject it to multilateral discipline. The GATT security exception in Article XXI, which permits measures "relating to fissionable materials or the materials from which they are derived" and measures "taken in time of war or other emergency in international relations", reinforced this sense of energy as a domain of sovereign prerogative.

The establishment of the WTO in 1995 and the accession of major energy producers — including Saudi Arabia in 2005 and the Russian Federation in 2012 — brought energy trade more firmly within the multilateral framework. WTO disputes have since addressed export restrictions on raw materials (*China — Raw Materials* and *China — Rare Earths*), local content requirements in renewable energy programmes (*Canada — Renewable Energy/Feed-in Tariff Program* and *India — Solar Cells*), and the European Union's energy market regulation (*EU — Energy Package*). These disputes are examined in Chapter 13.

Investment Law as the Primary Interface

If trade law has historically played a limited role in energy, investment law has played a central one. Energy and mining projects are capital-intensive, long-lived, and dependent on the stability of the host state's legal and fiscal regime; they are therefore particularly vulnerable to what economists call the "obsolescing bargain" — the tendency for the host state's bargaining power to increase once the investor's capital has been sunk. Investment treaties and investor–state arbitration were designed in large part to mitigate this risk. Statistics compiled by ICSID consistently show that the oil, gas, and mining sector, together with electric power

and other energy, account for a larger share of registered cases than any other economic sector.

The obsolescing bargain theory, associated with the work of Raymond Vernon, explains much of the dynamic of resource governance. At the time of the initial investment decision, the host state competes with other states for scarce capital and expertise and must offer attractive terms. Once the investment has been made, however, the investor's assets are immobile and the host state may seek to renegotiate the terms in its favour — through increased taxation, contract renegotiation, or expropriation. Investors anticipate this risk and demand protections: stabilisation clauses, international arbitration, investment treaty coverage, and political risk insurance. The law of international energy investment can be understood, in significant part, as an attempt to manage the obsolescing bargain.

Convergence and Tension

The relationship between energy law and international economic law is characterised by both convergence and tension. Convergence occurs where economic law supports energy policy objectives — for example, where investment protection encourages the capital flows necessary for the energy transition. Tension occurs where economic law constrains the regulatory autonomy that states need to pursue energy and climate objectives — for example, where investors challenge the phase-out of coal-fired power or the withdrawal of renewable energy subsidies, or where trade rules are invoked against local content requirements designed to build domestic manufacturing capacity. The phase-out of coal-fired power in the Netherlands gave rise to ECT claims by RWE and Uniper (the latter subsequently withdrawn), and the retroactive reduction of renewable energy incentives in Spain generated dozens of ECT arbitrations. The resulting debates about regulatory chill, the right to regulate, and the reform of investment treaties are examined in Chapter 14.

1.6 Legal Pluralism and Competing Regulatory Authorities

The Concept of Legal Pluralism

Legal pluralism describes situations in which two or more legal orders coexist in the same social field. Originally developed by legal anthropologists to describe the coexistence of state law and customary law in colonial and post-colonial societies, the concept has been extended to describe the coexistence of national, international, regional, and private normative orders in a globalised world. Energy and resource governance is a paradigmatic site of legal pluralism.

At least five layers of normative authority may be identified in a typical resource project:

1. **Customary and community norms.** In many resource-rich regions, land and resources are held under customary tenure systems that predate, and

coexist with, state law. In Nigeria, for example, customary land tenure was overlaid by the Land Use Act 1978, which vested land in each state in the Governor, while section 44(3) of the Constitution of the Federal Republic of Nigeria 1999 vests the entire property in and control of all minerals, mineral oils, and natural gas in the Government of the Federation. Communities in the Niger Delta nevertheless continue to assert claims to land and resources grounded in custom, and these claims shape the conduct of operations and the content of community development agreements.

2. **Subnational law.** In federal states, constitutional authority over resources may be divided between the federal government and the constituent units. In Canada, the provinces own and regulate most natural resources under section 92A of the Constitution Act 1867 (inserted in 1982); in Australia, the states own onshore minerals and petroleum; in the United States, mineral rights may be held privately, by the states, or by the federal government. In Nigeria, the Electricity Act 2023, enacted following a constitutional amendment, empowers states to regulate electricity markets within their territories.
3. **National law.** Constitutional provisions, petroleum and mining statutes, environmental legislation, tax law, and administrative law form the core of the regulatory framework.
4. **International and regional law.** Treaties, customary international law, and the law of regional organisations (such as the European Union, the Economic Community of West African States, and the African Union) impose obligations on states and, in some cases, confer rights on private parties.
5. **Transnational private regulation.** Industry standards, model contracts, voluntary codes of conduct, certification schemes, and the environmental and social performance standards of lenders — notably the International Finance Corporation (IFC) *Performance Standards on Environmental and Social Sustainability* and the Equator Principles adopted by commercial banks — impose requirements on companies that may exceed those of national law.

Sources of Conflict

The coexistence of multiple normative orders generates conflicts of several kinds. *Vertical* conflicts arise between different levels of authority — for example, between federal and state governments over the ownership of offshore resources, as in Nigeria's resource-control litigation, in which the Supreme Court held in *Attorney-General of the Federation v Attorney-General of Abia State* that for the purposes of revenue allocation the seaward boundary of a littoral state was the

low-water mark of its land surface.[9] *Horizontal* conflicts arise between different regimes at the same level — for example, between investment treaty obligations and environmental obligations. *Public–private* conflicts arise between state regulation and private standards or contracts. *Jurisdictional* conflicts arise between courts and tribunals asserting authority over the same dispute, as when an investor pursues parallel proceedings before domestic courts, commercial arbitration, and investment treaty arbitration.

Managing Pluralism

Legal systems have developed a range of techniques for managing these conflicts: hierarchical rules of priority (constitutional supremacy, the primacy of European Union law), principles of interpretation (systemic integration under Article 31(3)(c) of the Vienna Convention on the Law of Treaties, the presumption against conflict), doctrines of comity and deference (the margin of appreciation, the police powers doctrine in investment law), procedural mechanisms (fork-in-the-road and waiver clauses, *lis pendens*, *res judicata*), and institutional coordination (memoranda of understanding between regulators, joint commissions).

The International Law Commission's study of the fragmentation of international law, completed in 2006 under the chairmanship of Martti Koskenniemi, remains the most authoritative analysis of these techniques at the international level.[10] For energy and resource law, the key insight of the fragmentation debate is that specialised regimes — investment law, trade law, environmental law, human rights law — do not operate in isolation; they remain embedded in general international law, and their interpretation must take account of other relevant rules applicable between the parties.

> **COMPARATIVE PERSPECTIVE**
>
> **Federalism and resource authority in three jurisdictions.** In *Canada*, section 92A of the Constitution Act 1867 confirms provincial legislative authority over the exploration, development, conservation, and management of non-renewable natural resources, subject to federal paramountcy for interprovincial export. In *Nigeria*, section 44(3) of the 1999 Constitution vests ownership and control of minerals and petroleum in the Federal Government, and item 39 of the Exclusive Legislative List reserves "mines and minerals, including oil fields, oil mining, geological surveys and natural gas" to the National Assembly; the derivation principle in section 162(2) allocates not less than thirteen per cent of revenue accruing from natural resources

[9] *Attorney-General of the Federation v Attorney-General of Abia State & 35 Others* (2002) 6 NWLR (Pt 764) 542 (Supreme Court of Nigeria).

[10] International Law Commission, *Fragmentation of International Law: Difficulties Arising from the Diversification and Expansion of International Law*, Report of the Study Group finalised by Martti Koskenniemi, UN Doc A/CN.4/L.682 (13 April 2006).

to the states of origin. In the *United States*, subsurface minerals may be privately owned and severed from the surface estate, with federal and state governments owning substantial mineral estates in their own right and regulating activities on federal and state lands respectively. These differences shape the bargaining power of communities, the structure of fiscal regimes, and the incidence of resource conflicts.

1.7 Contemporary Challenges in Global Resource Governance

The field confronts a set of challenges that are without precedent in their scale and complexity. Six are of particular importance.

The Energy Transition and Stranded Assets

The first challenge is the transition from an energy system dominated by fossil fuels to one based predominantly on low-carbon sources. The legal implications are profound. Petroleum and coal licences granted for decades may become economically or legally unviable before their expiry, raising questions about compensation, decommissioning liability, and the protection of legitimate expectations. Investors in fossil fuel infrastructure may seek to rely on investment treaties to challenge phase-out measures. Producer states whose economies depend on hydrocarbon revenues face fiscal crises and must manage the social consequences of transition. The concept of "stranded assets" — assets that suffer from premature write-downs or devaluation as a result of the transition — has migrated from financial analysis into law, influencing disclosure obligations, directors' duties, and the interpretation of investment protections.

Critical Minerals and the New Extractivism

The second challenge arises from the paradox that decarbonisation requires more extraction, not less. The IEA has repeatedly projected that demand for minerals such as lithium, cobalt, nickel, copper, graphite, and rare earth elements will grow severalfold under scenarios consistent with the Paris Agreement. These minerals are geographically concentrated, and their processing is concentrated still further. States have responded with industrial policies, export restrictions, strategic stockpiles, and bilateral critical mineral agreements. Communities in mining regions confront the risk that the green transition will reproduce the environmental degradation and human rights abuses associated with earlier extractive booms. Chapter 8 addresses these issues in detail.

Energy Access and Energy Poverty

The third challenge is that a substantial proportion of humanity still lacks access to modern energy services. Hundreds of millions of people, the majority in sub-Saharan Africa, lack access to electricity, and far more lack access to clean cooking fuels. Sustainable Development Goal 7 commits states to ensure access

to affordable, reliable, sustainable, and modern energy for all by 2030. Achieving universal access while simultaneously decarbonising energy systems requires legal and regulatory frameworks that mobilise investment in decentralised and grid-based solutions, protect consumers, and allocate the costs of transition fairly. The concept of energy justice, explored in Chapter 23, provides a framework for analysing these distributional questions.

Geopolitics and Energy Security

The fourth challenge is the re-emergence of energy as an instrument of geopolitical competition. The disruption of gas supplies to Europe following the invasion of Ukraine, the use of sanctions against major energy exporters, attacks on energy infrastructure, and competition over critical mineral supply chains have demonstrated that energy security remains a central concern of state policy. The law governing sanctions, export controls, emergency powers, strategic reserves, and the protection of critical infrastructure — examined in Chapter 25 — has accordingly grown in importance.

Legitimacy of Investor–State Dispute Settlement

The fifth challenge concerns the legitimacy of the system of investor–state dispute settlement (ISDS). Critics argue that ISDS privileges foreign investors over domestic stakeholders, constrains the regulatory autonomy of states, produces inconsistent decisions, and lacks transparency and accountability. Reform efforts proceed in several forums: the United Nations Commission on International Trade Law (UNCITRAL) Working Group III, the European Union's proposal for a multilateral investment court, the modernisation of the ECT, and the renegotiation or termination of BITs by individual states. The outcome of these reforms will shape the legal environment for energy investment for decades.

Technological Change and Digitalisation

The sixth challenge arises from technological change. Hydrogen, advanced nuclear reactors, carbon capture, battery storage, offshore floating wind, and deep-sea mining all raise novel regulatory questions. Digitalisation — smart meters, distributed energy resources, automated trading, artificial intelligence in grid management — raises questions of data ownership, cybersecurity, and liability. Regulatory frameworks designed for centralised, fossil-fuel-based systems must be adapted to decentralised, digital, and decarbonised systems. Chapters 12 and 30 address these developments.

1.8 Doctoral Research Perspectives and Foundational Debates

The Nature of Doctoral Contribution

Doctoral research is distinguished from other forms of legal writing by the requirement of an original contribution to knowledge. In international energy

and natural resources law, such a contribution may take several forms: the identification and analysis of a new legal problem; a new interpretation of existing rules or cases; the application of a theoretical framework to a body of law; a comparative study revealing previously unrecognised patterns; an empirical investigation of how the law operates in practice; or a reasoned proposal for reform. Chapter 28 discusses research methodology in detail. Here, the purpose is to identify the foundational debates within which most doctoral research in the field is situated.

Debate One: Sovereignty versus Stability

The most enduring debate in the field concerns the tension between the sovereign right of states to regulate their natural resources in the public interest and the need of investors for legal and fiscal stability. The debate manifests itself in disputes over stabilisation clauses, the standard of compensation for expropriation, the scope of the fair and equitable treatment standard, and the legitimacy of contract renegotiation. Is the doctrine of permanent sovereignty compatible with binding long-term commitments to foreign investors? Can a state contract away its legislative power? How should the law balance the legitimate expectations of investors against the evolving needs of the host society?

Debate Two: Contract versus Regulation

A second debate concerns the relative roles of contract and regulation in governing resource activities. In some jurisdictions, the terms of resource development are set primarily by individually negotiated contracts; in others, primarily by general legislation and regulation applicable to all licensees. Each approach has advantages and drawbacks. Contracts offer flexibility and can be tailored to the circumstances of particular projects, but they are vulnerable to corruption, asymmetries of information, and lack of transparency. Regulation offers predictability, equality of treatment, and democratic accountability, but may be less responsive to project-specific risks. The trend in many jurisdictions, including Nigeria under the Petroleum Industry Act 2021, has been towards greater reliance on legislative frameworks, but contracts remain central.

Debate Three: Efficiency versus Equity

A third debate concerns the objectives of resource law. Economic approaches emphasise efficiency — the maximisation of resource rents, the minimisation of transaction costs, and the allocation of risk to the party best able to bear it. Equity-oriented approaches emphasise the fair distribution of benefits and burdens among states, investors, communities, and future generations, and the protection of the rights of affected persons. The "resource curse" literature, which examines why resource-rich states often experience slower growth, weaker institutions, and greater conflict than resource-poor states, has highlighted the importance of

institutional quality and has prompted reforms focused on transparency, revenue management, and accountability.

Debate Four: Development versus Sustainability

A fourth debate concerns the relationship between resource development and environmental sustainability. Developing states argue that they have a right to exploit their resources to finance development, as industrialised states did before them, and that the burden of climate mitigation should fall primarily on those states responsible for historical emissions. This argument is reflected in the principle of common but differentiated responsibilities and respective capabilities, articulated in the United Nations Framework Convention on Climate Change. At the same time, the physical limits of the carbon budget mean that a substantial proportion of known fossil fuel reserves cannot be burned if the temperature goals of the Paris Agreement are to be met. How should the right to development be reconciled with the imperative of decarbonisation? Who should bear the costs of leaving resources in the ground?

Debate Five: State-Centrism versus Transnational Governance

A fifth debate concerns the appropriate unit of analysis. Traditional approaches treat the state as the principal actor and source of law. Transnational approaches emphasise the role of corporations, international organisations, civil society, communities, and private standard-setters in creating and enforcing norms. The rise of corporate climate litigation, supply chain due diligence obligations, and private sustainability standards suggests that governance is increasingly polycentric. Doctoral research must decide how to account for this complexity without losing analytical clarity.

LEADING AUTHORITY

Texaco Overseas Petroleum Co v Libyan Arab Republic (1977)

Facts. Libya had granted deeds of concession to two American companies under its Petroleum Law of 1955. The concessions contained a stabilisation clause providing that the contractual rights expressly created by the concession should not be altered except by mutual consent, and an arbitration clause. In 1973 and 1974 Libya nationalised the companies' interests. The companies invoked arbitration; Libya refused to participate, and a sole arbitrator was appointed by the President of the International Court of Justice.

Holding. Sole Arbitrator Dupuy held that the concessions were internationalised contracts; that they were governed by principles of Libyan law common to international law and, in the absence of such common principles, by general principles of law; that the stabilisation clause was valid and binding; and that Libya had breached its obligations. Notably, he ordered *restitutio in integrum* — specific performance — as the primary remedy, although the parties subsequently settled.

Significance. The award is the high-water mark of the internationalisation theory. Its analysis of Resolution 1803 as reflecting customary law, and its treatment of the later Charter of Economic Rights and Duties of States as lacking that status, remain influential. Its remedy of restitution has rarely been followed; later tribunals, beginning with *LIAMCO* and *Aminoil*, preferred damages.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

The Kestrel Basin concession (hypothetical). The Republic of Aranda, a fictional coastal state, granted a thirty-year petroleum concession in 1998 to Northwind Energy plc, a company incorporated in a fictional state with which Aranda has a bilateral investment treaty. The concession contains (i) a choice of Arandan law "supplemented by such principles of international law as may be applicable"; (ii) a "freezing" stabilisation clause; and (iii) ICC arbitration seated in Geneva. In 2025, following discoveries of significant gas reserves and a change of government, Aranda enacts a Petroleum Revenue Act that raises the royalty rate from 8 to 15 per cent, imposes a windfall profits tax, and requires all existing concessionaires to convert to production sharing contracts within two years.

Prepare a memorandum of not more than 2,500 words that: (a) identifies each legal order potentially applicable to the dispute; (b) explains the competing characterisations of the measures as sovereign regulation or contractual breach; (c) assesses the legal effect of the stabilisation clause in the light of the doctrine of permanent sovereignty; and (d) recommends a dispute resolution strategy for Northwind, identifying the risks of parallel proceedings. *All parties, places, and instruments in this exercise are fictitious.*

CHAPTER SUMMARY

- International energy and natural resources law is a functional, pluralist, distributive, and transnational field concerned with the allocation of rights over energy sources and natural resources, the regulation of activities involving them, and the distribution of the resulting benefits and burdens.
- The field integrates energy law and natural resources law, a combination justified by shared history, doctrine, institutions, and — increasingly — the material interdependence created by the energy transition.
- Its history may be organised into five overlapping phases: the classical concession era, nationalisation, decolonisation and the New International Economic Order, liberalisation and investment protection, and the current era of resource nationalism, sustainability, and transition.
- Energy markets are highly internationalised, but energy regulation remains predominantly national; internationalisation proceeds through treaties, institutions, regulatory convergence, private ordering, and extraterritorial regulation.

- The dual character of the state as sovereign and contracting party is the source of many of the field's enduring problems, notably the status of state contracts and the distinction between contract and treaty claims.
- Legal pluralism is a defining feature of resource governance, requiring techniques of conflict management drawn from constitutional, international, and private law.
- Doctoral research is situated within five foundational debates: sovereignty versus stability, contract versus regulation, efficiency versus equity, development versus sustainability, and state-centrism versus transnational governance.

DOCTORAL RESEARCH QUESTIONS

1. To what extent does the concept of *lex petrolea* describe a genuine transnational legal order, and what are the implications of its existence or non-existence for the legitimacy of investor–state arbitration?
2. Has the energy transition rendered the integrated study of energy and natural resources law analytically indispensable, or does it obscure important differences between extractive and non-extractive energy sources?
3. How should the obsolescing bargain theory be revised to account for renewable energy projects, in which the host state's bargaining power may evolve differently from that in petroleum projects?
4. Can the techniques developed to manage the fragmentation of international law adequately address conflicts between investment protection and climate obligations, or is structural reform required?
5. What role should customary land tenure and community norms play in the allocation of resource rights in states, such as Nigeria, where the Constitution vests mineral ownership in the federal government?

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2 CHAPTER

Sources, Principles, and Institutions of International Energy Law

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Identify the formal sources of international energy and natural resources law and explain how each operates in the energy and resource context.
2. Analyse the content, legal status, and limits of the principle of permanent sovereignty over natural resources.
3. Apply the rules of state responsibility to conduct affecting energy investments, infrastructure, and the environment.
4. Explain the principles of jurisdiction, including the conditions under which states may regulate extraterritorially.
5. Evaluate the contribution of international organisations, soft law, industry standards, and corporate governance instruments to the field.
6. Assess the role of international courts and arbitral tribunals and analyse conflicts between domestic and international legal obligations.

Introduction

The preceding chapter described international energy and natural resources law as a pluralist field drawing upon multiple bodies of law. This chapter examines the materials from which that law is made — its sources — and the general principles and institutions through which it operates. It begins with the formal sources of international law identified in Article 38(1) of the Statute of the International Court of Justice and considers how each functions in the energy context. It then examines the principle that more than any other defines the field: permanent sovereignty over natural resources. The chapter proceeds to the law of state responsibility, the principles of jurisdiction, and the growing role of international organisations, soft law, and industry standards. It concludes with an analysis of

the institutions that adjudicate energy disputes and of the persistent problem of conflicts between domestic and international legal obligations.

A preliminary observation is in order. Students trained in a domestic legal system sometimes approach international law expecting a hierarchy of norms comparable to that of a national constitution, with clearly defined law-making institutions and a supreme court at the apex. International law does not work in that way. It is a decentralised system in which the subjects of the law — principally states — are also its makers, its interpreters, and, to a considerable extent, its enforcers. The sources of international law are therefore best understood not as a hierarchy but as a set of recognised processes through which states, and increasingly other actors, generate obligations. This decentralisation is particularly pronounced in the energy field, where there is no single governing treaty and where much of the normative work is done by contracts, standards, and the decisions of arbitral tribunals.

2.1 Treaties, Custom, and General Principles of International Law

Article 38 of the Statute of the International Court of Justice

Article 38(1) of the Statute of the International Court of Justice directs the Court, in deciding disputes in accordance with international law, to apply:

> (a) international conventions, whether general or particular, establishing rules expressly recognized by the contesting states;
> (b) international custom, as evidence of a general practice accepted as law;
> (c) the general principles of law recognized by civilized nations;
> (d) subject to the provisions of Article 59, judicial decisions and the teachings of the most highly qualified publicists of the various nations, as subsidiary means for the determination of rules of law.

Although Article 38 is formally a direction to the Court, it is universally regarded as an authoritative statement of the sources of international law. The archaic reference to "civilized nations" is today understood to refer to the community of nations generally. Article 38 does not establish a formal hierarchy among treaties, custom, and general principles, although in practice a treaty, as *lex specialis*, will usually prevail over a conflicting rule of custom as between the parties to it, subject always to the overriding force of peremptory norms (*jus cogens*).

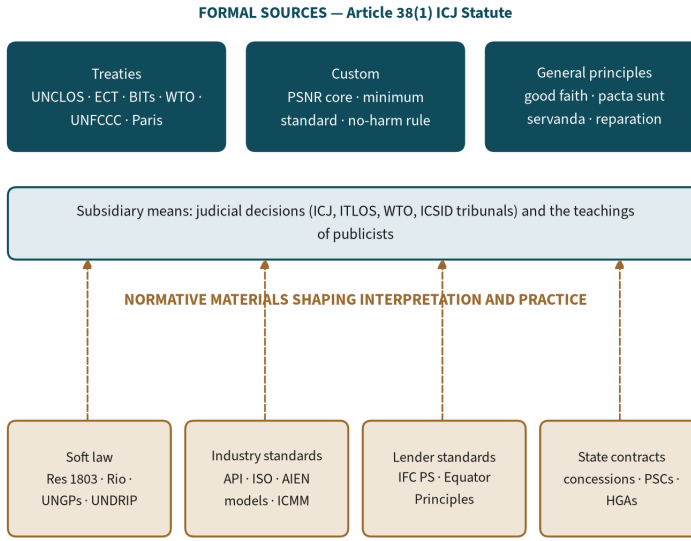


Figure 2.1 The sources of international energy and natural resources law. Formal sources under Article 38(1) of the ICJ Statute are supplemented by soft law, contracts, and transnational standards that shape practice and interpretation. *Source:* prepared for this edition.

Treaties

Treaties are the most important source of international energy law in terms of volume and specificity. The Vienna Convention on the Law of Treaties 1969 (VCLT) defines a treaty as "an international agreement concluded between States in written form and governed by international law, whether embodied in a single instrument or in two or more related instruments and whatever its particular designation". Treaties relevant to energy and natural resources may be classified as follows.

Multilateral treaties of general application that apply to energy and resource activities without being specifically directed at them. These include the United Nations Convention on the Law of the Sea 1982 (UNCLOS), the WTO agreements, the ICSID Convention 1965, the New York Convention on the Recognition and Enforcement of Foreign Arbitral Awards 1958, the United Nations Framework Convention on Climate Change 1992 (UNFCCC), the Paris Agreement 2015, and the Convention on Biological Diversity 1992.

Multilateral treaties specific to energy. The Energy Charter Treaty 1994 is the principal example of a multilateral treaty dedicated to energy. Others include the Statute of the International Atomic Energy Agency 1956, the Agreement on an International Energy Program 1974 (which established the IEA), the Statute of the International Renewable Energy Agency 2009, and the suite of nuclear

conventions on safety, security, liability, and waste management considered in Chapter 11.

Regional treaties, including the Treaty Establishing the Energy Community 2005, the ECOWAS Energy Protocol 2003 (modelled on the ECT), and the energy provisions of the Treaty on the Functioning of the European Union, notably Article 194.

Bilateral treaties, including more than two thousand bilateral investment treaties, tax treaties, maritime boundary agreements, joint development agreements, unitisation agreements for transboundary reservoirs, and intergovernmental agreements for cross-border pipelines and interconnectors.

Treaty Interpretation

Because so much of the field turns on the interpretation of treaties — investment treaties in particular — a doctoral candidate must have a thorough command of the rules of interpretation in Articles 31 to 33 of the VCLT, which reflect customary international law. Article 31(1) provides that a treaty "shall be interpreted in good faith in accordance with the ordinary meaning to be given to the terms of the treaty in their context and in the light of its object and purpose". Article 31(3) requires account to be taken, together with the context, of subsequent agreements between the parties regarding interpretation, subsequent practice establishing agreement regarding interpretation, and "any relevant rules of international law applicable in the relations between the parties". Article 32 permits recourse to supplementary means of interpretation, including the preparatory work of the treaty and the circumstances of its conclusion, to confirm the meaning resulting from Article 31 or to determine the meaning when that interpretation leaves it ambiguous or obscure or leads to a manifestly absurd or unreasonable result.

Three interpretive issues recur in energy disputes. The first is the weight to be given to the *object and purpose* of investment treaties: tribunals that emphasise the promotion and protection of investment, often stated in treaty preambles, have tended to adopt expansive interpretations of investor protections, while others insist that the object and purpose include a balance between investment protection and the host state's right to regulate. The second is the use of *subsequent agreements and practice*, including joint interpretive statements issued by treaty parties, such as the interpretive note issued by the NAFTA Free Trade Commission in 2001 limiting the minimum standard of treatment to the customary international law minimum standard. The third is *systemic integration* under Article 31(3)(c), which permits tribunals to take account of environmental, human rights, and climate obligations in interpreting investment treaties.

Customary International Law

Customary international law arises from a general practice of states accepted as law (*opinio juris*). Both elements must be established: practice that is sufficiently widespread, representative, and consistent, and the acceptance of that practice as legally required or permitted. The International Law Commission's *Draft Conclusions on Identification of Customary International Law*, adopted in 2018, provide authoritative guidance on the evidence of each element.

Customary international law performs several functions in the energy field. It supplies rules of general application where no treaty exists, such as the rules on the protection of foreign property and the standard of compensation for expropriation, the obligation to prevent significant transboundary harm, and the rules on state responsibility. It fills gaps in treaty regimes and informs their interpretation. And in some areas — notably the minimum standard of treatment of aliens — treaty provisions expressly incorporate customary standards.

The identification of custom in the energy field is often difficult. Investment treaties are numerous and broadly similar in content, prompting debate over whether they have generated or codified customary rules. In *Barcelona Traction*, the International Court of Justice observed that, in the field of foreign investment, the law had developed through bilateral and multilateral agreements rather than general rules,[1] and this remains a contested question. The dominant view is that the proliferation of BITs has not of itself transformed their substantive standards into custom, because states conclude such treaties to obtain reciprocal benefits rather than out of a sense of legal obligation, although the treaties may serve as evidence of the content of the customary minimum standard.

General Principles of Law

General principles of law recognised by the community of nations serve primarily to fill gaps in treaty and customary law. The International Law Commission adopted *Draft Conclusions on General Principles of Law* in 2023, distinguishing between principles derived from national legal systems and principles that may be formed within the international legal system. General principles of particular importance in the energy field include good faith, *pacta sunt servanda*, the prohibition of abuse of rights, the principle of unjust enrichment, estoppel, the principle that no one may profit from his own wrong, the obligation to make reparation for injury caused by a wrongful act, and the procedural principles of due process and equality of arms.

General principles played a central role in the arbitral jurisprudence of the concession era. In the *Abu Dhabi* arbitration of 1951, Lord Asquith of Bishopstone,

[1] *Barcelona Traction, Light and Power Company, Limited (Belgium v Spain)*, Second Phase, Judgment, ICJ Reports 1970, p. 3, paras 89–90.

acting as umpire, declined to apply the law of Abu Dhabi on the ground that no settled body of legal principles existed there that could be applied to the construction of modern commercial instruments, and instead applied principles "rooted in the good sense and common practice of the generality of civilised nations — a sort of 'modern law of nature'".[2] In *Sapphire International Petroleums v National Iranian Oil Company* (1963), the arbitrator similarly applied general principles of law.[3] Such reasoning reflected attitudes towards non-Western legal systems that are now rightly regarded as unacceptable, and modern tribunals generally apply the law chosen by the parties or the law of the host state, supplemented where appropriate by international law.

Subsidiary Means: Judicial Decisions and Teachings

Judicial decisions and the teachings of highly qualified publicists are subsidiary means for the determination of rules of law. There is no formal doctrine of binding precedent in international law; Article 59 of the ICJ Statute provides that decisions of the Court have no binding force except between the parties and in respect of the particular case. Nevertheless, the decisions of the ICJ, the International Tribunal for the Law of the Sea, the WTO Appellate Body, and investment tribunals carry significant persuasive authority, and tribunals routinely cite earlier awards. In investment arbitration, a *jurisprudence constante* has emerged on many issues, although significant divergences persist. The influence of scholarly writing is also considerable: the works of authors such as Nico Schrijver on permanent sovereignty, Christoph Schreuer on the ICSID Convention, and Philippe Sands on environmental law are frequently cited by tribunals.

2.2 Permanent Sovereignty over Natural Resources

Origins and Development

The principle of permanent sovereignty over natural resources (PSNR) is the cornerstone of international resource law. It emerged in the 1950s from the efforts of newly independent and developing states to secure control over their natural wealth, which in many cases remained subject to concessions granted by colonial administrations or on terms that the host states regarded as inequitable. The principle was first articulated in General Assembly Resolution 523 (VI) of 1952 on integrated economic development and commercial agreements, and Resolution 626 (VII) of the same year, which affirmed the right of peoples "freely to use and exploit their natural wealth and resources". A Commission on Permanent Sover-

[2] *Petroleum Development (Trucial Coast) Ltd v Sheikh of Abu Dhabi*, Award of September 1951, (1952) 1 ICLQ 247.

[3] *Sapphire International Petroleums Ltd v National Iranian Oil Company*, Award of 15 March 1963, (1967) 35 ILR 136.

eignty over Natural Resources was established in 1958, and its work culminated in Resolution 1803 (XVII) of 1962.

The principle was incorporated into common Article 1(2) of the International Covenant on Civil and Political Rights and the International Covenant on Economic, Social and Cultural Rights 1966, which provides:

> All peoples may, for their own ends, freely dispose of their natural wealth and resources without prejudice to any obligations arising out of international economic co-operation, based upon the principle of mutual benefit, and international law. In no case may a people be deprived of its own means of subsistence.

Article 21 of the African Charter on Human and Peoples' Rights 1981 is more emphatic still, providing that "all peoples shall freely dispose of their wealth and natural resources", that this right "shall be exercised in the exclusive interest of the people", and that "in no case shall a people be deprived of it". The provision also obliges states parties to "undertake to eliminate all forms of foreign economic exploitation particularly that practised by international monopolies so as to enable their peoples to fully benefit from the advantages derived from their national resources".

Content of the Principle

PSNR comprises a bundle of rights and, increasingly, duties. The principal rights include:

- the right to possess, use, and dispose of natural resources;
- the right to determine freely the conditions of exploration, development, and disposition of resources;
- the right to regulate foreign investment and to supervise the activities of foreign investors;
- the right to nationalise or expropriate foreign property, subject to the payment of appropriate compensation;
- the right to settle disputes in accordance with national law, subject to agreements to the contrary;
- the right to manage resources in pursuance of national environmental and developmental policy.

Schrijver's influential study demonstrated that the principle has evolved from a primarily rights-based claim asserted by developing states against foreign interests into a more balanced concept that also encompasses duties: the duty to exercise sovereignty in the interest of national development and the well-being of the people; the duty to respect the rights and interests of indigenous peoples; the duty to observe international agreements in good faith; the duty to treat foreign

investors fairly; and the duty to manage resources sustainably and to avoid transboundary environmental harm.[4]

Legal Status

The legal status of PSNR has been considered by the International Court of Justice. In *Armed Activities on the Territory of the Congo (Democratic Republic of the Congo v Uganda)*, the Court recognised that Resolution 1803 and the related instruments reflected customary international law in respect of the principle of permanent sovereignty, although it held that the principle was not applicable to the specific situation of looting, pillage, and exploitation of natural resources by members of the army of a state militarily intervening in another state, which fell instead to be addressed under the law of belligerent occupation and international humanitarian law.[5] In its 2022 judgment on reparations in the same case, the Court awarded the Democratic Republic of the Congo a global sum in respect of damage to natural resources.

It is generally accepted that the core of PSNR — the right of states to control and regulate their natural resources, and to expropriate foreign property for a public purpose on payment of compensation — forms part of customary international law. More contested are the claims, associated with the Charter of Economic Rights and Duties of States, that compensation is to be determined solely by domestic law and that disputes are to be resolved exclusively by domestic courts. The arbitral tribunal in *Texaco v Libya* held that Resolution 1803, adopted by a large majority including both developed and developing states, reflected customary law, whereas the relevant provisions of the Charter, adopted over the opposition of the principal capital-exporting states, did not.

PSNR and Contractual Commitments

The relationship between PSNR and contractual commitments is one of the field's central problems. Can a state, in the exercise of its sovereignty, bind itself not to exercise its sovereign powers in the future — for example, by agreeing to a stabilisation clause that freezes the applicable law? The arbitral jurisprudence has generally answered this question in the affirmative, on the ground that the power to enter into binding commitments is itself an attribute of sovereignty. As the sole arbitrator in *Texaco v Libya* put it, "the State, by entering into an international agreement with a private person, has not alienated its sovereignty but has made use of it".[6] On this view, PSNR does not authorise a state to disregard its contrac-

[4] Nico Schrijver, *Sovereignty over Natural Resources: Balancing Rights and Duties* (Cambridge University Press 1997), especially chs 9–10.

[5] *Armed Activities on the Territory of the Congo (Democratic Republic of the Congo v Uganda)*, Judgment, ICJ Reports 2005, p. 168, para 244; Reparations, Judgment, ICJ Reports 2022, p. 13.

[6] *Texaco Overseas Petroleum Company v Libyan Arab Republic*, Award of 19 January 1977, (1979) 53 ILR 389, para 68.

tual obligations; it merely affirms the state's ultimate authority over its resources, which must be exercised in accordance with international law, including the obligation to pay compensation for any taking.

The *Aminoil* tribunal adopted a more nuanced position. It accepted that stabilisation clauses were valid but held that, in the absence of an express prohibition on nationalisation of particular duration, the clause in the Aminoil concession did not prohibit nationalisation as such; rather, its effect was to be taken into account in determining the compensation payable. The tribunal also emphasised that the concession had undergone significant evolution over time, through renegotiation and the acceptance by the company of changes in the fiscal regime, such that the contract had been transformed and its original stabilisation clause could not be read in isolation from that evolution. The approach of *Aminoil* — treating stabilisation clauses as relevant to compensation rather than as an absolute bar to sovereign action — has been influential in later practice and is examined further in Chapters 14 and 15.

KEY DOCTRINAL POINT

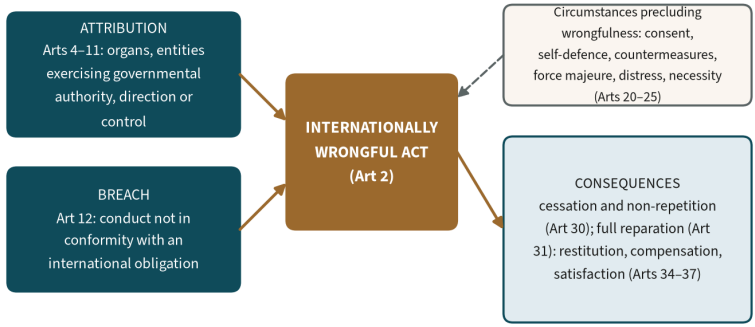
Permanent sovereignty over natural resources is not an unlimited licence to disregard commitments. In its modern form it is a principle of balanced rights and duties: the state controls its resources and may regulate and expropriate them in the public interest, but it must exercise that control in good faith, for the benefit of its people, in accordance with its international obligations, and subject to the payment of compensation where property is taken.

2.3 State Responsibility and International Legal Obligations

The Articles on State Responsibility

The law of state responsibility determines when a state has committed an internationally wrongful act and what consequences follow. Its authoritative statement is the International Law Commission's *Articles on Responsibility of States for Internationally Wrongful Acts* (ARSIWA), adopted in 2001 and commended to governments by the General Assembly. Although ARSIWA is not a treaty, many of its provisions are recognised as reflecting customary international law, and they are routinely applied by the ICJ and by investment tribunals.

Article 1 of ARSIWA states the basic principle: "Every internationally wrongful act of a State entails the international responsibility of that State." Under Article 2, an internationally wrongful act exists when conduct consisting of an action or omission (a) is attributable to the state under international law and (b) constitutes a breach of an international obligation of the state.



Chorzów Factory (1928): reparation must, as far as possible, wipe out all the consequences of the illegal act

Figure 2.2 Elements and consequences of state responsibility under the ILC Articles (2001). *Source:* prepared for this edition.

Attribution

Attribution is of particular importance in the energy sector, where states frequently act through state-owned enterprises, national oil companies, regulatory agencies, and subnational authorities. ARSIWA provides several grounds of attribution:

- **Organs of the state** (Article 4). The conduct of any state organ — legislative, executive, judicial, or other, whether central or territorial — is attributable to the state. A federal state is therefore responsible for the conduct of its constituent units, and a state is responsible for the decisions of its courts and regulators.
- **Persons or entities exercising elements of governmental authority** (Article 5). The conduct of an entity that is not a state organ but is empowered by law to exercise elements of governmental authority is attributable to the state when the entity acts in that capacity. A national oil company that is empowered to grant licences, collect royalties, or regulate operations exercises governmental authority in performing those functions.
- **Conduct directed or controlled by the state** (Article 8). The conduct of a person or group is attributable to the state if it acts on the instructions of, or under the direction or control of, the state in carrying out the conduct.
- **Conduct acknowledged and adopted by the state** (Article 11).

The attribution of the conduct of national oil companies and state-owned utilities has been extensively litigated. In *Noble Ventures v Romania*, the tribunal held that acts of a state privatisation agency were attributable to Romania because the

agency exercised governmental authority.[7] In *EDF (Services) Limited v Romania*, the tribunal examined whether the conduct of state-owned companies controlled by the state was attributable under Articles 5 and 8.[8] The general principle that emerges from the jurisprudence is that the separate legal personality of a state enterprise does not of itself shield the state from responsibility, but neither does state ownership of itself render the enterprise's commercial conduct attributable. The analysis turns on the function performed and the degree of state direction or control over the specific conduct at issue. Investment treaties sometimes contain specific provisions on the treatment of state enterprises, and the ECT, in Article 22, requires each contracting party to ensure that any state enterprise that it maintains or establishes conducts its activities in a manner consistent with the party's obligations under Part III of the treaty.

Breach of an International Obligation

A breach occurs when an act of a state is not in conformity with what is required of it by an international obligation, regardless of the origin or character of the obligation (Article 12). In the energy field, relevant obligations may arise under investment treaties, trade agreements, environmental treaties, human rights treaties, the law of the sea, and customary international law. A state may also incur responsibility for failing to exercise due diligence to prevent harm caused by private actors — for example, failing to prevent pollution from private oil operations from causing transboundary harm, or failing to provide full protection and security to a foreign investor's installations against attack.

The relationship between breach of contract and breach of international law is a recurring issue. A mere breach of a contract between a state and a foreign investor does not, of itself, constitute a breach of international law. Something more is required: an exercise of sovereign power (*puissance publique*) that goes beyond the conduct of an ordinary contracting party, a denial of justice in the resolution of the contractual dispute, or the operation of an "umbrella clause" in an investment treaty elevating contractual undertakings to treaty obligations. These issues are examined in Chapter 14.

Circumstances Precluding Wrongfulness

ARSIWA identifies circumstances that preclude the wrongfulness of conduct that would otherwise breach an international obligation: consent (Article 20), self-defence (Article 21), countermeasures (Article 22), *force majeure* (Article 23), distress (Article 24), and necessity (Article 25).

[7] *Noble Ventures, Inc v Romania*, ICSID Case No ARB/01/11, Award, 12 October 2005, paras 68–86.

[8] *EDF (Services) Limited v Romania*, ICSID Case No ARB/05/13, Award, 8 October 2009, paras 185–213.

The defence of **necessity** has been extensively litigated in the energy sector, particularly in the arbitrations arising from Argentina's economic crisis of 2001–2002, during which Argentina froze utility tariffs, converted dollar-denominated tariffs into pesos, and suspended tariff adjustment mechanisms that foreign investors in its gas and electricity sectors had been guaranteed. Article 25 permits a state to invoke necessity only if the act "is the only way for the State to safeguard an essential interest against a grave and imminent peril" and "does not seriously impair an essential interest of the State or States towards which the obligation exists, or of the international community as a whole"; necessity may not be invoked if the obligation excludes it or if the state has contributed to the situation of necessity.

The Argentine cases produced divergent results. In *CMS Gas Transmission Co v Argentina*, the tribunal rejected Argentina's necessity defence, finding that the measures were not the only way to respond to the crisis and that Argentina had contributed to the situation.[9] In *LG&E Energy Corp v Argentina*, by contrast, the tribunal accepted that a state of necessity existed for a defined period between December 2001 and April 2003, during which Argentina was exempt from liability.[10] The *ad hoc* annulment committee in *CMS* criticised the tribunal for conflating the customary necessity defence with the non-precluded measures clause in Article XI of the United States–Argentina BIT, while declining to annul the award on that ground.[11] The inconsistency of these outcomes on substantially similar facts became a prominent illustration of the legitimacy concerns surrounding investment arbitration.

Consequences: Cessation and Reparation

A state responsible for an internationally wrongful act is under an obligation to cease the act, if it is continuing, and to offer appropriate assurances and guarantees of non-repetition (Article 30). It is also under an obligation to make full reparation for the injury caused (Article 31). The classic statement of the reparation principle is that of the Permanent Court of International Justice in the *Factory at Chorzów* case:

> The essential principle contained in the actual notion of an illegal act ... is that reparation must, as far as possible, wipe out all the consequences of the illegal act

[9] *CMS Gas Transmission Company v Argentine Republic*, ICSID Case No ARB/01/8, Award, 12 May 2005, paras 315–331.

[10] *LG&E Energy Corp, LG&E Capital Corp and LG&E International Inc v Argentine Republic*, ICSID Case No ARB/02/1, Decision on Liability, 3 October 2006, paras 226–266.

[11] *CMS Gas Transmission Company v Argentine Republic*, ICSID Case No ARB/01/8, Decision of the ad hoc Committee on the Application for Annulment, 25 September 2007, paras 128–136.

> and re-establish the situation which would, in all probability, have existed if that act had not been committed.[12]

Reparation may take the form of restitution, compensation, or satisfaction, either singly or in combination (Article 34). Restitution — re-establishing the situation that existed before the wrongful act — is the primary form, but in practice it is rarely ordered in energy disputes, both because of the practical difficulty of compelling a sovereign to reverse a nationalisation and because of respect for sovereignty. Compensation, covering financially assessable damage including loss of profits insofar as established (Article 36), is the usual remedy. The valuation of compensation for energy assets — using discounted cash flow, comparable transactions, or asset-based methods — is one of the most technically demanding aspects of energy arbitration and is examined in Chapter 26.

2.4 Jurisdiction and Extraterritorial Regulation

Principles of Jurisdiction

Jurisdiction in international law refers to the authority of a state to prescribe rules (prescriptive or legislative jurisdiction), to apply those rules through its courts (adjudicative jurisdiction), and to enforce them (enforcement jurisdiction). The bases of prescriptive jurisdiction are well established:

- **Territoriality.** A state may regulate conduct occurring within its territory. The subjective territorial principle covers conduct initiated within the territory; the objective territorial principle covers conduct completed or producing its effects within it.
- **Nationality.** A state may regulate the conduct of its nationals, including corporations incorporated under its law, wherever they act.
- **Protective principle.** A state may regulate conduct abroad that threatens its security or essential governmental functions.
- **Universality.** For certain grave international crimes, any state may exercise jurisdiction.
- **Passive personality.** A state may, in limited circumstances, regulate conduct abroad that harms its nationals.

The so-called **effects doctrine**, developed in United States antitrust law, permits the regulation of conduct abroad that has substantial, direct, and foreseeable effects within the regulating state. Its application has been controversial, particularly when used to regulate the conduct of foreign companies abroad.

Enforcement jurisdiction is strictly territorial. In the *Lotus* case, the Permanent Court of International Justice held that a state "may not exercise its power in any

[12] *Factory at Chorzów (Germany v Poland)*, Merits, 1928 PCIJ Series A, No 17, p. 47.

form in the territory of another State" without a permissive rule to that effect, while affirming broad discretion as to prescriptive jurisdiction.[13]

Jurisdiction over Maritime Resources

Jurisdiction over offshore energy and resources is governed by UNCLOS, which establishes a graduated system of maritime zones. The coastal state has full sovereignty over its internal waters and territorial sea (up to 12 nautical miles from the baselines), subject to the right of innocent passage. In the exclusive economic zone (EEZ), extending to 200 nautical miles, the coastal state has sovereign rights for the purpose of exploring and exploiting, conserving, and managing natural resources, and with regard to other activities for the economic exploitation and exploration of the zone, "such as the production of energy from the water, currents and winds" (Article 56). On the continental shelf, the coastal state exercises sovereign rights for the purpose of exploring it and exploiting its natural resources (Article 77); these rights are exclusive and do not depend on occupation or express proclamation. Beyond national jurisdiction, the mineral resources of the seabed — "the Area" — are the common heritage of mankind and are administered by the International Seabed Authority. These rules are examined in Chapter 20.

Extraterritorial Regulation in the Energy Sector

Extraterritorial regulation has become increasingly important in the energy sector. Four categories may be distinguished.

Anti-corruption legislation. The United States Foreign Corrupt Practices Act 1977 prohibits the bribery of foreign officials by issuers of securities listed in the United States, domestic concerns, and persons acting within United States territory. The United Kingdom Bribery Act 2010 creates a corporate offence of failure to prevent bribery that applies to any organisation that carries on a business or part of a business in the United Kingdom, regardless of where the bribery occurs. Both statutes have been enforced extensively against energy and mining companies. The OECD Anti-Bribery Convention 1997 requires parties to criminalise the bribery of foreign public officials.

Sanctions. Economic sanctions targeting the energy sectors of particular states — including Iran, Russia, and Venezuela — are a prominent instrument of foreign policy. Some sanctions regimes have "secondary" effects, threatening penalties against foreign persons who engage in specified transactions with sanctioned parties even where there is no connection with the sanctioning state's territory. The European Union has adopted a "Blocking Statute" (Council Regulation (EC) No 2271/96) prohibiting EU persons from complying with certain extraterritorial sanctions. Sanctions are examined in Chapters 22 and 25.

[13] *SS Lotus (France v Turkey)*, 1927 PCIJ Series A, No 10, pp. 18–19.

Supply chain due diligence. A growing body of legislation imposes due diligence obligations on companies with respect to their global supply chains. Section 1502 of the United States Dodd–Frank Act 2010 required disclosure of the use of conflict minerals from the Democratic Republic of the Congo and adjoining countries. The European Union's Conflict Minerals Regulation (EU) 2017/821 imposes due diligence obligations on importers of tin, tantalum, tungsten, and gold. The EU Batteries Regulation (EU) 2023/1542 requires due diligence regarding raw materials used in batteries, and the Corporate Sustainability Due Diligence Directive (EU) 2024/1760 establishes broader human rights and environmental due diligence obligations for large companies, subject to amendments under the European Union's 2025 simplification initiatives.

Climate and environmental measures. The European Union's Carbon Border Adjustment Mechanism (CBAM), established by Regulation (EU) 2023/956, requires importers of specified carbon-intensive goods — including iron and steel, aluminium, cement, fertilisers, electricity, and hydrogen — to purchase certificates reflecting the embedded emissions of those goods, with the definitive regime applying from 2026. The EU Methane Regulation (EU) 2024/1787 imposes monitoring, reporting, and verification requirements on fossil fuel imports. Such measures regulate production processes occurring abroad by attaching conditions to market access, and their compatibility with international trade law is examined in Chapter 13.

PRACTICE NOTE

When advising a multinational energy company on extraterritorial exposure, counsel should map, for each jurisdiction in which the company is incorporated, listed, or carries on business, the anti-corruption, sanctions, export-control, and due diligence regimes that may apply to conduct anywhere in the group. Conflicts between regimes — for example, between a home-state sanction and a host-state contractual obligation to supply — should be identified at the contract negotiation stage and addressed through sanctions clauses, termination rights, and *force majeure* provisions.

2.5 International Organizations and Specialized Energy Institutions

The Fragmented Institutional Landscape

There is no single international organisation with comprehensive competence over energy. Instead, the institutional landscape is fragmented among organisations with specialised mandates, overlapping memberships, and varying legal powers. This fragmentation reflects the political sensitivity of energy, the divergent interests of producers and consumers, and the historical development of

energy governance through issue-specific responses to particular crises. Chapter 4 provides a detailed analysis of the principal energy institutions. This section introduces the main categories and their legal significance as sources and agents of international energy law.

Categories of Institutions

Consumer-oriented institutions. The International Energy Agency was established in 1974 within the framework of the OECD in response to the 1973–1974 oil crisis. Its founding treaty, the Agreement on an International Energy Program, obliges members to maintain emergency oil stocks equivalent to at least ninety days of net imports and to participate in coordinated emergency response measures.

Producer-oriented institutions. OPEC, established in 1960, and the Gas Exporting Countries Forum, established as an intergovernmental organisation in 2008, coordinate the policies of producing states. Since 2016, OPEC and a group of non-OPEC producers including the Russian Federation have coordinated production through the "OPEC+" arrangements under the Declaration of Cooperation.

Technology- and sector-specific institutions. The International Atomic Energy Agency (IAEA), established in 1957, promotes the peaceful uses of nuclear energy and administers safeguards to verify that nuclear material is not diverted to weapons. The International Renewable Energy Agency (IRENA), established in 2009 with headquarters in Abu Dhabi, promotes the adoption of renewable energy.

Treaty-based energy organisations. The Energy Charter Conference is the governing body of the ECT. The Energy Community, established by treaty in 2005, extends the European Union's internal energy market rules to contracting parties in South-East Europe and the Black Sea region, including Ukraine.

Dialogue forums. The International Energy Forum, headquartered in Riyadh, provides a forum for dialogue between producers and consumers, and administers the Joint Organisations Data Initiative (JODI) on oil and gas statistics.

General international organisations with energy competences. The United Nations and its agencies (including the United Nations Environment Programme, the UNFCCC secretariat, and UN-Energy), the World Bank Group (including the IFC and the Multilateral Investment Guarantee Agency), regional development banks, the WTO, and the OECD all exercise significant influence over energy policy.

Regional organisations. The European Union exercises extensive legislative competence over energy under Article 194 of the Treaty on the Functioning of

the European Union. In Africa, the African Union's Agenda 2063, the African Energy Commission (AFREC), the ECOWAS Regional Electricity Regulatory Authority, the West African Power Pool, and the Southern African Power Pool promote regional cooperation.

Legal Powers and Outputs

The legal powers of these institutions vary greatly. Most energy organisations lack the power to adopt decisions binding on their members, and their influence derives from information, expertise, coordination, and soft law. Exceptions include the IEA's emergency response mechanism, the IAEA's safeguards agreements (which are binding treaties between the Agency and individual states), the Energy Community's dispute settlement procedure, and, most significantly, the European Union, whose regulations and directives are binding on member states and, in the case of regulations, directly applicable.

The outputs of international organisations contribute to international energy law in several ways: they may be the forum for the negotiation of treaties; their resolutions and declarations may constitute evidence of customary international law or contribute to its formation; their standards and guidelines may be incorporated into domestic law, contracts, or investment conditions; and their data and analysis frame policy debates.

2.6 Soft Law, Industry Standards, and Corporate Governance

The Concept of Soft Law

"Soft law" refers to normative instruments that are not legally binding in themselves but that nonetheless influence the conduct of states and other actors and may acquire legal effect through incorporation, interpretation, or the formation of custom. Examples include General Assembly resolutions, declarations adopted at international conferences, codes of conduct, guidelines, recommendations of international organisations, and voluntary standards adopted by industry or multi-stakeholder initiatives. The concept is contested: some scholars reject it as a contradiction in terms, insisting on a binary distinction between law and non-law, while others regard it as an indispensable description of the graduated normativity characteristic of contemporary international governance.

Soft law is especially significant in energy and resource governance for several reasons. States are often reluctant to accept binding obligations in a field so closely tied to sovereignty and security. Many relevant actors — corporations, financial institutions, communities — are not subjects of international law in the traditional sense and cannot be parties to treaties. Technological and market conditions change rapidly, making flexible instruments preferable to treaties that are difficult to amend. And soft law can serve as a laboratory for norms that may later harden into binding law.

Principal Soft Law Instruments

The principal soft law instruments relevant to energy and natural resources include:

- **UN General Assembly Resolution 1803 (XVII) (1962)** on permanent sovereignty over natural resources, which, although a resolution, is widely regarded as reflecting customary international law.
- **The Rio Declaration on Environment and Development (1992)**, whose principles — including prevention (Principle 2), precaution (Principle 15), the polluter-pays principle (Principle 16), and environmental impact assessment (Principle 17) — have been incorporated into numerous treaties and domestic laws.
- **The UN Guiding Principles on Business and Human Rights (2011)**, establishing the "Protect, Respect and Remedy" framework.
- **The OECD Guidelines for Multinational Enterprises on Responsible Business Conduct**, revised most recently in 2023, which are backed by a system of National Contact Points that handle complaints ("specific instances") against enterprises.
- **The UN Declaration on the Rights of Indigenous Peoples (2007)**, which articulates the principle of free, prior, and informed consent.
- **The Voluntary Principles on Security and Human Rights (2000)**, a multi-stakeholder initiative addressing the use of public and private security forces by extractive companies.
- **The Natural Resource Charter**, setting out principles for the governance of natural resources for the benefit of citizens.
- **The Santiago Principles (2008)**, Generally Accepted Principles and Practices for sovereign wealth funds, many of which are funded by resource revenues.

Industry Standards and Transnational Private Regulation

Beyond intergovernmental soft law lies a dense body of industry standards and private regulation. Technical standards developed by the American Petroleum Institute (API), the International Organization for Standardization (ISO), and the International Electrotechnical Commission (IEC) govern the design, construction, and operation of energy installations and are frequently incorporated by reference into regulation and contracts. Model contracts developed by industry associations, including the AIEN Model Joint Operating Agreement and Model Unitization and Unit Operating Agreement, standardise commercial relations. The International Association of Oil & Gas Producers (IOGP) publishes guidance on health, safety, and environmental management. In mining, the International Council on Mining and Metals (ICMM) has adopted Mining Principles

that bind its member companies, and the Global Industry Standard on Tailings Management (2020) was developed following the catastrophic collapse of the Brumadinho tailings dam in Brazil in 2019.

Financial institutions exercise considerable regulatory influence through the conditions they attach to project finance. The IFC Performance Standards (2012) establish requirements on environmental and social risk management, labour, resource efficiency, community health and safety, land acquisition and involuntary resettlement, biodiversity, indigenous peoples, and cultural heritage. The Equator Principles, adopted by commercial banks, apply the IFC Performance Standards to project finance transactions above specified thresholds. Because many major energy and mining projects depend on international finance, these standards often impose more demanding requirements than domestic law.

Corporate Governance as a Source of Norms

Corporate governance instruments — codes of conduct, sustainability policies, human rights policies, and climate commitments — are increasingly treated as sources of legal obligation. In *Milieudefensie v Royal Dutch Shell*, the District Court of The Hague in 2021 relied on the UN Guiding Principles and other soft law instruments to give content to the unwritten standard of care under Dutch tort law and ordered Shell to reduce the net carbon dioxide emissions of the Shell group by 45 per cent by 2030 relative to 2019 levels.[14] On appeal, the Hague Court of Appeal in November 2024 set aside the specific reduction order, holding that there was insufficient consensus in climate science on a specific reduction percentage applicable to an individual company, while affirming that companies such as Shell have an obligation to contribute to the reduction of greenhouse gas emissions and that this obligation can be derived from the standard of care informed by human rights and soft law instruments.[15] The case illustrates how soft law can acquire legal force through domestic private law.

The Hardening of Soft Law

Soft law norms can harden into binding law through several pathways: incorporation into treaties (the Rio principles in the UNFCCC and the Convention on Biological Diversity); incorporation into domestic legislation (the UN Guiding Principles in mandatory human rights due diligence statutes); incorporation into contracts (the IFC Performance Standards in loan agreements and host government agreements); judicial reliance (as in the *Shell* litigation); and the formation of customary international law. The study of these pathways is a rich field for doctoral research.

[14] *Milieudefensie et al v Royal Dutch Shell plc*, District Court of The Hague, 26 May 2021, ECLI:NL:RBDHA:2021:5339.

[15] *Shell plc v Milieudefensie et al*, Court of Appeal of The Hague, Judgment of 12 November 2024.

2.7 International Judicial and Arbitral Institutions

The International Court of Justice

The International Court of Justice, the principal judicial organ of the United Nations, has decided numerous cases bearing on energy and natural resources, including maritime delimitation disputes involving offshore petroleum resources (*North Sea Continental Shelf*, 1969; *Land and Maritime Boundary between Cameroon and Nigeria*, 2002; *Maritime Delimitation in the Black Sea*, 2009; *Territorial and Maritime Dispute (Nicaragua v Colombia)*, 2012; *Maritime Delimitation in the Indian Ocean (Somalia v Kenya)*, 2021), disputes over shared watercourses and hydropower (*Gabčíkovo-Nagymaros Project*, 1997; *Pulp Mills on the River Uruguay*, 2010; *Dispute over the Status and Use of the Waters of the Silala*, 2022), and disputes concerning the protection of foreign investment (*Barcelona Traction*, 1970; *Elettronica Sicula SpA (ELSI)*, 1989). Its advisory opinion of 23 July 2025 on the *Obligations of States in respect of Climate Change* is of profound significance for energy law and is analysed in Chapter 18.

The Court's jurisdiction in contentious cases depends on the consent of states, given through special agreement, compromissory clauses in treaties, or optional clause declarations under Article 36(2) of its Statute. Because private parties have no standing, the Court plays a limited role in investor–state disputes, which are resolved principally through arbitration.

The International Tribunal for the Law of the Sea and Annex VII Tribunals

The International Tribunal for the Law of the Sea (ITLOS), established under Annex VI of UNCLOS and seated in Hamburg, has jurisdiction over disputes concerning the interpretation or application of the Convention. Its Special Chamber decided the maritime boundary dispute between Ghana and Côte d'Ivoire in 2017, a case driven by offshore petroleum discoveries, in which the Chamber also rejected Côte d'Ivoire's claim that Ghana had violated its sovereign rights by conducting oil activities in the disputed area.[16] Its Seabed Disputes Chamber delivered an important advisory opinion in 2011 on the responsibilities of states sponsoring activities in the Area, holding that sponsoring states have obligations of due diligence and direct obligations, including the obligation to apply a precautionary approach.[17] In 2024 the full Tribunal delivered an advisory

[16] *Dispute Concerning Delimitation of the Maritime Boundary between Ghana and Côte d'Ivoire in the Atlantic Ocean (Ghana/Côte d'Ivoire)*, ITLOS Case No 23, Judgment of the Special Chamber, 23 September 2017.

[17] *Responsibilities and Obligations of States Sponsoring Persons and Entities with respect to Activities in the Area*, Advisory Opinion, ITLOS Seabed Disputes Chamber, 1 February 2011, ITLOS Reports 2011, p. 10.

opinion holding that anthropogenic greenhouse gas emissions constitute pollution of the marine environment within the meaning of UNCLOS.

Arbitral tribunals constituted under Annex VII of UNCLOS have decided several disputes with energy dimensions, including the *South China Sea Arbitration (Philippines v China)* of 2016, which addressed the legal status of maritime features and the lawfulness of Chinese activities, including interference with Philippine petroleum exploration.

The WTO Dispute Settlement System

The WTO Dispute Settlement Body, comprising panels and the Appellate Body, has decided several disputes concerning energy goods and renewable energy policies. Since December 2019, the Appellate Body has been unable to hear appeals because the United States has blocked the appointment of new members. Many WTO members have joined the Multi-Party Interim Appeal Arbitration Arrangement (MPIA) under Article 25 of the Dispute Settlement Understanding as an interim substitute, but appeals "into the void" by members not party to the MPIA can prevent panel reports from being adopted. This crisis has weakened the enforcement of trade disciplines in the energy sector.

Investment Arbitration Institutions

Investor–state disputes in the energy sector are resolved principally through arbitration under the auspices of:

- **ICSID**, established by the Convention on the Settlement of Investment Disputes between States and Nationals of Other States 1965, with its own self-contained system of annulment rather than review by national courts, and enforcement provisions requiring contracting states to recognise awards as if they were final judgments of their own courts (Article 54);
- **the Permanent Court of Arbitration (PCA)** in The Hague, which administers many arbitrations conducted under the UNCITRAL Arbitration Rules, including the *Yukos* arbitrations and numerous ECT cases;
- **the Arbitration Institute of the Stockholm Chamber of Commerce (SCC)**, designated as one of the forums for investor–state disputes under Article 26 of the ECT;
- **the International Chamber of Commerce (ICC) International Court of Arbitration**, which administers many commercial and some investment disputes;
- **the London Court of International Arbitration (LCIA)** and regional centres, including the Singapore International Arbitration Centre, the Cairo Regional Centre for International Commercial Arbitration, and the Lagos Court of Arbitration.

The institutional framework, procedural rules, and practice of energy arbitration are examined in Chapter 26.

Regional Courts and Human Rights Bodies

Regional courts and human rights bodies have become increasingly important forums for energy-related disputes. The Court of Justice of the European Union has decided cases of enormous significance for intra-EU investment arbitration (*Achmea*, 2018; *Komstroy*, 2021; *PL Holdings*, 2021) and energy market regulation. The European Court of Human Rights, in *Verein KlimaSeniorinnen Schweiz v Switzerland* (2024), held that Article 8 of the European Convention on Human Rights encompasses a right to effective protection by state authorities from the serious adverse effects of climate change.[18] The African Commission on Human and Peoples' Rights, in *Social and Economic Rights Action Center (SERAC) and Center for Economic and Social Rights v Nigeria* (2001), found that Nigeria had violated the African Charter through the conduct of the state oil company and its failure to regulate the activities of oil companies in Ogoniland.[19] The ECOWAS Community Court of Justice, in *SERAP v Federal Republic of Nigeria* (2012), held Nigeria responsible for failing to protect the right to a satisfactory environment in the Niger Delta against the activities of oil companies.

2.8 Conflicts Between Domestic and International Legal Obligations

Monism, Dualism, and Their Limits

The relationship between international law and domestic law is traditionally described in terms of monism and dualism. Monist systems treat international law and domestic law as parts of a single legal order, such that treaties may have direct effect in domestic law upon ratification. Dualist systems treat the two as separate, such that treaties must be incorporated by legislation before they can be applied by domestic courts. Nigeria follows a dualist approach for treaties: section 12(1) of the Constitution of the Federal Republic of Nigeria 1999 provides that "no treaty between the Federation and any other country shall have the force of law except to the extent to which any such treaty has been enacted into law by the National Assembly". In *Abacha v Fawehinmi*, the Supreme Court of Nigeria held that the African Charter on Human and Peoples' Rights, having been domesticated by the African Charter on Human and Peoples' Rights (Ratification and Enforcement)

[18] *Verein KlimaSeniorinnen Schweiz and Others v Switzerland*, App No 53600/20, European Court of Human Rights (Grand Chamber), Judgment of 9 April 2024.

[19] *Social and Economic Rights Action Center (SERAC) and Center for Economic and Social Rights (CESR) v Nigeria*, Communication No 155/96, African Commission on Human and Peoples' Rights, Decision of 27 October 2001.

Act, was part of Nigerian law, although subordinate to the Constitution.[20] The United Kingdom is likewise dualist, whereas the Netherlands and many civil law states are closer to the monist model.

The International Law Perspective

From the perspective of international law, the classification of a state's domestic system is irrelevant to its international obligations. Article 27 of the VCLT provides that "a party may not invoke the provisions of its internal law as justification for its failure to perform a treaty". Article 3 of ARSIWA provides that the characterisation of an act of a state as internationally wrongful is governed by international law and is not affected by its characterisation as lawful by internal law. Accordingly, a state cannot escape liability under an investment treaty by pointing to the lawfulness of its measures under its own constitution, nor can it rely on a domestic court decision annulling a contract as a defence if the decision itself amounts to a denial of justice.

Typical Conflicts in the Energy Sector

Conflicts between domestic and international obligations arise in several recurring situations in the energy sector:

- **Legislative change and stabilisation.** A state enacts new fiscal or environmental legislation that is valid under its constitution but conflicts with stabilisation commitments in contracts or with investment treaty protections.
- **Judicial decisions.** Domestic courts annul or refuse to enforce contracts or arbitral awards, or issue anti-arbitration injunctions, in circumstances that international tribunals may regard as breaches of treaty obligations or of the New York Convention.
- **Constitutional limits on contracting.** Contracts concluded by state entities may be void under domestic law for failure to obtain required approvals, raising questions about whether the state can rely on its own law to avoid its obligations. Article 46 of the VCLT, by analogy, permits a state to invoke a violation of internal law regarding competence to conclude treaties only where the violation was manifest and concerned a rule of fundamental importance.
- **Climate and environmental obligations.** Domestic courts may order a state to adopt more stringent climate measures, as the Supreme Court of the Netherlands did in *State of the Netherlands v Urgenda Foundation* (2019),[21] potentially generating claims by investors whose investments are affected by the resulting measures.

[20] *Abacha v Fawehinmi* (2000) 6 NWLR (Pt 660) 228 (Supreme Court of Nigeria).

[21] *State of the Netherlands v Urgenda Foundation*, Supreme Court of the Netherlands, 20 December 2019, ECLI:NL:HR:2019:2007.

- **Human rights obligations and investment protection.** A state's obligations to protect the rights of indigenous peoples or the right to water may conflict with commitments to investors.

The Intra-EU Investment Arbitration Conflict

The most dramatic recent example of conflict between legal orders arose from intra-EU investment arbitration under the ECT. In *Slovak Republic v Achmea BV* (2018), the Court of Justice of the European Union held that arbitration clauses in intra-EU BITs were incompatible with EU law because they removed disputes potentially concerning the interpretation or application of EU law from the EU judicial system.[22] In *Republic of Moldova v Komstroy LLC* (2021), the Court extended this reasoning to the ECT, holding that Article 26 of the ECT must be interpreted as not applicable to disputes between a member state and an investor of another member state concerning an investment made by the latter in the first member state.[23]

Investment tribunals constituted under the ECT have overwhelmingly rejected the intra-EU objection, holding that they derive their jurisdiction from the ECT as a treaty of international law and that EU law cannot override the consent expressed in it. The result has been a direct conflict between legal orders: awards rendered against EU member states in favour of EU investors (many arising from Spain's withdrawal of renewable energy incentives) have been upheld by arbitral tribunals and enforced or recognised by courts in some non-EU jurisdictions such as the United States, Australia, and the United Kingdom, while courts within the European Union have refused enforcement and the European Commission has treated payment of such awards as potentially constituting unlawful state aid. The European Union, Euratom, and a number of member states subsequently withdrew from the ECT. This saga is a vivid case study in legal pluralism and is examined further in Chapters 14 and 26.

LEADING AUTHORITY

Factory at Chorzów (Germany v Poland), PCIJ, 1928

Facts. Following the transfer of Upper Silesia to Poland after the First World War, Poland took possession of a nitrate factory at Chorzów owned by German companies, in breach of the Geneva Convention of 1922 concerning Upper Silesia. Germany brought proceedings seeking reparation.

Holding. The Permanent Court held that it is a principle of international law that the breach of an engagement involves an obligation to make reparation in an adequate

[22] Case C-284/16, *Slowakische Republik v Achmea BV*, EU:C:2018:158 (CJEU, Grand Chamber, 6 March 2018).

[23] Case C-741/19, *République de Moldavie v Komstroy LLC*, EU:C:2021:655 (CJEU, Grand Chamber, 2 September 2021).

form, and that reparation must, as far as possible, wipe out all the consequences of the illegal act. It distinguished between lawful expropriation, which requires payment of the value of the undertaking at the time of dispossession plus interest, and unlawful expropriation, for which restitution in kind, or payment of a sum corresponding to the value that restitution would bear, is required — potentially including the value of the property at the date of the judgment rather than the date of the taking.

Significance for energy law. *Chorzów* remains the foundation of the law of reparation. Investment tribunals in energy cases — including *ADC v Hungary* (2006), *ConocoPhillips v Venezuela* (2019), and *Yukos Universal v Russia* (2014) — have relied on the distinction between lawful and unlawful expropriation to determine the valuation date and the measure of compensation, with major financial consequences.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Attributing the conduct of a national oil company (hypothetical). The fictional Federal Republic of Calvaria owns 100 per cent of Calvaria National Petroleum Company Ltd (CNPC), a company incorporated under the Calvarian Companies Act. Under the Calvarian Petroleum Act, CNPC holds all state participating interests in joint ventures and is empowered to approve work programmes and budgets on behalf of the state. In 2024, CNPC refused to approve a joint venture work programme submitted by its foreign partner, Delta Ridge Resources Inc, citing "national energy security priorities" communicated in a letter from the Minister of Petroleum. Delta Ridge alleges that the refusal was designed to force it to sell its interest at an undervalue to a company connected to senior officials.

Advise Delta Ridge on (a) whether CNPC's conduct is attributable to Calvaria under ARSIWA Articles 4, 5, and 8; (b) whether the refusal could constitute a breach of an investment treaty obligation as opposed to a mere contractual breach; and (c) what evidence should be gathered to support attribution. *All entities and facts are fictitious.*

CHAPTER SUMMARY

- The formal sources of international energy law are treaties, custom, and general principles, with judicial decisions and scholarly writings as subsidiary means of determining the law; no formal hierarchy exists, but treaties operate as *lex specialis* and peremptory norms prevail over all.
- Treaty interpretation under VCLT Articles 31–33 is central to the field, particularly in investment disputes; systemic integration under Article 31(3)(c) allows environmental and human rights obligations to inform the interpretation of investment treaties.
- Permanent sovereignty over natural resources is a principle of customary international law encompassing both rights and duties; it does not authorise a state

to disregard its commitments, but affirms the state's ultimate authority subject to international law.

- The ILC Articles on State Responsibility govern attribution, breach, defences, and reparation; the attribution of national oil company conduct and the necessity defence have generated extensive energy jurisprudence.
- Jurisdiction is primarily territorial, but extraterritorial regulation through anti-corruption, sanctions, supply chain, and carbon border measures increasingly shapes energy activities.
- International organisations, soft law, industry standards, lender requirements, and corporate governance instruments constitute a significant layer of normative authority, which may harden into binding law.
- Conflicts between domestic and international obligations are resolved, from the international perspective, by the rule that internal law cannot justify non-performance; the intra-EU arbitration saga illustrates the limits of that rule when legal orders collide.

DOCTORAL RESEARCH QUESTIONS

1. Has the proliferation of investment treaties containing similar substantive standards contributed to the formation of customary international law on the treatment of foreign energy investments? What methodology should be used to answer this question after the ILC's 2018 Draft Conclusions?
2. To what extent does the modern, duty-oriented conception of permanent sovereignty over natural resources constrain the freedom of states to grant long-term fossil fuel licences in the light of climate obligations?
3. Should the conduct of national oil companies be subject to a special rule of attribution, given their hybrid commercial and governmental functions?
4. What are the conditions under which soft law instruments such as the UN Guiding Principles acquire binding force through domestic tort law, and is this process democratically legitimate?
5. Does the intra-EU arbitration conflict demonstrate the failure of the techniques for managing fragmentation identified by the International Law Commission?

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3 CHAPTER

Resource Sovereignty, State Ownership, and Constitutional Governance

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the legal foundations of resource sovereignty at the international and constitutional levels.
2. Compare the principal models of subsurface resource ownership — state ownership, the accession (landowner) system, and hybrid regimes — and their consequences for licensing and revenue.
3. Analyse the public trust doctrine and related concepts of state stewardship over natural resources.
4. Evaluate the allocation of resource authority in federal systems, with particular reference to Nigeria, Canada, Australia, and the United States.
5. Assess the rights of indigenous peoples and local communities in resource governance.
6. Analyse resource nationalism, expropriation, nationalisation, and the standards of compensation under international law.

Introduction

Chapter 2 examined permanent sovereignty over natural resources as a principle of international law. This chapter turns to the ways in which that sovereignty is organised and exercised within states. International law recognises that a state controls the natural resources within its territory and continental shelf, but it is silent on the question of how ownership and authority are to be distributed internally: whether resources belong to the state, to the owners of the surface, to subnational units, or to communities; which organs of government may grant rights to explore and exploit them; and how the revenues they generate are to be shared. These questions are answered by constitutional law, statute, and custom, and the answers vary greatly among states.

The internal organisation of resource sovereignty has profound practical consequences. It determines who an investor must negotiate with, what kind of title the investor obtains, who receives royalties, and which courts have jurisdiction over disputes. It shapes the political economy of resource-rich states, influencing the incidence of conflict, the quality of governance, and the distribution of wealth. And it determines the extent to which affected communities can participate in decisions about resource development and share in its benefits.

The chapter proceeds in eight sections. It begins with the legal foundations of resource sovereignty, then compares models of constitutional ownership of subsurface resources. It examines the public trust doctrine, federalism and subnational authority, and the position of indigenous peoples. It analyses resource nationalism and the law of expropriation and compensation, and concludes with a comparative survey of constitutional approaches.

3.1 Legal Foundations of Resource Sovereignty

Territorial Sovereignty and Resource Control

Resource sovereignty is, at its root, an incident of territorial sovereignty. A state's sovereignty over its land territory extends to the subsoil beneath it, without any fixed depth limit, and to the airspace above it. Sovereignty over the territorial sea likewise extends to the seabed and subsoil (UNCLOS, Article 2). Beyond the territorial sea, the coastal state enjoys not sovereignty but "sovereign rights" over the natural resources of the EEZ and the continental shelf — functional rights that are exclusive but limited to the purposes specified in the Convention. The distinction between sovereignty and sovereign rights is important: in the EEZ and on the continental shelf, other states enjoy freedoms of navigation, overflight, and the laying of submarine cables and pipelines, subject to the coastal state's rights.

In the classical conception, the state's territorial sovereignty entitled it to deal with its resources as it saw fit, subject only to obligations it had freely assumed and to the general rules protecting the property of aliens. The doctrine of permanent sovereignty over natural resources, discussed in Chapter 2, reinforced this conception and gave it a new political meaning in the context of decolonisation, emphasising that sovereignty over resources belongs to the people of the state and must be exercised for their development and well-being.

The People as Bearers of Sovereignty

A significant development in the modern law of resource sovereignty is the recognition that sovereignty over natural resources is vested not merely in the state as an abstract entity but in the people. Common Article 1(2) of the International Covenants refers to the right of "all peoples" to freely dispose of their natural wealth and resources. Article 21 of the African Charter on Human and Peoples' Rights similarly vests the right in "peoples" and requires that it be exercised "in the

exclusive interest of the people". Many national constitutions declare that natural resources belong to the people, with the state acting as trustee or custodian. Article 257(6) of the Constitution of Ghana 1992, for example, vests every mineral in its natural state in the President "on behalf of, and in trust for, the people of Ghana". Article 62(1)(f) of the Constitution of Kenya 2010 classifies all minerals and mineral oils as public land, and Article 62(3) provides that such public land vests in and is held by the national government in trust for the people of Kenya.

The recognition of the people as bearers of resource sovereignty has several legal implications. It supports arguments that resource revenues must be managed transparently and for the public benefit. It provides a normative basis for the participation of citizens and communities in resource decisions. And it has been invoked to challenge contracts concluded by unrepresentative or corrupt governments, raising difficult questions about the continuity of state obligations following changes of government — an issue examined in the context of odious debt and illegitimate contracts.

The Limits of Sovereignty

Resource sovereignty is subject to limits imposed by international law. These include:

- obligations assumed by treaty, including investment treaties, trade agreements, and environmental agreements;
- the customary obligation to prevent significant transboundary harm, recognised in *Trail Smelter* and affirmed by the International Court of Justice in its *Legality of the Threat or Use of Nuclear Weapons* advisory opinion and in *Pulp Mills*;
- the obligation to cooperate in the management of shared natural resources, such as transboundary watercourses and straddling petroleum deposits;
- the international minimum standard of treatment of aliens and the rules governing expropriation;
- human rights obligations, including the rights of indigenous peoples;
- obligations under the law of armed conflict regarding the exploitation of resources in occupied territory.

These limits do not negate sovereignty; they define the conditions under which it must be exercised.

3.2 Constitutional Ownership of Subsurface Resources

Three Models of Ownership

National legal systems allocate ownership of subsurface resources according to three broad models.

The regalian or domanial system (state ownership). Under this model, which has its origins in Roman law, medieval European royal prerogatives over precious metals, and the Spanish colonial *derecho minero*, subsurface minerals and petroleum belong to the state (historically, the Crown or the sovereign) irrespective of the ownership of the surface. The state grants rights to explore and exploit through licences, concessions, or contracts. The regalian system prevails in the great majority of states, including virtually all of Africa, Latin America, the Middle East, and Asia, and in Europe. In the United Kingdom, petroleum in its natural condition in strata was vested in the Crown by the Petroleum (Production) Act 1934, a vesting now continued by section 2 of the Petroleum Act 1998; coal was nationalised in 1946 and is now vested in the Coal Authority; and gold and silver have long been "royal mines" belonging to the Crown.

The accession or landowner system. Under this model, derived from the common law maxim *cuius est solum, eius est usque ad caelum et ad inferos* ("whoever owns the soil owns it up to the heavens and down to the depths"), the owner of the surface also owns the minerals beneath it, unless they have been severed by conveyance or reserved by the state. The accession system prevails in the United States (for privately owned land) and was historically important in parts of Canada. It is the principal reason why private mineral ownership, mineral leasing between private parties, and royalty interests owned by individuals are characteristic features of American oil and gas law.

The *res nullius* and claim systems. Historically, some systems treated minerals as unowned until discovered and appropriated, giving rise to "free mining" or claim-staking regimes under which the first person to discover and stake a claim acquired rights. The United States General Mining Law of 1872 retains elements of this approach for hard-rock minerals on federal public lands, permitting citizens to locate claims on valuable mineral deposits. Free entry systems have been modified in many Canadian provinces to require consultation with indigenous peoples before claims are registered.

	Regalian / state ownership	Accession / landowner	Claim / free entry
Owner of minerals	State, nation or people (e.g. Nigeria s 44(3), UK, Mexico art 27)	Surface owner unless severed (US private land)	Unowned until located and claimed (US Mining Law 1872)
Grant instrument	Licence, lease, concession, PSC, service contract	Private lease negotiated with mineral owner	Location and registration of claim; patent
Rent capture	Royalty, tax, profit oil, state participation	Bonus and royalty to owner; state taxes	Limited royalty; taxation
Regulator's role	Owner and regulator: opens acreage, sets terms	Prevent waste, protect correlative rights	Record claims, environmental control
Typical issues	Contract stability, transparency, community benefit	Rule of capture, pooling, unitisation	Conflicts with indigenous rights, environment

Figure 3.1 Models of subsurface resource ownership and their typical legal consequences. *Source:* prepared for this edition.

The Rule of Capture and Its Consequences

The accession system, applied to fugacious resources such as oil and gas that migrate through porous rock, gave rise in the United States to the "rule of capture": a landowner owns all the oil and gas produced from wells drilled on his or her land, even if some of that oil and gas migrated from beneath neighbouring land. In *Westmoreland & Cambria Natural Gas Co v De Witt* (1889), the Supreme Court of Pennsylvania analogised oil and gas to wild animals (*ferae naturae*), which belong to the person who captures them. The rule of capture encouraged a race to drill, leading to wasteful over-production, the premature loss of reservoir pressure, and the dissipation of resources.

Terence Daintith's study of the rule of capture demonstrates how this legal doctrine shaped the organisation of the world oil industry, prompting the development of conservation regulation (well spacing, production prorationing, and compulsory pooling and unitisation) in the United States, and influencing the decision of many other states to adopt state ownership and licensing systems that would avoid the inefficiencies of fragmented ownership.[1] The doctrine of correlative rights, developed to mitigate the rule of capture, recognises that each owner

[1] Terence Daintith, *Finders Keepers? How the Law of Capture Shaped the World Oil Industry* (RFF Press 2010).

of land overlying a common reservoir has a right to a fair opportunity to produce its just and equitable share of the oil and gas, and a duty not to injure the reservoir or waste its resources.

Constitutional Vesting Clauses

In states following the regalian model, ownership is typically vested in the state by a constitutional provision. Some illustrative examples follow.

Nigeria. Section 44(3) of the Constitution of the Federal Republic of Nigeria 1999 provides:

> Notwithstanding the foregoing provisions of this section, the entire property in and control of all minerals, mineral oils and natural gas in under or upon any land in Nigeria or in, under or upon the territorial waters and the Exclusive Economic Zone of Nigeria shall vest in the Government of the Federation and shall be managed in such manner as may be prescribed by the National Assembly.

This vesting is reiterated in section 1 of the Petroleum Industry Act 2021 and section 1 of the Nigerian Minerals and Mining Act 2007. The Land Use Act 1978 separately vests all land comprised in the territory of each state in the Governor of that state, to be held in trust and administered for the use and common benefit of all Nigerians, with holders of land enjoying rights of occupancy rather than ownership. The combination of federal ownership of minerals and state control of land, overlaid on customary tenure, generates complex legal relationships that are examined further in section 3.4.

Mexico. Article 27 of the Mexican Constitution of 1917 vests in the Nation direct ownership of all minerals and hydrocarbons, declares that this ownership is inalienable and imprescriptible, and — following the constitutional reforms of 2013 and their partial reversal in 2024–2025 — regulates the participation of private parties in the hydrocarbon and electricity sectors.

Bolivia. Article 349 of the Constitution of Bolivia 2009 declares that natural resources are the property and direct domain, indivisible and imprescriptible, of the Bolivian people, and that their administration corresponds to the state in the collective interest.

Norway. Norway, which has no comparable constitutional clause, relies on statute: section 1-1 of the Petroleum Act 1996 provides that the Norwegian state has the proprietary right to subsea petroleum deposits and the exclusive right to resource management.

Consequences of the Ownership Model

The choice of ownership model has significant consequences. Under state ownership, the state is both owner and regulator; it captures the resource rent

through royalties, taxes, and production shares; and it decides which areas will be opened for exploration and on what terms. The relationship between the state and the investor is typically governed by a licence, concession, or contract, and the investor's rights are those conferred by that instrument. Under the accession system, the landowner captures a substantial portion of the rent through lease bonuses and royalties; the state's revenues derive principally from taxation; and the regulator's role is primarily to prevent waste, protect correlative rights, and safeguard the environment. Political economy research suggests that the distribution of resource rents to private landowners in the United States has helped sustain public support for oil and gas development in producing regions, whereas the concentration of rents in the state under the regalian model may contribute to the disconnection between resource extraction and local benefits that characterises many producer states.

COMPARATIVE PERSPECTIVE

Ownership and community benefit. In the United States, a farmer in North Dakota who owns the mineral estate beneath his land may receive royalties of 12.5 to 20 per cent or more of production value from a shale well, a direct private share of the resource rent. In Nigeria, a farmer in the Niger Delta whose land overlies a producing field owns neither the petroleum nor, under the Land Use Act, the land itself in the full sense; he is entitled to compensation for surface rights and crops affected by operations, and his community may benefit from the host community development trust established under Chapter 3 of the Petroleum Industry Act 2021. The contrast illustrates how the ownership model shapes the distribution of benefits and, consequently, local attitudes towards resource development.

3.3 Public Trust Doctrine and State Stewardship

Origins of the Public Trust Doctrine

The public trust doctrine holds that certain natural resources are held by the state in trust for the public and may not be alienated or managed in a manner inconsistent with the public interest. The doctrine traces its origins to Roman law, which recognised that certain things — the air, running water, the sea, and the shores of the sea — were common to all (*res communes*) under the Institutes of Justinian. It was received into English common law in relation to tidal waters and the foreshore, which the Crown held subject to the public rights of navigation and fishing, and it was developed most fully in the United States.

The leading American authority is *Illinois Central Railroad Co v Illinois* (1892), in which the United States Supreme Court held that the State of Illinois could not irrevocably convey to a private railroad company title to the submerged lands of Chicago's harbour on Lake Michigan, because those lands were held

in trust for the people for purposes of navigation, commerce, and fishing.[2] The Court reasoned that a state could no more abdicate its trust over property in which the whole people are interested than it could abdicate its police powers. The doctrine was revitalised by Joseph Sax's influential 1970 article, which proposed it as a tool for environmental protection, and was extended by the Supreme Court of California in *National Audubon Society v Superior Court* (1983) to require the state to consider the public trust in allocating water rights affecting Mono Lake.[3]

The Doctrine Beyond the United States

The public trust doctrine has been received and adapted in several other jurisdictions. The Supreme Court of India, in *M.C. Mehta v Kamal Nath* (1997), held that the public trust doctrine was part of Indian law and that the state, as trustee of natural resources, was under a legal duty to protect them; it quashed a lease of riparian forest land granted to a motel company.[4] Courts in Pakistan, the Philippines, Kenya, and South Africa have invoked the doctrine or analogous concepts. Section 3 of South Africa's Mineral and Petroleum Resources Development Act 28 of 2002 provides that "mineral and petroleum resources are the common heritage of all the people of South Africa and the State is the custodian thereof for the benefit of all South Africans", a formulation that replaced private mineral ownership with state custodianship. The Constitutional Court of South Africa, in *Agri South Africa v Minister for Minerals and Energy* (2013), held that this transition did not amount to an expropriation of private mineral rights requiring compensation, because the state had not acquired the rights for itself but had become their custodian.[5]

Stewardship and Intergenerational Equity

The public trust doctrine is closely related to the concept of intergenerational equity, which holds that each generation holds the natural and cultural resources of the planet in trust for future generations. Edith Brown Weiss's formulation of the principle of intergenerational equity — comprising conservation of options, conservation of quality, and conservation of access — has been influential in international environmental law.[6] In the context of non-renewable resources, intergenerational equity raises questions about the rate of extraction, the investment of resource revenues, and the obligation to leave future generations with comparable

[2] *Illinois Central Railroad Co v Illinois*, 146 US 387 (1892).

[3] *National Audubon Society v Superior Court of Alpine County*, 33 Cal 3d 419 (1983); see Joseph L. Sax, "The Public Trust Doctrine in Natural Resource Law: Effective Judicial Intervention" (1970) 68 *Michigan Law Review* 471.

[4] *M.C. Mehta v Kamal Nath* (1997) 1 SCC 388 (Supreme Court of India).

[5] *Agri South Africa v Minister for Minerals and Energy* [2013] ZACC 9; 2013 (4) SA 1 (CC).

[6] Edith Brown Weiss, *In Fairness to Future Generations: International Law, Common Patrimony, and Intergenerational Equity* (Transnational/United Nations University 1989).

opportunities. Sovereign wealth funds, such as Norway's Government Pension Fund Global and Nigeria's Sovereign Investment Authority, established by the Nigeria Sovereign Investment Authority (Establishment, etc.) Act 2011 with a Future Generations Fund among its constituent funds, represent institutional responses to this concern.

Climate litigants have invoked the public trust doctrine to argue that governments hold the atmosphere in trust and are obliged to protect it. In *Juliana v United States*, young plaintiffs relied in part on the public trust doctrine to claim that the federal government's promotion of fossil fuels violated their constitutional rights; the Ninth Circuit Court of Appeals held in 2020 that the plaintiffs lacked standing because the relief sought was beyond the power of an Article III court to provide, and the litigation was ultimately terminated after the Supreme Court declined review in 2025.[7] Courts in other jurisdictions have been more receptive to stewardship-based reasoning, as discussed in Chapter 18.

Legal Consequences of Stewardship

Where resources are held in trust, several legal consequences may follow. The state may be prohibited from alienating trust resources entirely or from granting rights that substantially impair the public interest. Grants of rights may be subject to implied conditions requiring their exercise consistently with the trust. Courts may review state decisions affecting trust resources with heightened scrutiny. And citizens may have standing to enforce the trust. These consequences can create tension with the security of tenure sought by investors, particularly where a court later concludes that a grant was inconsistent with the trust. Investors and their advisers should therefore assess, as part of legal due diligence, whether the host jurisdiction recognises a public trust doctrine or constitutional stewardship obligations that could affect the validity or scope of resource rights.

3.4 Federalism and Subnational Resource Authority

The Federal Dilemma

In federal states, the allocation of authority over natural resources between the federal government and the constituent units is among the most politically charged constitutional questions. Resources are often geographically concentrated, so that their ownership and revenues may disproportionately benefit particular regions. Federal ownership may be perceived by producing regions as the appropriation of their wealth by the centre; subnational ownership may produce stark inequalities between resource-rich and resource-poor regions and undermine national cohesion. The resolution of this dilemma varies widely.

[7] *Juliana v United States*, 947 F 3d 1159 (9th Cir 2020).

Nigeria: Federal Ownership, Derivation, and Resource Control

Nigeria provides a paradigmatic example of the federal dilemma. Before independence and during the First Republic, the regions enjoyed significant fiscal autonomy and the principle of derivation — the allocation of revenue to the region from which it was derived — played a major role in revenue sharing. Following the civil war and the rise of petroleum as the dominant source of public revenue, the federal government progressively centralised control. The Petroleum Act 1969 vested ownership of petroleum in the state (the Federal Military Government), and successive constitutions confirmed federal ownership. The derivation percentage fell sharply in the 1970s and 1980s.

The 1999 Constitution restored a minimum derivation component: section 162(2) provides that "the principle of derivation shall be constantly reflected in any approved formula as being not less than thirteen per cent of the revenue accruing to the Federation Account directly from any natural resources". The question of whether offshore revenues should be counted as derived from littoral states gave rise to the resource control litigation. In *Attorney-General of the Federation v Attorney-General of Abia State & 35 Others* (2002), the Supreme Court held that the seaward boundary of a littoral state for the purposes of the derivation principle was the low-water mark of its land surface (or the seaward limit of inland waters), so that revenues from offshore production on the continental shelf and in the EEZ accrued to the Federation rather than to the states. The decision was politically controversial and was effectively reversed in part by the Allocation of Revenue (Abolition of Dichotomy in the Application of the Principle of Derivation) Act 2004, which provided that the 200-metre water depth isobath would be the limit for the purposes of derivation, thereby treating shallow-water offshore production as derived from the adjoining littoral states.

The demand for "resource control" — greater control by producing states and communities over the resources found in their territories and the revenues derived from them — remains a central theme of Nigerian politics. It has been associated with the Ogoni struggle of the 1990s, the Kaiama Declaration of 1998, militancy in the Niger Delta in the 2000s, and the amnesty programme of 2009. The Petroleum Industry Act 2021 responded in part by establishing host community development trusts funded by an annual contribution of three per cent of the actual annual operating expenditure of settlors (holders of interests in petroleum prospecting licences and petroleum mining leases) in the preceding financial year (section 240(2)), although critics argue that this falls far short of meaningful resource control.

Electricity Federalism in Nigeria

In contrast to petroleum, electricity in Nigeria has undergone significant decentralisation. The Constitution (Fifth Alteration) (No. 33) Act 2023 amended the Exclusive and Concurrent Legislative Lists to permit states to legislate on electricity generation, transmission, and distribution within their territories. The Electricity Act 2023, which repealed the Electric Power Sector Reform Act 2005, provides a framework under which states may establish their own electricity markets and regulators, with the Nigerian Electricity Regulatory Commission transferring regulatory oversight of intrastate markets to state regulators once established. A number of states, including Enugu, Ekiti, Ondo, Imo, Oyo, Edo, Kogi, and Lagos, have enacted electricity laws and established regulatory commissions, and regulatory oversight has been transferred to several of them. This process of electricity federalism is examined in Chapter 9.

Canada: Provincial Ownership

In Canada, the provinces own the natural resources within their boundaries by virtue of section 109 of the Constitution Act 1867 (extended to the prairie provinces by the Natural Resources Transfer Agreements of 1930). Section 92A, added by the Constitution Act 1982 following prolonged federal–provincial conflict over energy pricing and taxation (notably the National Energy Program of 1980), confirms exclusive provincial legislative authority over exploration, development, conservation, and management of non-renewable natural resources, forestry resources, and electrical energy, and permits provinces to legislate on the export of such resources to other parts of Canada, subject to federal paramountcy. The federal government retains authority over offshore resources (confirmed by the Supreme Court of Canada in *Reference re Newfoundland Continental Shelf* (1984)), interprovincial and international pipelines, nuclear energy, and matters of national concern. Offshore resources off Newfoundland and Labrador and Nova Scotia are managed jointly through accords implemented by federal and provincial legislation.

The Supreme Court of Canada in *References re Greenhouse Gas Pollution Pricing Act* (2021) upheld federal carbon pricing legislation under the "peace, order and good government" power as a matter of national concern,[8] while in *Reference re Impact Assessment Act* (2023) it held that significant parts of the federal impact assessment regime were unconstitutional because they reached beyond federal jurisdiction into matters of provincial competence.[9]

[8] *References re Greenhouse Gas Pollution Pricing Act*, 2021 SCC 11.

[9] *Reference re Impact Assessment Act*, 2023 SCC 23.

Australia: State Ownership Onshore and Offshore Arrangements

In Australia, the states own onshore minerals and petroleum and regulate their development. Offshore, following the High Court's decision in *New South Wales v Commonwealth* (the *Seas and Submerged Lands Case*) (1975), which upheld Commonwealth sovereignty over the territorial sea and continental shelf, the Offshore Constitutional Settlement of 1979 allocated jurisdiction over the first three nautical miles of the territorial sea to the states, with the Commonwealth retaining authority beyond. Petroleum activities in Commonwealth waters are governed by the Offshore Petroleum and Greenhouse Gas Storage Act 2006, administered by the National Offshore Petroleum Titles Administrator and regulated for safety and environmental matters by the National Offshore Petroleum Safety and Environmental Management Authority (NOPSEMA).

The United States: Layered Ownership

In the United States, mineral ownership is layered among private parties, states, the federal government, and Native American tribes. The federal government owns the mineral estate under approximately 700 million acres of land and administers leasing under the Mineral Leasing Act 1920 and, for the outer continental shelf beyond state waters (generally three nautical miles), under the Outer Continental Shelf Lands Act 1953. The Submerged Lands Act 1953 confirmed state title to submerged lands within three nautical miles of the coast (three marine leagues for Texas and the Gulf coast of Florida). Regulation of oil and gas production on private and state lands is primarily a matter of state law, administered by bodies such as the Railroad Commission of Texas.

3.5 Indigenous Peoples and Resource Governance

The Encounter between Resource Development and Indigenous Rights

A large proportion of the world's remaining mineral, petroleum, and renewable energy resources lies on or beneath lands traditionally owned, occupied, or used by indigenous peoples. The encounter between resource development and indigenous rights has been one of the most significant sources of legal conflict in the field, and the recognition of indigenous rights has fundamentally altered the legal landscape for resource projects in many jurisdictions. Chapter 21 provides a detailed treatment of the international legal framework; this section focuses on the constitutional and property law dimensions.

Recognition of Indigenous Title and Rights

The recognition of indigenous land rights has proceeded through constitutional provisions, legislation, judicial decisions, and treaty arrangements. In Australia, the High Court in *Mabo v Queensland (No 2)* (1992) rejected the doctrine of *terra*

nullius and recognised native title at common law,[10] a recognition subsequently regulated by the Native Title Act 1993, which confers upon native title holders a "right to negotiate" with respect to certain future acts, including the grant of mining tenements. In *Wik Peoples v Queensland* (1996), the High Court held that pastoral leases did not necessarily extinguish native title.

In Canada, section 35 of the Constitution Act 1982 recognises and affirms existing aboriginal and treaty rights. The Supreme Court of Canada recognised aboriginal title in *Delgamuukw v British Columbia* (1997) and, for the first time, made a declaration of aboriginal title over a specific territory in *Tsilhqot'in Nation v British Columbia* (2014), holding that aboriginal title confers the right to decide how the land will be used and to enjoy its economic benefits, and that governments seeking to use titled land without consent must justify the infringement.[11] The Court has also developed a duty to consult and, where appropriate, accommodate aboriginal peoples when the Crown contemplates conduct that might adversely affect asserted or established rights (*Haida Nation v British Columbia (Minister of Forests)*, 2004). These duties have major implications for pipeline and mining projects; in *Clyde River (Hamlet) v Petroleum Geo-Services Inc* (2017), the Supreme Court quashed an authorisation for offshore seismic testing because the consultation with the Inuit had been inadequate.

In Latin America, the Inter-American Court of Human Rights has developed a substantial jurisprudence on indigenous and tribal peoples' property rights under Article 21 of the American Convention on Human Rights. In *Saramaka People v Suriname* (2007), the Court held that the state must ensure the effective participation of the Saramaka people in development or investment plans within their territory, guarantee that they receive a reasonable benefit, and ensure that no concession is issued without a prior environmental and social impact assessment; for large-scale projects with a major impact, the state has a duty to obtain their free, prior, and informed consent.[12] In *Kichwa Indigenous People of Sarayaku v Ecuador* (2012), the Court found Ecuador responsible for permitting a petroleum company to carry out seismic exploration on Sarayaku territory without prior consultation.

In Africa, the African Commission on Human and Peoples' Rights in *Centre for Minority Rights Development (Kenya) and Minority Rights Group International on behalf of Endorois Welfare Council v Kenya* (2010) found that Kenya had violated the rights of the Endorois community by evicting them from their ancestral land around Lake Bogoria without adequate consultation or compensation,

[10] *Mabo v Queensland (No 2)* (1992) 175 CLR 1 (High Court of Australia).

[11] *Tsilhqot'in Nation v British Columbia*, 2014 SCC 44, [2014] 2 SCR 257.

[12] *Saramaka People v Suriname*, Inter-American Court of Human Rights, Judgment of 28 November 2007, Series C No 172, paras 129–137.

and the African Court on Human and Peoples' Rights in *African Commission on Human and Peoples' Rights v Republic of Kenya* (the *Ogiek* case, 2017) found violations of the Ogiek people's rights to their ancestral land in the Mau Forest.

Indigenous Ownership of Subsurface Resources

A distinct question is whether indigenous peoples own the subsurface resources beneath their lands. In most regalian systems, the answer is no: the state's ownership of minerals extends beneath indigenous lands, and indigenous rights are limited to the surface and to rights of consultation, compensation, and benefit-sharing. There are exceptions. In the United States, tribes own the mineral estate beneath many reservation lands, and mineral development is governed by the Indian Mineral Leasing Act 1938 and the Indian Mineral Development Act 1982. In Australia's Northern Territory, the Aboriginal Land Rights (Northern Territory) Act 1976 confers a veto over exploration on Aboriginal land, subject to a national interest override. In Canada, the content of aboriginal title, as described in *Delgamuukw*, includes subsurface mineral rights.

The question has particular resonance in Africa, where the distinction between indigenous peoples and other local communities is often contested and where customary communities assert claims to resources beneath their lands. Under Nigerian law, such claims are foreclosed by section 44(3) of the Constitution, but communities may assert rights to compensation, environmental protection, and participation in benefit-sharing arrangements.

3.6 Resource Nationalism and Regulatory Reform

Defining Resource Nationalism

"Resource nationalism" describes the tendency of governments to assert greater control over natural resources and a larger share of the economic benefits they generate. The term is often used pejoratively by investors and commentators to describe measures perceived as opportunistic or confiscatory, but it can also describe legitimate efforts by states to correct inequitable bargains, to capture a fair share of windfall profits, and to promote national development. A more neutral analytical definition treats resource nationalism as a spectrum of state measures designed to increase state control over, or state revenue from, the resource sector, ranging from fiscal adjustments at one end to outright expropriation at the other.

Resource nationalism is cyclical. It tends to intensify during periods of high commodity prices, when the gap between the returns earned by investors and those anticipated at the time of investment widens and governments come under political pressure to capture windfall gains. It tends to recede during periods of low prices, when states compete to attract investment and are more willing to offer favourable terms. This cyclicality reflects the obsolescing bargain described in Chapter 1.

Forms of Resource Nationalism

Resource nationalism takes a variety of forms, which may be classified by increasing intensity:

1. **Fiscal measures**: increases in royalty rates, income tax rates, or special resource taxes; the introduction of windfall profit taxes or resource rent taxes; restrictions on cost recovery or deductions; and changes to the treatment of transfer pricing.
2. **Local content and beneficiation requirements**: obligations to employ nationals, procure goods and services locally, process minerals domestically before export, or transfer technology. Nigeria's Nigerian Oil and Gas Industry Content Development Act 2010 is a prominent example; Indonesia's ban on the export of unprocessed nickel ore, in force since January 2020, illustrates the beneficiation model.
3. **Export restrictions and domestic supply obligations**: requirements to supply a portion of production to the domestic market, often at below-market prices (domestic market obligations), or restrictions on exports. Nigeria's Petroleum Industry Act 2021 establishes a domestic crude oil supply obligation and a domestic gas delivery obligation.
4. **Increased state participation**: mandatory carried or paid interests for the state or national resource company in new or existing projects; requirements that a minimum percentage of shares be held by nationals. Tanzania's Natural Wealth and Resources (Permanent Sovereignty) Act 2017 and Natural Wealth and Resources Contracts (Review and Re-Negotiation of Unconscionable Terms) Act 2017 combined participation requirements with a statutory power to renegotiate "unconscionable" terms, and the Mining (Amendment) Act 2017 provided for a non-dilutable free carried interest of not less than sixteen per cent for the government in mining companies.
5. **Contract renegotiation and conversion**: unilateral or coerced renegotiation of contracts, or the mandatory conversion of existing contracts to new forms (for example, from concessions to production sharing contracts or service contracts).
6. **Expropriation and nationalisation**: the outright taking of investor assets, with or without compensation.

Drivers and Justifications

Several factors drive resource nationalism: high commodity prices and perceived windfall profits; the perception that existing contracts were negotiated on unfair terms, under duress, or as a result of corruption; fiscal pressures on governments; the political salience of resource sovereignty; the desire to build national industrial capacity; and, increasingly, concerns about energy and resource security. Governments frequently justify their measures by reference to permanent sovereignty,

the public interest, and the need to correct asymmetries of information and bargaining power at the time of contracting.

Regulatory Reform: The Nigerian Petroleum Industry Act 2021

Not all reform of resource governance is properly characterised as resource nationalism. Many states have undertaken comprehensive reforms designed to modernise their legal frameworks, improve governance, and attract investment. Nigeria's Petroleum Industry Act 2021 (PIA), the product of nearly two decades of legislative effort, is a leading example. The PIA:

- restructured the institutional framework by establishing two regulators — the Nigerian Upstream Petroleum Regulatory Commission (NUPRC) and the Nigerian Midstream and Downstream Petroleum Regulatory Authority (NMDPRA) — and providing for the incorporation of the Nigerian National Petroleum Corporation as a limited liability company, Nigerian National Petroleum Company Limited (NNPC Ltd), operating on commercial principles;
- replaced the oil prospecting licence and oil mining lease with petroleum prospecting licences and petroleum mining leases, with provisions on relinquishment, work commitments, and the conversion of existing licences;
- introduced a new fiscal regime comprising a hydrocarbon tax on upstream crude oil profits, companies income tax, and production-based royalties, with incentives for gas and frontier exploration;
- established host community development trusts;
- created frameworks for environmental remediation and decommissioning funds; and
- established the Frontier Exploration Fund and the Midstream and Downstream Gas Infrastructure Fund.

The PIA illustrates the trend towards replacing discretionary, contract-based governance with a comprehensive statutory framework. Its implementation and the conversion of existing licences to the new regime are examined in Chapters 5 and 24.

3.7 Expropriation, Nationalization, and Compensation

Terminology

"Expropriation" refers to the taking of property by the state. "Nationalisation" usually refers to large-scale takings of entire industries or sectors as part of a programme of economic or social reform. "Confiscation" refers to a taking without compensation. "Indirect expropriation" or "creeping expropriation" refers to measures that do not formally transfer title but substantially deprive the owner of the use, enjoyment, or economic value of the property. The distinction

between indirect expropriation and non-compensable regulation is one of the most difficult questions in international investment law and is examined in detail in Chapter 14. This section focuses on the general international law framework, with particular reference to its development through energy disputes.

Conditions of Lawful Expropriation

Customary international law, as reflected in Resolution 1803 and in the practice of investment treaties, permits a state to expropriate foreign property provided that the expropriation is:

1. for a public purpose;
2. non-discriminatory;
3. carried out in accordance with due process of law; and
4. accompanied by the payment of compensation.

These conditions are reflected, with variations, in virtually every investment treaty. Article 13 of the ECT, for example, prohibits expropriation of investments of investors of another contracting party except where the expropriation is "(a) for a purpose which is in the public interest; (b) not discriminatory; (c) carried out under due process of law; and (d) accompanied by the payment of prompt, adequate and effective compensation".

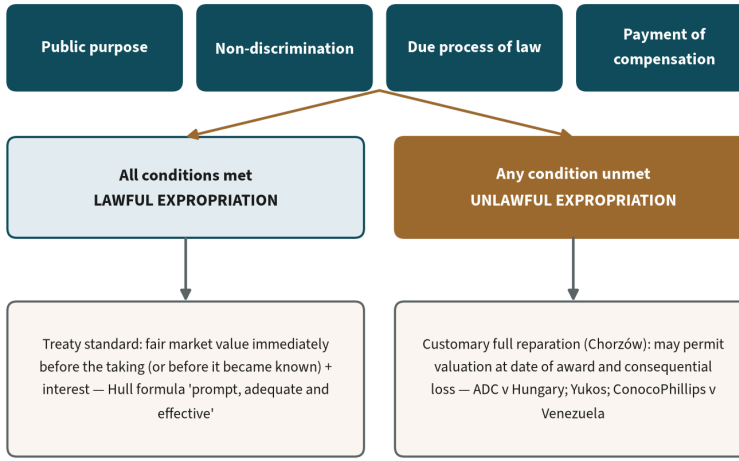


Figure 3.2 The four conditions of lawful expropriation and their consequences for compensation. *Source:* prepared for this edition.

The Standard of Compensation

The standard of compensation has been the most contested aspect of the law of expropriation. Three formulations compete.

The Hull standard — "prompt, adequate and effective" compensation — requires compensation equivalent to the full market value of the expropriated property, paid without delay and in convertible currency. It was championed by capital-exporting states and is reflected in most modern investment treaties, often with the further specification that compensation shall be equivalent to the fair market value of the investment immediately before the expropriation or before it became publicly known, whichever is earlier.

The "appropriate compensation" standard of Resolution 1803 and the Charter of Economic Rights and Duties of States permits account to be taken of a range of circumstances, potentially justifying compensation below full market value. The *Aminoil* tribunal applied this standard, holding that the "legitimate expectations" of the parties, as reflected in the evolution of their contractual relationship, informed the determination of appropriate compensation, and that compensation should take account of a reasonable rate of return rather than speculative profits.

Partial or no compensation has been asserted in the context of large-scale nationalisations forming part of fundamental social and economic reform, on the ground that full compensation would make such reform impossible. This position has not commanded general acceptance as a rule of international law, although lump-sum settlements negotiated between states have frequently provided for compensation at substantially less than full value.

In treaty-based arbitration today, the debate has largely been resolved in favour of the full market value standard for lawful expropriation, because the treaties themselves specify it. The more significant debates now concern valuation methodology, the valuation date, and the distinction between lawful and unlawful expropriation.

Lawful and Unlawful Expropriation

Following *Chorzów Factory*, tribunals distinguish between lawful expropriation, for which compensation is due under the treaty standard (typically fair market value at the date of taking), and unlawful expropriation, for which full reparation is due under customary international law. The distinction matters because full reparation may require valuation at the date of the award rather than the date of taking, allowing the claimant to benefit from subsequent increases in value — for example, increases in oil prices.

In *ADC Affiliate Ltd v Hungary* (2006), the tribunal held that because Hungary's expropriation was unlawful, the claimants were entitled to the value of their investment at the date of the award, which was considerably higher

than at the date of the taking.[13] In *Yukos Universal Ltd v Russian Federation* (2014), the tribunal found that Russia's measures against Yukos amounted to an unlawful expropriation and awarded damages of approximately USD 50 billion to the former majority shareholders, using a valuation date chosen to give the claimants the benefit of the higher value.[14] In *ConocoPhillips v Venezuela* (2019), the tribunal, having previously held that Venezuela's taking of ConocoPhillips' interests in heavy oil projects was unlawful because Venezuela had failed to negotiate compensation in good faith, awarded approximately USD 8.5 billion plus interest.[15] By contrast, in *Venezuela Holdings BV (Mobil) v Venezuela* (2014), the tribunal found that the claimants had not established that the expropriation was unlawful merely because the compensation offered was inadequate, and accordingly applied the treaty standard of compensation for lawful expropriation, valuing the investment as of the date of the taking.

The Yukos awards were set aside by the District Court of The Hague in 2016 on jurisdictional grounds relating to the provisional application of the ECT, reinstated by the Hague Court of Appeal in 2020, and subjected to further proceedings, with the Supreme Court of the Netherlands in 2021 remitting the question of alleged fraud in the arbitration to the Amsterdam Court of Appeal, which rejected Russia's fraud arguments in 2024. The saga illustrates the complexity and duration of enforcement proceedings against a sovereign that resists payment.

Compensation for Natural Resource Projects: Valuation Issues

The valuation of expropriated natural resource projects presents distinctive difficulties. The value of a petroleum or mining project depends on the volume of reserves, future commodity prices, production costs, fiscal terms, and the risk profile of the host state, each of which is uncertain. Tribunals have generally accepted the discounted cash flow (DCF) method for producing assets with an established record of profitability, while being sceptical of DCF valuations for projects at an early stage of development. In *Tethyan Copper Company Pty Limited v Pakistan* (2019), the tribunal awarded approximately USD 4 billion in damages and interest for the denial of a mining lease for the Reko Diq copper-gold deposit, valuing the project by DCF despite the fact that it had not yet entered production — a decision that attracted criticism and was followed by a negotiated

[13] *ADC Affiliate Limited and ADC & ADMC Management Limited v Republic of Hungary*, ICSID Case No ARB/03/16, Award, 2 October 2006, paras 493–499.

[14] *Yukos Universal Limited (Isle of Man) v Russian Federation*, PCA Case No AA 227, Final Award, 18 July 2014.

[15] *ConocoPhillips Petrozuata BV, ConocoPhillips Hamaca BV and ConocoPhillips Gulf of Paria BV v Bolivarian Republic of Venezuela*, ICSID Case No ARB/07/30, Award, 8 March 2019.

settlement in 2022 involving the reconstitution of the project with new participation arrangements.[16]

The Iran–United States Claims Tribunal and Lump-Sum Settlements

A substantial body of jurisprudence on the expropriation of energy assets was generated by the Iran–United States Claims Tribunal, established by the Algiers Accords of January 1981 following the Iranian revolution and the hostage crisis. Many of the claims arose from the nationalisation of the Iranian oil industry and the termination of petroleum agreements after 1979. In *Amoco International Finance Corporation v Iran* (1987), Chamber Three distinguished between lawful and unlawful expropriation and held that, where the taking was lawful, the compensation owed was the value of the undertaking as a going concern at the date of the taking, excluding the speculative future profits that would have been recoverable as damages for an unlawful taking.[17] In *Phillips Petroleum Company Iran v Iran* (1989), the Tribunal held that the claimant was entitled to compensation for the expropriation of its contractual rights under a joint structure agreement with the National Iranian Oil Company, and that the full value of those rights was to be determined using a discounted cash flow analysis informed by contemporaneous price expectations.[18] In *Starrett Housing Corporation v Iran* (1983), the Tribunal recognised that measures taken by a state can interfere with property rights to such an extent that the rights are rendered so useless that they must be deemed to have been expropriated, even though the state does not purport to have expropriated them and legal title remains with the original owner — a foundational statement of the concept of indirect expropriation.

Outside institutional adjudication, many nationalisation disputes have been resolved by lump-sum agreements between the expropriating state and the home state of the investors, under which the expropriating state pays a global sum that the home state distributes among its nationals through a domestic claims commission. The United States Foreign Claims Settlement Commission and the United Kingdom Foreign Compensation Commission administered such distributions in respect of nationalisations in Eastern Europe, Cuba, and elsewhere. The compensation provided under lump-sum agreements has frequently been considerably less than the full value of the property taken. Whether such settlements constitute state practice supporting a standard of less-than-full compensation, or merely represent negotiated compromises without *opinio juris*, remains debated;

[16] *Tethyan Copper Company Pty Limited v Islamic Republic of Pakistan*, ICSID Case No ARB/12/1, Award, 12 July 2019.

[17] *Amoco International Finance Corporation v Islamic Republic of Iran*, Partial Award No 310-56-3, 14 July 1987, 15 Iran-US CTR 189.

[18] *Phillips Petroleum Company Iran v Islamic Republic of Iran*, Award No 425-39-2, 29 June 1989, 21 Iran-US CTR 79.

the prevailing view is that they reflect pragmatic settlement rather than a belief that international law requires no more.

Due Process and Judicial Review of Expropriation

The due process condition requires that an expropriation be carried out in accordance with the procedures established by domestic law and with the fundamental international standards of procedural fairness, including reasonable advance notice, a fair hearing, and the opportunity to challenge the measure before an independent and impartial body. In *ADC v Hungary*, the tribunal held that due process "demands an actual and substantive legal procedure for a foreign investor to raise its claims against the depriving actions already taken or about to be taken against it", including a reasonable chance within a reasonable time to claim its legitimate rights and have its claims heard.[19] Domestic constitutional law frequently imposes additional requirements. Section 44(1) of the Nigerian Constitution, for instance, prohibits the compulsory acquisition of any interest in moveable or immoveable property except in the manner and for the purposes prescribed by a law that, among other things, requires the prompt payment of compensation and gives the person claiming compensation a right of access to a court or tribunal for the determination of his interest and the amount of compensation. Section 25 of the Nigerian Investment Promotion Commission Act (Cap N117, Laws of the Federation of Nigeria 2004) further provides that no enterprise shall be nationalised or expropriated by any government of the Federation, and that no person who owns the enterprise shall be compelled by law to surrender his interest to any other person, save where the acquisition is in the national interest or for a public purpose and under a law that provides for payment of fair and adequate compensation and a right of access to the courts.

3.8 Comparative Constitutional Approaches

A Typology of Constitutional Approaches

Constitutions address natural resources in strikingly different ways. A comparative survey reveals at least five approaches:

1. **Silence.** Some constitutions, including the Constitution of the United States, do not address natural resource ownership at all, leaving the matter to property law and legislation.
2. **Vesting clauses.** Many constitutions vest ownership of minerals and petroleum in the state, the nation, or the people (Nigeria, Mexico, Ghana).

[19] *ADC Affiliate Limited v Republic of Hungary*, ICSID Case No ARB/03/16, Award, 2 October 2006, para 435.

3. **Federal allocation clauses.** Federal constitutions allocate authority between levels of government (Canada's sections 92A and 109; Nigeria's Exclusive Legislative List and derivation clause).
4. **Stewardship and sustainability clauses.** Some constitutions impose duties of sustainable management. Article 112 of the Constitution of Norway (formerly Article 110b) provides that natural resources shall be managed on the basis of comprehensive long-term considerations that will safeguard this right for future generations as well. The Supreme Court of Norway in *Greenpeace Nordic and Nature and Youth v Ministry of Petroleum and Energy* (2020) held that Article 112 is a substantive right but that the threshold for judicial intervention in parliamentary decisions is high, and it declined to invalidate Arctic petroleum licences; the subsequent *Greenpeace Nordic v Norway* proceedings before the Oslo District Court in 2024 held certain field development approvals invalid for failure to assess combustion emissions.
5. **Rights of nature.** The Constitution of Ecuador 2008 (Articles 71–74) recognises the rights of nature (*Pacha Mama*) to exist, persist, maintain, and regenerate its vital cycles. The Constitutional Court of Ecuador in the *Los Cedros* case (2021) held that mining concessions in the Los Cedros protected forest violated the rights of nature.

Revenue-Sharing Clauses

Constitutions in resource-rich federations frequently prescribe revenue-sharing arrangements. Nigeria's thirteen per cent derivation principle has been noted. The Constitution of Iraq 2005, Article 112, provides that the federal government, with the producing governorates and regional governments, shall undertake the management of oil and gas extracted from present fields, with revenues distributed in a fair manner in proportion to the population distribution — a provision whose ambiguity has fuelled protracted disputes between Baghdad and the Kurdistan Regional Government, including the 2022 ruling of Iraq's Federal Supreme Court declaring the Kurdistan Region's oil and gas law unconstitutional, and the 2023 ICC arbitral award in Iraq's favour concerning exports through the Iraq–Türkiye pipeline. The Constitution of Kenya 2010 requires legislation providing for the sharing of benefits from natural resources between the national government, counties, and local communities, implemented in part by section 58 of the Petroleum Act 2019, which allocates 20 per cent of the national government's share of profit petroleum to the county government and 5 per cent to the local community.

Constitutional Governance of Resource Revenues

A further dimension of constitutional governance concerns the management of the revenues generated by resource extraction. Because petroleum and mineral revenues are volatile, finite, and often very large relative to the rest of the economy, they create risks of macroeconomic instability, pro-cyclical spending, and the diversion of public funds. Constitutional and quasi-constitutional responses fall into four categories.

Fiscal rules limit the use of resource revenues in the annual budget. Norway's fiscal rule, adopted in 2001, provides that the structural non-oil deficit should over time correspond to the expected real return on the Government Pension Fund Global, now estimated at three per cent. Chile's structural balance rule adjusts fiscal targets for the long-term price of copper. Nigeria's Fiscal Responsibility Act 2007 introduced a budget framework based on a reference oil price, with revenues above that price saved in the Excess Crude Account, an arrangement whose legal basis was repeatedly contested between the federal and state governments.

Sovereign wealth funds set aside resource revenues for stabilisation, saving, or development purposes. Their governance is guided by the Santiago Principles. The Nigeria Sovereign Investment Authority Act 2011 established three funds — the Stabilisation Fund, the Future Generations Fund, and the Nigeria Infrastructure Fund — each with defined investment mandates. Ghana's Petroleum Revenue Management Act 2011 (Act 815) established the Ghana Stabilisation Fund and the Ghana Heritage Fund and created a Public Interest and Accountability Committee to monitor compliance, an innovative example of statutory civil society oversight.

Earmarking and distribution rules allocate revenues to particular purposes or populations. Some states have experimented with direct distribution of resource revenues to citizens, as in the Alaska Permanent Fund Dividend, paid annually to qualifying residents since 1982.

Transparency and audit obligations require the publication of resource revenues and contracts and their independent audit. The Nigeria Extractive Industries Transparency Initiative (NEITI) Act 2007 made Nigeria the first EITI implementing country to give the initiative a statutory basis, requiring periodic audits of the extractive sector and the publication of the results. Section 3 of the Act sets out NEITI's objectives, including ensuring due process and transparency in payments made by extractive industry companies to the federal government and its agencies.

These arrangements form part of the constitutional governance of resources in a broad sense, and their effectiveness depends on enforcement mechanisms — legislative oversight, independent audit institutions, judicial review, and civil

society monitoring. Doctoral research on the resource curse has increasingly focused on the design and performance of such arrangements.

Lessons from Comparison

Comparative analysis suggests several lessons. First, the formal ownership model matters less than the quality of the institutions that administer it; state ownership can be associated with excellent governance (Norway) or with chronic mismanagement. Second, revenue-sharing arrangements that leave producing regions and communities with a visible and meaningful share of benefits tend to reduce conflict, but must be designed to avoid creating new inequalities or local governance failures. Third, constitutional entrenchment of sustainability and stewardship duties can provide a basis for judicial review of resource decisions, although courts have generally been cautious about intervening in what they regard as policy matters. Fourth, the recognition of indigenous and community rights through constitutional, judicial, and treaty mechanisms has become a defining feature of resource governance, with profound implications for project development.

LEADING AUTHORITY

Government of the State of Kuwait v American Independent Oil Company (Aminoil), 1982

Facts. Aminoil held a sixty-year concession granted in 1948 by the Ruler of Kuwait over the Kuwaiti share of the Neutral Zone. The concession contained stabilisation clauses providing that the Ruler would not annul the agreement by general or special legislation or administrative measures, and that no alteration should be made except by mutual agreement. Over the years, the fiscal terms were revised several times. In 1977, Kuwait terminated the concession by decree and took over Aminoil's assets.

Holding. The tribunal (Professor Paul Reuter, President; Professor Hamed Sultan; Sir Gerald Fitzmaurice) held that the nationalisation was lawful. The stabilisation clauses did not prohibit nationalisation because such a far-reaching limitation on sovereignty would require express stipulation and would have to cover a relatively limited period; moreover, the contract had evolved through renegotiation. However, the stabilisation clauses gave rise to legitimate expectations relevant to the assessment of compensation, which was to be "appropriate" and was calculated by reference to the value of the undertaking and a reasonable rate of return. Kuwait was ordered to pay approximately USD 179 million, including interest.

Significance. Aminoil is the leading authority for the proposition that stabilisation clauses operate primarily to affect compensation rather than to invalidate sovereign acts. Its emphasis on the evolving nature of long-term contracts and on legitimate expectations anticipated modern debates over fair and equitable treatment.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Constitutional design for a new petroleum producer (hypothetical). The fictional Republic of Tamara, a federal state with five provinces, has recently discovered substantial onshore and offshore petroleum reserves. Its constitution is silent on resource ownership. A constitutional review commission has asked you to prepare a 3,000-word advisory paper addressing: (a) the choice between federal and provincial ownership of onshore and offshore resources; (b) the design of a revenue-sharing formula, including any derivation component and its treatment of offshore production; (c) the incorporation of stewardship and intergenerational equity duties; (d) the recognition of the rights of customary landholding communities; and (e) mechanisms to reduce the risk of resource-related conflict. Draw on comparative experience from at least four jurisdictions. *Tamara and its circumstances are fictitious.*

CHAPTER SUMMARY

- Resource sovereignty is an incident of territorial sovereignty, extended by UNCLOS to sovereign rights over the resources of the EEZ and continental shelf; modern instruments increasingly locate that sovereignty in the people, with the state as trustee.
- Subsurface ownership follows three broad models — regalian (state), accession (landowner), and claim-based systems — with major consequences for licensing, revenue distribution, and community attitudes.
- The public trust doctrine and related stewardship concepts impose fiduciary-type duties on states, enforceable in some jurisdictions by citizens and capable of affecting the security of investor rights.
- Federal systems allocate resource authority in widely differing ways; Nigeria's combination of federal ownership, a thirteen per cent derivation floor, the onshore/offshore dichotomy litigation, and recent electricity decentralisation illustrates the political sensitivity of the federal dilemma.
- Indigenous rights to land, consultation, consent, and benefit-sharing have been recognised through constitutional provisions, judicial decisions, and regional human rights jurisprudence, transforming the legal environment for resource projects.
- Resource nationalism is cyclical and takes forms ranging from fiscal adjustments to expropriation; it must be distinguished from good-faith regulatory reform such as Nigeria's Petroleum Industry Act 2021.
- Lawful expropriation requires public purpose, non-discrimination, due process, and compensation; the distinction between lawful and unlawful expropriation determines the measure and valuation date of compensation, with very large financial consequences in energy cases.

DOCTORAL RESEARCH QUESTIONS

1. Does the constitutional recognition of the people as bearers of resource sovereignty create enforceable obligations regarding the management of resource revenues, and how might such obligations be judicially enforced?
2. To what extent has the public trust doctrine been, or could it be, received into Nigerian law as a constraint on the grant of petroleum and mining rights?
3. Evaluate whether host community development trusts under the Petroleum Industry Act 2021 represent a meaningful response to demands for resource control, using comparative evidence from benefit-sharing regimes elsewhere.
4. Is the distinction between lawful and unlawful expropriation, with its consequences for valuation date, theoretically justified, or does it produce windfalls for claimants inconsistent with the compensatory purpose of reparation?
5. How should the valuation of undeveloped resource projects in investment arbitration account for the risk that climate policy will render reserves unburnable?

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4 CHAPTER

International Energy Institutions, Regulatory Systems, and Governance

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Describe the architecture of global energy governance and explain why it is fragmented and polycentric.
2. Analyse the legal foundations, mandates, powers, and limitations of the International Energy Agency, OPEC, the International Renewable Energy Agency, and the International Atomic Energy Agency.
3. Evaluate the role of regional energy regulatory institutions, including those of the European Union, the Energy Community, and African regional bodies.
4. Explain the rationale for independent regulatory agencies and the mechanisms by which their accountability is secured.
5. Assess the prospects for regulatory cooperation and institutional reform in global energy governance.

Introduction

Institutions matter. The rules of international energy law described in the previous chapters do not apply themselves; they are negotiated, interpreted, monitored, and enforced by institutions — international organisations, regional bodies, national regulators, courts, and tribunals. The design of these institutions shapes the content and effectiveness of the law. An institution with a narrow mandate, limited membership, and no binding powers will produce different outcomes from one with broad competence, universal membership, and the authority to adopt binding decisions.

Global energy governance is distinctive for the absence of a central institution. Unlike trade, which is governed by the WTO, or international civil aviation, which is governed by the International Civil Aviation Organization, energy has no single global organisation with comprehensive competence. Instead, governance

is distributed among a constellation of institutions with specialised mandates, overlapping memberships, and divergent interests. This chapter maps that constellation, examines the principal institutions in detail, and considers the role of national regulators and the prospects for reform.

4.1 Global Energy Governance Architecture

The Concept of Global Energy Governance

"Global energy governance" refers to the collective efforts of states and non-state actors to manage energy-related problems that transcend national borders through rules, institutions, and processes. The problems that global energy governance addresses include the security of supply for importing states and security of demand for exporting states, the stability of energy prices, the prevention of nuclear proliferation, the mitigation of climate change, the promotion of energy access in developing countries, the governance of transboundary infrastructure, and the protection of foreign investment.

Scholars have characterised the architecture of global energy governance as a "regime complex" — a set of partially overlapping and non-hierarchical institutions governing a particular issue area. The concept, developed by Kal Raustiala, David Victor, Robert Keohane, and others, captures the absence of a single overarching regime and the presence of multiple institutions whose rules and activities interact, sometimes reinforcing and sometimes conflicting with one another.

Why Is Energy Governance Fragmented?

Several factors explain the fragmentation of global energy governance.

Divergent interests. Producers and consumers have fundamentally opposed interests regarding prices: producers seek high and stable prices, consumers seek low and stable prices. These opposing interests have produced separate institutions — OPEC for producers, the IEA for consumers — rather than a single inclusive organisation.

Sovereignty sensitivities. Energy resources are closely tied to national sovereignty and security. States have been reluctant to cede authority over energy policy to international institutions, preferring arrangements that preserve national discretion.

Path dependence. Energy institutions were created in response to specific crises — the nuclear arms race (IAEA), the oil shocks of the 1970s (IEA), the collapse of the Soviet Union (Energy Charter), the climate crisis (UNFCCC, IRENA) — and each bears the imprint of the circumstances of its creation.

Technological diversity. Different energy sources raise different governance challenges: nuclear energy raises proliferation and safety concerns; oil raises supply security and price stability concerns; renewables raise technology diffusion and investment concerns. Specialised institutions have emerged for each.

Shifting centres of gravity. The growth of energy demand in emerging economies, particularly China and India, has shifted the centre of gravity of global energy markets away from the OECD states that dominated the institutions created in the twentieth century. Existing institutions have struggled to accommodate these new actors.

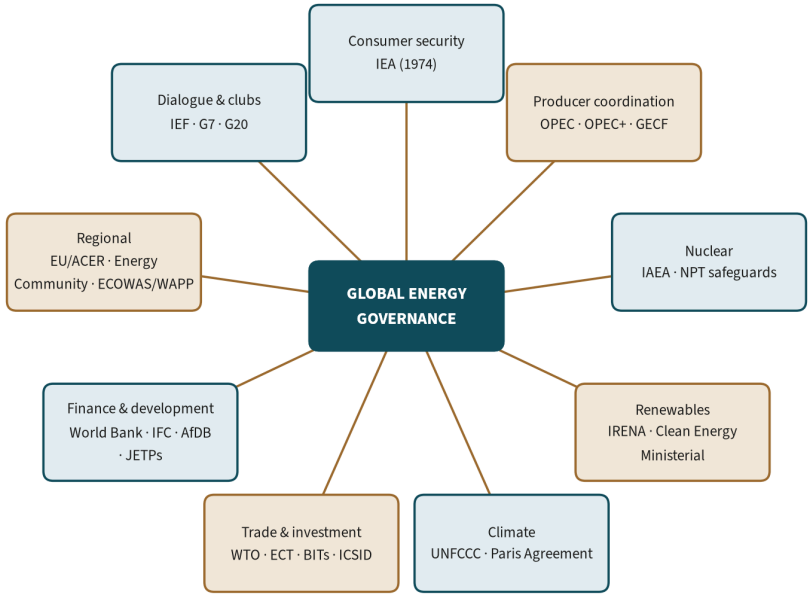


Figure 4.1 The polycentric architecture of global energy governance. *Source:* prepared for this edition.

Mapping the Architecture

The architecture of global energy governance may be mapped along two dimensions: the *functional* dimension, which identifies the governance function performed (security, markets, investment, technology, environment, development, nuclear non-proliferation), and the *membership* dimension, which distinguishes between universal, plurilateral, and regional institutions. Table 4.1 summarises the principal institutions.

Table 4.1 Principal international energy institutions. Membership figures are approximate and change over time; readers should consult the institutions' official sources for current data.

Institution	Founded	Legal basis	Member-ship	Core function	Binding pow-ers
International Energy Agency (IEA)	1974	Agreement on an International En-ergy Program	32 mem-ber coun-tries (OECD states) plus as-sociation countries	Oil supply secu-rity; energy data and analysis; clean energy tran-sitions	Emergency stockholding and response obligations
OPEC	1960	OPEC Statute	12 mem-ber coun-tries (2025)	Coordination of petroleum poli-cies; market sta-bilisation	Production decisions po-litically bind-ing; no exter-nal enforce-ment
International Renewable Energy Agency (IRENA)	2009	IRENA Statute	Approx. 170 members	Promotion of re-newable energy	None (advi-sory)
International Atomic En-ergy Agency (IAEA)	1957	IAEA Statute	Approx. 180 member states	Peaceful uses of nuclear energy; safeguards; safety standards	Safeguards agreements; standards binding when incorporated
Energy Char-ter Confer-ence	1994	Energy Charter Treaty	Contract-ing par-ties (con-tracted after EU with-drawal)	Investment pro-tection; transit; trade	Treaty obliga-tions; in-vestor–state arbitration
International Energy Forum (IEF)	1991	IEF Charter (char-ter 2011)	Approx. 70 coun-tries	Producer–con-sumer dialogue; JODI data	None
Gas Exporting Countries Fo-rum (GECF)	2008	GECF Statute	Gas-ex-porting states	Coordination of gas producer in-terests	None
UNFCCC / Paris Agree-ment	1992 / 2015	Treaties	Near-uni-versal	Climate mitiga-tion and adapta-tion	Procedural obligations; NDC cycle

Informal Governance: G7, G20, and Clubs

In addition to formal international organisations, informal groupings play a significant role in energy governance. The Group of Seven (G7) and the Group of Twenty (G20) have repeatedly addressed energy security, fossil fuel subsidies, and clean energy transitions. At the G20 summit in Pittsburgh in 2009, leaders committed to "rationalize and phase out over the medium term inefficient fossil

fuel subsidies that encourage wasteful consumption". The G7's Climate Club, launched in 2022–2023, seeks to accelerate the decarbonisation of industry. The Clean Energy Ministerial and Mission Innovation bring together energy ministers to promote clean energy technologies. Just Energy Transition Partnerships (JETPs), concluded between groups of donor states and South Africa (2021), Indonesia (2022), Vietnam (2022), and Senegal (2023), combine public and private finance to support the transition away from coal. Such informal arrangements lack legal personality and binding powers but can mobilise political commitment and finance.

4.2 The International Energy Agency and Its Institutional Role

Origins and Legal Foundation

The International Energy Agency was created in November 1974 in response to the oil embargo imposed by Arab members of OPEC in October 1973 following the Yom Kippur War and the associated quadrupling of oil prices. The embargo exposed the vulnerability of industrialised oil-importing states and the absence of any mechanism for coordinating their response. At the initiative of the United States, sixteen OECD states signed the Agreement on an International Energy Program (IEP Agreement) on 18 November 1974, establishing the IEA as an autonomous agency within the framework of the OECD.

The IEP Agreement is a treaty. Membership of the IEA is open only to members of the OECD that are willing and able to meet the requirements of the emergency response mechanism, including the holding of oil stocks. This requirement has been a significant barrier to the membership of major emerging economies such as China and India, which are not OECD members. To address this limitation, the IEA launched an "association" initiative in 2015, under which China, India, Indonesia, Brazil, South Africa, and other states have become association countries participating in IEA work without assuming the obligations of full membership.

Emergency Response Mechanism

The core legal obligation imposed by the IEP Agreement is the maintenance of emergency oil reserves. Article 2 requires each participating country to maintain emergency reserves sufficient to sustain consumption for at least sixty days with no net oil imports, a requirement subsequently increased to ninety days of net imports. The reserves may be held by governments (as in the United States Strategic Petroleum Reserve), by dedicated stockholding agencies, or by industry under obligation.

The IEP Agreement also established an emergency sharing system, triggered when the group or an individual participating country sustains or can reasonably be expected to sustain a reduction in the daily rate of its oil supplies of at least seven

per cent of the average daily rate of its final consumption during a base period. Upon activation, participating countries are required to implement demand restraint measures and to share available oil. In practice, the formal emergency sharing system has never been activated. Instead, the IEA has relied on "coordinated emergency response measures" (CERM), a more flexible mechanism under which members make stocks available in coordinated releases. Coordinated stock releases were undertaken in 1991 (Gulf War), 2005 (Hurricanes Katrina and Rita), 2011 (Libyan civil war), and twice in 2022 (following the Russian invasion of Ukraine), the latter being the largest in the Agency's history.

Evolution of the IEA's Mandate

Although established principally to address oil supply security, the IEA's mandate has broadened considerably. Its *World Energy Outlook*, published annually, is among the most influential sources of energy data and scenarios. In 2021, the IEA published *Net Zero by 2050: A Roadmap for the Global Energy Sector*, which concluded that achieving net zero emissions by 2050 would require no new oil and gas fields approved for development beyond those already committed as of 2021, and no new coal mines or mine extensions. The report was controversial among producers and some IEA members, illustrating the tensions between the Agency's historical mission of supply security and its newer role in promoting clean energy transitions. In 2022, IEA ministers mandated the Agency to help countries achieve net zero emissions, and the Agency has developed significant work programmes on critical minerals security, including the Critical Minerals Security Programme.

The IEA and Critical Minerals Security

The IEA's recent expansion into critical minerals illustrates how institutions created for one security problem adapt to another. The Agency's 2021 special report, *The Role of Critical Minerals in Clean Energy Transitions*, and its subsequent annual *Global Critical Minerals Outlook* documented the high geographical concentration of the extraction and, especially, the processing of minerals such as lithium, cobalt, nickel, graphite, and rare earth elements, and warned that supply disruptions could slow the energy transition and raise its cost. In 2024, IEA members launched the Critical Minerals Security Programme, which includes work on supply diversification, market transparency, and the feasibility of stockholding arrangements for critical minerals analogous in some respects to emergency oil stocks.

The analogy with oil is, however, imperfect. Oil is a fungible commodity traded on deep and liquid markets, whereas many critical minerals are traded in thin, opaque markets with product specifications tailored to individual buyers; stockholding is therefore more complex and potentially more expensive. More-

over, the security risks associated with critical minerals arise less from sudden supply interruptions than from structural concentration and the use of export controls as instruments of industrial and geopolitical strategy. Whether the IEA's legal toolkit, built around the IEP Agreement's oil-specific obligations, can be extended to minerals without treaty amendment is a significant institutional question. Chapter 8 examines the substantive law of critical minerals.

Legal Assessment

The IEA is a hybrid institution. Its emergency response obligations are legally binding treaty commitments, but its broader influence derives from its analytical authority, convening power, and technical expertise rather than from legal compulsion. Its limited membership constrains its legitimacy as a global institution, and its identification with consuming states limits its acceptance by producers. Nevertheless, it remains the most important institution for consumer-country cooperation and an indispensable source of the data on which energy policy and legal analysis depend.

4.3 OPEC and Petroleum Market Coordination

Origins and Legal Foundation

The Organization of the Petroleum Exporting Countries was founded at the Baghdad Conference of 10–14 September 1960 by Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela. Its Statute, adopted in 1961 and subsequently amended, defines its principal aims as "the co-ordination and unification of the petroleum policies of Member Countries and the determination of the best means for safeguarding their interests, individually and collectively" and the devising of "ways and means of ensuring the stabilization of prices in international oil markets with a view to eliminating harmful and unnecessary fluctuations" (Article 2). OPEC is headquartered in Vienna. Its membership has varied over time; Nigeria joined in 1971. Qatar left in 2019, Ecuador in 2020, and Angola in 2024.

The supreme authority of OPEC is the Conference, composed of delegations of the member countries, which meets at least twice a year and takes decisions by unanimity of full members (Article 11). The Board of Governors directs the management of the Organization, and the Secretariat, headed by the Secretary General, carries out executive functions.

Production Quotas and Market Management

OPEC's most visible function is the setting of production ceilings for its members. The system of production quotas was formally introduced in 1982 in response to falling prices and has been the principal instrument of OPEC market management since then. Quota decisions are adopted by the Conference and are politically binding on members, but the Statute provides no mechanism

for enforcing compliance, and over-production by members has been a recurring feature of OPEC history.

Since December 2016, OPEC members have coordinated production with ten non-OPEC producers, including the Russian Federation, Kazakhstan, Mexico, and Oman, under the Declaration of Cooperation, in an arrangement known as "OPEC+". In July 2019, the participants adopted the Charter of Cooperation, providing a framework for continued cooperation. OPEC+ has implemented significant production cuts, including an unprecedented cut of 9.7 million barrels per day agreed in April 2020 in response to the collapse in demand caused by the COVID-19 pandemic.

Quota Allocation and Baseline Disputes

The allocation of production quotas among members is among the most contentious aspects of OPEC's work, because the "baseline" from which each member's reductions are calculated determines its share of the collective market. Disputes over baselines have repeatedly strained the organisation. In 2023, OPEC+ participants agreed that the maximum sustainable production capacities of several members would be assessed by independent sources for the purpose of setting baselines for 2025, reflecting disagreement over whether some African producers, which had been persistently unable to produce up to their quotas, should have their baselines reduced. Angola announced its withdrawal from OPEC in December 2023, citing disagreement with its assigned production level.

For Nigeria, the quota system interacts with domestic production constraints. For extended periods in the early 2020s, Nigeria produced well below its OPEC allocation as a result of crude oil theft, pipeline vandalism, underinvestment, and the divestment of onshore assets by international oil companies. These constraints meant that the binding limit on Nigerian production was operational rather than legal, and they focused domestic policy attention on the security of pipeline infrastructure and the regulatory framework for onshore asset transfers under the Petroleum Industry Act 2021. The episode demonstrates that the legal significance of an international institution's decisions depends on the material circumstances of its members.

The Legal Character of OPEC Decisions

The legal character of OPEC Conference resolutions has never been authoritatively determined. They are adopted by the organ of an international organisation established by treaty, and in that sense they are acts of international law. Yet the Statute does not expressly confer on the Conference the power to adopt decisions binding on members regarding production, and members have consistently treated quota decisions as political commitments rather than legal obligations enforceable by sanctions. The OPEC+ Declaration of Cooperation and Charter

of Cooperation are, by their terms and the practice of the participants, non-binding instruments. The absence of legal enforcement is consistent with the general preference of states for flexibility in matters touching resource sovereignty, and it explains why OPEC's effectiveness has historically depended on the willingness of one or more members — usually Saudi Arabia — to act as "swing producer" by adjusting its own output to stabilise the market.

Legal Issues Surrounding OPEC

OPEC's activities raise several legal questions. The first concerns competition law. OPEC's coordination of production would, if carried out by private companies, constitute a cartel prohibited by the competition laws of most states. Private plaintiffs in the United States have sought to sue OPEC and its members under the Sherman Act. In *International Association of Machinists and Aerospace Workers v OPEC* (1981), the Ninth Circuit Court of Appeals affirmed the dismissal of such a suit on the basis of the act of state doctrine.[1] Legislation known as the "No Oil Producing and Exporting Cartels Act" (NOPEC), which would remove sovereign immunity and the act of state defence for such claims, has been introduced repeatedly in the United States Congress but has not been enacted.

The second concerns trade law. Production quotas are, in effect, restrictions on the production and therefore the export of oil, raising questions of compatibility with Article XI of the GATT, which prohibits quantitative restrictions on exports. Most OPEC members are members of the WTO, but no WTO challenge to OPEC production quotas has been brought. Arguments that quotas could be justified under Article XX(g) as measures relating to the conservation of exhaustible natural resources, or that production restrictions differ legally from export restrictions, have been advanced but not tested.

The third concerns the relationship between OPEC decisions and contractual obligations. Production quotas may require the restriction of production by joint ventures involving foreign companies, raising questions about whether quota compliance constitutes *force majeure* or a change in law under production sharing contracts and joint operating agreements. Many Nigerian production sharing contracts and joint venture arrangements contain provisions allocating the burden of OPEC-mandated production restrictions.

OPEC and the Energy Transition

OPEC faces a fundamental strategic challenge in the energy transition. If global oil demand peaks and declines, the value of members' reserves will fall, and competition for market share may intensify. OPEC has consistently argued that oil will remain a significant part of the global energy mix for decades and has warned

[1] *International Association of Machinists and Aerospace Workers v Organization of Petroleum Exporting Countries*, 649 F 2d 1354 (9th Cir 1981).

against underinvestment in oil supply. Within the UNFCCC process, OPEC members — particularly Saudi Arabia — have often resisted language calling for a phase-out of fossil fuels, and the formulation adopted at COP28, calling for "transitioning away from fossil fuels in energy systems", reflected a compromise in which producer-state resistance played a central role.

4.4 The International Renewable Energy Agency

Establishment and Legal Foundation

The International Renewable Energy Agency was established by the Statute of IRENA, opened for signature in Bonn on 26 January 2009 and entering into force on 8 July 2010. Its headquarters are in Masdar City, Abu Dhabi, in the United Arab Emirates — a symbolically significant choice, locating the principal international institution for renewable energy in a major oil-exporting state. Membership is open to all states members of the United Nations and to regional intergovernmental economic integration organisations; the European Union is a member. With approximately 170 members, IRENA has one of the broadest memberships of any energy institution.

Article II of the Statute defines IRENA's objective as promoting "the widespread and increased adoption and the sustainable use of all forms of renewable energy", taking into account national and domestic priorities and benefits derived from a combined approach of renewable energy and energy efficiency measures, and the contribution of renewable energy to environmental preservation, climate protection, economic growth and social cohesion, access to and security of energy supply, regional development, and intergenerational responsibility. Article III defines renewable energy to include bioenergy, geothermal energy, hydropower, ocean energy (including tidal, wave, and ocean thermal energy), solar energy, and wind energy.

Functions and Activities

IRENA's functions, set out in Article IV, are advisory and facilitative. They include analysing and systematising renewable energy practices, policy instruments, investment mechanisms, and best practices; providing policy advice and assistance to members on request; promoting knowledge and technology transfer; providing capacity building and education; and stimulating research and networking. IRENA publishes influential reports, including the *World Energy Transitions Outlook*, annual *Renewable Capacity Statistics*, and *Renewable Power Generation Costs*, the last of which has documented the dramatic decline in the costs of solar photovoltaic and wind power over the past decade.

IRENA has no power to adopt binding decisions. Its influence derives from its analytical work, its convening role, and its support for member states in designing renewable energy policies. In the context of COP28 in 2023, IRENA played a

central role in tracking progress towards the pledge, endorsed in the global stocktake decision, to triple global renewable energy capacity and double the global average annual rate of energy efficiency improvements by 2030.

Assessment

IRENA represents a distinctive model of international energy governance: a universal-membership institution with a promotional rather than regulatory mandate. Its strengths lie in its legitimacy, its expertise, and its capacity to support developing states. Its limitations lie in the absence of binding powers and of mechanisms for addressing the legal barriers to renewable energy deployment — such as trade disputes over local content requirements and investment disputes over the withdrawal of incentives — which are governed by other regimes.

4.5 The International Atomic Energy Agency

Establishment and Mandate

The International Atomic Energy Agency was established by its Statute, approved on 23 October 1956 and entering into force on 29 July 1957. Its creation followed President Eisenhower's "Atoms for Peace" address to the United Nations General Assembly in December 1953, which proposed an international agency to promote the peaceful uses of atomic energy while preventing the diversion of nuclear materials to military purposes. The IAEA is headquartered in Vienna and is an autonomous organisation within the United Nations system, reporting annually to the General Assembly and, where appropriate, to the Security Council.

Article II of the Statute defines the Agency's dual objective: to "seek to accelerate and enlarge the contribution of atomic energy to peace, health and prosperity throughout the world" and to "ensure, so far as it is able, that assistance provided by it or at its request or under its supervision or control is not used in such a way as to further any military purpose". This dual mandate — promotion and control — has shaped the Agency's work and generated tensions between its developmental and verification roles.

Safeguards

The IAEA's most significant legal function is the application of safeguards to verify that nuclear material is not diverted from peaceful uses to nuclear weapons or other nuclear explosive devices. Under Article III of the Treaty on the Non-Proliferation of Nuclear Weapons 1968 (NPT), each non-nuclear-weapon state party undertakes to accept safeguards, as set forth in an agreement to be negotiated and concluded with the IAEA, for the exclusive purpose of verification of the fulfilment of its obligations under the Treaty. These comprehensive safeguards agreements, based on a model document known as INFCIRC/153 (Corrected),

apply to all source or special fissionable material in all peaceful nuclear activities within the territory of the state.

Following the discovery in 1991 of Iraq's clandestine nuclear weapons programme, which comprehensive safeguards had failed to detect, the IAEA Board of Governors approved in 1997 the Model Additional Protocol (INFCIRC/540), which significantly expands the information that states must provide and the access that inspectors enjoy, including complementary access to sites. The Additional Protocol is voluntary but has been concluded by a large majority of states with comprehensive safeguards agreements.

Where the Agency finds that a state is in non-compliance with its safeguards agreement, the Board of Governors may report the non-compliance to all members and to the Security Council and General Assembly (Statute, Article XII.C). The Board found Iran in non-compliance in 2005, leading to referral to the Security Council and the adoption of sanctions resolutions, and later to the Joint Comprehensive Plan of Action of 2015, under which the IAEA was entrusted with extensive monitoring and verification responsibilities. The Board again found Iran in breach of its safeguards obligations in June 2025.

Safety Standards

Article III.A.6 of the Statute authorises the Agency "to establish or adopt ... standards of safety for protection of health and minimization of danger to life and property". The IAEA Safety Standards Series — comprising Safety Fundamentals, Safety Requirements, and Safety Guides — is not binding in itself, but the standards are binding on the Agency in relation to its own operations and on states in relation to operations assisted by the Agency, and they are widely incorporated into national regulation. The Convention on Nuclear Safety 1994 and the Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management 1997, both adopted under IAEA auspices, establish binding obligations reinforced by peer review mechanisms. These instruments are examined in Chapter 11.

Nuclear Newcomers and the Milestones Approach

A growing number of states without existing nuclear power programmes — including several in Africa — have expressed interest in developing nuclear energy. The IAEA supports such "newcomer" states through its Milestones Approach, which identifies nineteen infrastructure issues (including the national position, nuclear safety, management, funding and financing, the legal framework, safeguards, the regulatory framework, radiation protection, the electrical grid, human resource development, stakeholder involvement, site and supporting facilities, environmental protection, emergency planning, nuclear security, the nuclear fuel cycle, radioactive waste management, industrial involvement, and procurement)

that must be addressed across three phases of programme development, each concluding with a milestone. The Agency conducts Integrated Nuclear Infrastructure Review (INIR) missions to assess progress.

The legal framework is an explicit component of the Milestones Approach. A newcomer state is expected to adhere to the principal international nuclear instruments and to enact comprehensive national nuclear legislation establishing an independent regulatory body, licensing requirements, nuclear liability rules, and provisions on safeguards, security, and waste management. In Nigeria, the Nuclear Safety and Radiation Protection Act 1995 established the Nigerian Nuclear Regulatory Authority as the regulator for nuclear safety and radiological protection, while the Nigeria Atomic Energy Commission, established under the Nigeria Atomic Energy Commission Act 1976, is responsible for promoting the development of atomic energy. Ghana, Kenya, and Uganda are among the other African states that have engaged extensively with the IAEA on nuclear power infrastructure. The interaction between IAEA guidance and national legislation is a good illustration of how the soft law outputs of an international organisation shape domestic legal frameworks.

Assessment

The IAEA is the most legally powerful of the energy-specific international organisations, by virtue of its safeguards function and its role in the non-proliferation regime. Its experience demonstrates both the possibilities and the limits of international verification: safeguards have been strengthened in response to failures, but the Agency depends on the cooperation of states and on the Security Council for enforcement. Its promotional role has expanded with renewed interest in nuclear energy as a low-carbon source, including small modular reactors.

4.6 Regional Energy Regulatory Institutions

The European Union

The European Union possesses the most developed regional energy governance system in the world. Article 194 of the Treaty on the Functioning of the European Union, introduced by the Treaty of Lisbon in 2009, provides that Union policy on energy shall aim, in a spirit of solidarity between member states, to ensure the functioning of the energy market, ensure security of energy supply, promote energy efficiency and energy saving and the development of new and renewable forms of energy, and promote the interconnection of energy networks. Article 194(2) preserves "a Member State's right to determine the conditions for exploiting its energy resources, its choice between different energy sources and the general structure of its energy supply", a significant limitation on Union competence that reflects the sovereignty sensitivities discussed above.

EU energy law has developed through successive legislative packages. The Third Energy Package of 2009 required the unbundling of transmission system operators from generation and supply activities, strengthened the independence of national regulatory authorities, and established the Agency for the Cooperation of Energy Regulators (ACER) and the European Networks of Transmission System Operators for Electricity and Gas (ENTSO-E and ENTSOG). The Clean Energy for All Europeans Package of 2019 updated the framework for electricity markets, renewable energy, energy efficiency, and governance, including the Regulation on the Governance of the Energy Union and Climate Action (EU) 2018/1999, which requires member states to adopt integrated national energy and climate plans. The "Fit for 55" package, adopted between 2021 and 2023, aligned EU energy law with the target of reducing net greenhouse gas emissions by at least 55 per cent by 2030 relative to 1990 levels, established by the European Climate Law, Regulation (EU) 2021/1119, which also enshrines the objective of climate neutrality by 2050. The REPowerEU plan of 2022 sought to end dependence on Russian fossil fuels, and in 2024 the Union adopted reforms of electricity market design and a hydrogen and decarbonised gas market package.

The Court of Justice of the European Union has played a major role in shaping EU energy law, including through cases on unbundling, the independence of regulators, state aid for renewable energy (*PreussenElektra*, 2001), and the principle of energy solidarity. In *Germany v Poland* (2021), the Court held that the principle of energy solidarity in Article 194(1) is a legally binding principle that requires the EU institutions and member states to take into account the interests of other member states in energy security, annulling a Commission decision on the OPAL gas pipeline that had failed to examine the effects on Poland's security of supply.[2]

The Energy Community

The Energy Community, established by the Treaty Establishing the Energy Community signed in Athens on 25 October 2005, extends the EU's internal energy market rules to contracting parties in South-East Europe and the Black Sea region, currently comprising Albania, Bosnia and Herzegovina, Georgia, Kosovo (designation without prejudice to positions on status), Moldova, Montenegro, North Macedonia, Serbia, and Ukraine. Contracting parties undertake to implement the relevant EU *acquis* in energy, environment, competition, and renewables. The Energy Community has a Secretariat in Vienna and a dispute settlement procedure under which the Ministerial Council may declare that a contracting party has failed to comply with its obligations. The Energy Community has been

[2] Case C-848/19 P, *Federal Republic of Germany v Republic of Poland*, EU:C:2021:598 (CJEU, 15 July 2021).

a significant vehicle for the integration of Ukraine's energy system with that of the European Union, which culminated in the synchronisation of the Ukrainian and Moldovan electricity grids with the continental European grid in March 2022.

African Regional Institutions

Africa has developed a growing network of regional energy institutions, reflecting both the continent's vast energy resources and the imperative of expanding energy access.

The African Union. The AU's Agenda 2063 identifies energy as a priority, and the African Energy Commission (AFREC), established by a Convention adopted in 2001 and in force since 2008, is the AU's specialised agency for energy. The Programme for Infrastructure Development in Africa (PIDA) identifies priority regional energy projects. The African Single Electricity Market (AfSEM), launched in 2021, aims to create the world's largest electricity market by integrating Africa's regional power pools. The Agreement Establishing the African Continental Free Trade Area (AfCFTA), signed in 2018 and in force since 2019, has implications for trade in energy goods and services.

ECOWAS. In West Africa, the Economic Community of West African States has established the ECOWAS Energy Protocol (2003), modelled on the ECT, which provides for investment protection, trade, and transit in the energy sector; the ECOWAS Regional Electricity Regulatory Authority (ERERA), established in 2008 and based in Accra, which regulates cross-border electricity exchanges; the West African Power Pool (WAPP), a specialised institution of ECOWAS established in 1999 and given its current status in 2006, which coordinates the development of regional transmission infrastructure and the regional electricity market; and the ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), based in Praia. The West African Gas Pipeline, which transports Nigerian gas to Benin, Togo, and Ghana, is governed by a treaty concluded among the four states in 2003 and by an International Project Agreement with the pipeline company. The proposed Nigeria–Morocco Gas Pipeline and the Trans-Saharan Gas Pipeline would extend this infrastructure substantially.

Southern and Eastern Africa. The Southern African Power Pool (SAPP), established in 1995, is the most mature of the African power pools and operates a competitive regional electricity market. The Regional Electricity Regulators Association of Southern Africa (RERA) supports regulatory cooperation. The Eastern Africa Power Pool (EAPP) and the Central African Power Pool (CAPP) pursue similar objectives in their regions.

Development Finance Institutions as Energy Governors

Multilateral development banks and other development finance institutions are among the most influential actors in energy governance in developing countries,

although they are rarely described as energy institutions. Their influence operates through several channels: the conditions attached to policy-based lending, which have historically encouraged power sector unbundling, privatisation, and the establishment of independent regulators; the environmental and social standards attached to project finance, notably the World Bank's Environmental and Social Framework (2018) and the IFC Performance Standards; the provision of guarantees and political risk insurance, including through the Multilateral Investment Guarantee Agency; and the provision of technical assistance for the drafting of legislation and model contracts.

Development banks have also become central to the financing of energy access. Mission 300, launched by the World Bank Group and the African Development Bank in 2024, aims to connect 300 million people in Africa to electricity by 2030 through a combination of grid extension, mini-grids, and off-grid solutions. At the Mission 300 Africa Energy Summit held in Dar es Salaam in January 2025, participating governments presented national energy compacts setting out policy and regulatory reforms intended to attract investment. The African Development Bank's Desert to Power initiative seeks to develop solar generation across the Sahel. Such initiatives blend finance with policy conditionality, and their legal dimension — the reforms to tariff regulation, licensing, and procurement that they require — is a significant but understudied aspect of energy law in Africa.

The energy lending policies of development banks have also become a site of contestation over fossil fuel finance. The World Bank Group announced in 2017 that it would cease financing upstream oil and gas after 2019, save in exceptional circumstances, and the European Investment Bank's 2019 energy lending policy ended financing for most unabated fossil fuel projects. Developing-country governments, including Nigeria, have argued that such policies unfairly constrain the use of natural gas as a transition fuel in countries with low per capita emissions and large energy access deficits, an argument that connects institutional policy to the principle of common but differentiated responsibilities discussed in Chapter 18.

Other Regional Arrangements

Elsewhere, regional cooperation in energy takes a variety of forms. The Association of Southeast Asian Nations (ASEAN) has pursued the ASEAN Power Grid and the Trans-ASEAN Gas Pipeline projects through memoranda of understanding. In Latin America, the Central American Electrical Interconnection System (SIEPAC) is underpinned by the Framework Treaty for the Electricity Market of Central America (1996), which established a regional regulator, the Comisión Regional de Interconexión Eléctrica (CRIE). The Gulf Cooperation Council Interconnection Authority operates a regional transmission grid linking the six GCC states.

COMPARATIVE PERSPECTIVE

Regional integration models. The European Union represents a *supranational* model, in which binding legislation, an independent regional regulator (ACER), and a supreme court (the CJEU) drive deep market integration. The Energy Community represents an *acquis-export* model, extending EU rules to neighbouring states through treaty commitments. ECOWAS and SADC represent *intergovernmental* models, in which regional regulators and power pools coordinate national systems with more limited binding authority, and in which the pace of integration depends heavily on the construction of physical infrastructure and on the financial health of national utilities. The choice of model reflects not only political preferences regarding sovereignty but also the maturity of national markets and institutions.

4.7 Independent Regulators and Administrative Accountability

The Rise of the Independent Regulator

At the national level, the most significant institutional development in energy governance over the past four decades has been the spread of independent regulatory agencies. Before the liberalisation era, the electricity and gas sectors in most countries were dominated by vertically integrated, state-owned monopolies that were, in effect, self-regulating, subject to ministerial oversight. The petroleum sector was typically regulated by a ministry, sometimes acting through a national oil company that combined commercial and regulatory functions.

The liberalisation and privatisation of energy markets, beginning in the United Kingdom, Chile, and New Zealand in the 1980s and spreading globally in the 1990s, required new regulatory institutions. Where private companies own and operate natural monopoly networks, or where competition is introduced into generation and supply, an independent regulator is needed to set tariffs, ensure non-discriminatory network access, license market participants, monitor market conduct, and protect consumers. The model of the independent regulatory agency, long established in the United States in the form of state public utility commissions and the Federal Power Commission (later the Federal Energy Regulatory Commission), was transplanted across the world, frequently with the encouragement of the World Bank and other development institutions.

In Nigeria, the Electric Power Sector Reform Act 2005 established the Nigerian Electricity Regulatory Commission (NERC), now continued under the Electricity Act 2023. The Petroleum Industry Act 2021 established the NUPRC and the NMDPRA as regulators separate from the commercial NNPC Ltd, replacing the Department of Petroleum Resources, the Petroleum Products Pricing Regulatory Agency, and the Petroleum Equalisation Fund (Management) Board.

The Rationale for Independence

Several justifications have been advanced for the independence of energy regulators.

Credible commitment. Energy infrastructure is capital-intensive and long-lived, and investors require assurance that tariffs will allow them to recover their investment and earn a reasonable return. Political control of tariffs creates a risk that governments will hold down prices for electoral reasons, expropriating investors' sunk capital. Delegating tariff-setting to an independent regulator with a statutory mandate and secure tenure is a means by which governments can credibly commit not to behave opportunistically.

Expertise. Energy regulation is technically complex, requiring expertise in economics, engineering, accounting, and law. Specialised agencies can develop and retain such expertise more effectively than ministries.

Separation of roles. Where the state owns energy companies, independence separates the state's role as owner from its role as regulator, reducing conflicts of interest and promoting a level playing field between state-owned and private participants. The separation of the regulatory and commercial functions of national oil companies is a central theme of petroleum sector reform.

Insulation from capture. Independence is also intended to protect regulators from capture by the industries they regulate, although independence from government does not of itself guarantee independence from industry.

Elements of Independence

The independence of a regulator depends on several institutional features:

- **Legal foundation**: establishment by statute with a clearly defined mandate, rather than by executive decision;
- **Appointment and tenure**: appointment through a transparent process, fixed terms of office, and protection against removal except for defined cause;
- **Financial autonomy**: funding through levies on regulated entities or secure budgetary allocations, rather than discretionary appropriations;
- **Decision-making autonomy**: authority to take decisions without ministerial direction on individual matters, subject to general policy guidance;
- **Organisational autonomy**: control over staffing and internal organisation.

EU law imposes detailed independence requirements on national regulatory authorities. Under Article 57 of Directive (EU) 2019/944 on common rules for the internal market for electricity, member states must guarantee the independence of regulatory authorities and ensure that they exercise their powers impartially and transparently; regulatory staff and management must act independently from any market interest and must not seek or take direct instructions from any government or other public or private entity when carrying out regulatory tasks.

In *Commission v Germany* (2021), the Court of Justice held that Germany had failed to fulfil its obligations by conferring on the federal government powers to determine tariff methodologies that EU law reserved exclusively to the national regulatory authority.[3]

The Governance of National Oil Companies

National oil companies (NOCs) control the majority of the world's proven oil and gas reserves and are therefore among the most significant institutions in global energy governance, even though they are corporate entities rather than regulators. Their governance raises distinctive legal questions concerning the separation of commercial, policy, and regulatory functions; the relationship between the NOC and the state treasury; transparency and accountability; and the application of corporate law to entities whose shareholder is the state.

The "Norwegian model" is frequently cited as a benchmark. Under this model, policy functions are exercised by the Ministry of Energy, regulatory and resource management functions by a specialised agency (the Norwegian Offshore Directorate, as the Norwegian Petroleum Directorate was renamed in 2024), and commercial functions by a company operating on market terms (Equinor, formerly Statoil, partially listed since 2001), with the state's direct financial interests in petroleum licences managed separately by a wholly state-owned company, Petoro. Comparative research has questioned whether the Norwegian separation of functions is a necessary condition of good NOC performance, observing that it presupposes strong institutional capacity, and that some NOCs perform well under integrated models while some separated models perform poorly.

Nigeria's Petroleum Industry Act 2021 moved decisively towards a separation of functions. Section 53 provides for the incorporation of NNPC Limited under the Companies and Allied Matters Act, with ownership of its shares vested in the Federal Government and held by the Ministry of Finance Incorporated and the Ministry of Petroleum Resources on behalf of the Federation. NNPC Ltd is to operate as a commercial entity without recourse to government funding and to pay dividends to its shareholders, and the Act contemplates that its shares may in future be offered to the public. Regulatory functions previously exercised by NNPC divisions and the Department of Petroleum Resources were transferred to the NUPRC and the NMDPRA. Whether this formal separation will produce the commercial discipline and transparency its drafters intended depends on implementation, including the robustness of corporate governance at NNPC Ltd, the publication of audited accounts, and the treatment of legacy obligations such as fuel subsidy arrears.

[3] Case C-718/18, *European Commission v Federal Republic of Germany*, EU:C:2021:662 (CJEU, 2 September 2021).

Administrative Accountability

Independence must be balanced by accountability. An independent regulator exercises significant public power — it sets prices paid by millions of consumers, grants and revokes licences, and imposes penalties — and in a democratic system such power must be subject to mechanisms of accountability. These mechanisms include:

- **Legislative oversight**: reporting obligations to the legislature, appearances before parliamentary committees, and legislative approval of budgets;
- **Judicial review**: the right of affected persons to challenge regulatory decisions before courts or specialised tribunals on grounds of illegality, irrationality, and procedural impropriety, examined further in Chapter 27;
- **Procedural requirements**: obligations to consult on proposed decisions, to give reasons, to publish decisions, and to act transparently;
- **Appeals**: specialised appeal mechanisms, such as the Competition and Markets Authority's role in hearing appeals against certain decisions of Ofgem in the United Kingdom;
- **Performance review**: periodic review of regulatory performance by audit institutions or independent bodies.

The relationship between independence and accountability is a subject of ongoing scholarly debate. Some scholars emphasise the democratic deficit of independent agencies and argue for stronger political control; others emphasise the importance of independence for investment and argue that accountability is best secured through procedural transparency and judicial review. In the United States, the Supreme Court's decision in *Loper Bright Enterprises v Raimondo* (2024), overruling the *Chevron* doctrine of judicial deference to reasonable agency interpretations of ambiguous statutes, and its articulation of the "major questions doctrine" in *West Virginia v Environmental Protection Agency* (2022), have significantly altered the balance between regulatory agencies and courts, with major implications for energy and environmental regulation.[4]

Regulatory Independence in Practice: The Developing-Country Experience

The transplantation of the independent regulator model to developing countries has produced mixed results. Formal independence enshrined in legislation has often been undermined in practice by political interference in appointments, ministerial pressure on tariff decisions, inadequate funding, and limited technical capacity. Studies of electricity regulation in Africa have found that regulators frequently lack the authority to set cost-reflective tariffs in the face of political

[4] *Loper Bright Enterprises v Raimondo*, 603 US 369 (2024); *West Virginia v Environmental Protection Agency*, 597 US 697 (2022).

resistance, leading to utility financial deficits, underinvestment, and poor service quality. The Nigerian experience illustrates these challenges: NERC's efforts to implement cost-reflective tariffs under the Multi-Year Tariff Order framework have repeatedly been constrained by political considerations, and the accumulation of tariff shortfalls has required substantial government subsidies. The introduction of a "Band A" tariff structure in 2024, under which customers receiving at least twenty hours of supply daily pay higher, more cost-reflective tariffs, represents an attempt to align tariffs with service quality.

These experiences suggest that formal independence is a necessary but not sufficient condition for effective regulation. It must be supported by political commitment, adequate resources, technical capacity, and a broader governance environment that respects the rule of law. Scholars such as Brian Levy and Pablo Spiller have emphasised that regulatory design must be adapted to the institutional endowment of each country, and that in countries with weak checks and balances, contractual commitments and international guarantees may be needed to supplement regulatory independence.[5]

4.8 Regulatory Cooperation and Institutional Reform

Forms of Regulatory Cooperation

Regulatory cooperation in the energy sector takes several forms. At the bilateral level, regulators conclude memoranda of understanding to exchange information, coordinate decisions on cross-border infrastructure, and share expertise. At the regional level, regulators participate in associations such as the Council of European Energy Regulators (CEER), the Energy Regulators Regional Association (ERRA) for Central and Eastern Europe and beyond, the African Forum for Utility Regulators (AFUR), and RERA. At the global level, the International Confederation of Energy Regulators (ICER) provides a forum for cooperation among regional associations.

Regulatory cooperation is particularly important for cross-border infrastructure. The development of an interconnector or cross-border pipeline requires decisions by regulators in each jurisdiction on matters such as cost allocation, access rules, and tariffs. Within the European Union, Regulation (EU) 2022/869 on guidelines for trans-European energy infrastructure provides for the identification of projects of common interest and a framework for cross-border cost allocation decisions, with ACER empowered to decide where national regulators cannot agree.

[5] Brian Levy and Pablo T. Spiller, "The Institutional Foundations of Regulatory Commitment: A Comparative Analysis of Telecommunications Regulation" (1994) 10 *Journal of Law, Economics, and Organization* 201.

Proposals for Institutional Reform

The fragmentation of global energy governance has prompted numerous proposals for reform. These include:

Expanding the IEA. Proposals to open IEA membership to non-OECD states, particularly China and India, would require amendment of the IEP Agreement and the OECD framework. The association initiative represents an incremental step in this direction.

A World Energy Organization. Some scholars have proposed the creation of a comprehensive world energy organisation, analogous to the WTO, with universal membership and competence over energy trade, investment, and security. Such proposals face formidable political obstacles, given the divergence of producer and consumer interests and the reluctance of states to cede sovereignty over energy.

Strengthening the energy dimension of the climate regime. The UNFCCC and Paris Agreement are not energy institutions, but the decisions of the Conference of the Parties increasingly address energy directly, as in the COP28 global stocktake decision. Some scholars argue that the climate regime should become the primary forum for global energy governance; others caution that it lacks the expertise and instruments to address energy security and market issues.

Reforming the Energy Charter Treaty. The modernisation of the ECT, negotiated between 2020 and 2022 and adopted by the Energy Charter Conference in December 2024, sought to align the treaty with the Paris Agreement by permitting parties to exclude protection for fossil fuel investments, clarifying the right to regulate, and refining investment protection standards. The withdrawals of the European Union, Euratom, and several European states, and the continuing participation of other contracting parties, have left the treaty's future uncertain. These developments are examined in Chapter 14.

Networked governance. An alternative to institutional consolidation is to improve coordination among existing institutions through networked governance: joint initiatives, memoranda of understanding, shared data platforms, and the alignment of mandates. The joint work of the IEA, IRENA, the World Bank, the World Health Organization, and UN statistical bodies on tracking progress towards Sustainable Development Goal 7 illustrates this approach.

Data, Transparency, and Institutional Legitimacy

A final dimension of institutional reform concerns data and transparency. Energy governance depends on reliable data on production, consumption, stocks, prices, reserves, emissions, and investment. Much of that data is held by governments and companies that may have strategic or commercial reasons to withhold it, and the quality of data varies widely among states. The Joint Organisations Data Initiative, administered by the International Energy Forum with the participation

of APEC, Eurostat, GECF, IEA, OLADE, OPEC, and the United Nations Statistics Division, seeks to improve the transparency of oil and gas statistics. The EITI Standard requires implementing countries to disclose information along the extractive industry value chain, including contracts, licences, beneficial ownership, production, revenues, and the allocation of revenues. Satellite monitoring of methane emissions, gas flaring, and infrastructure is increasingly supplementing official data, as in the United Nations Environment Programme's International Methane Emissions Observatory.

Data has legal significance. Regulatory decisions, arbitral damages assessments, and climate litigation all depend on evidence, and the authority of institutions such as the IEA rests heavily on the credibility of their data and scenarios. Contests over data — over the reliability of emissions inventories, the assumptions of energy scenarios, or the reserves on which valuations depend — are therefore contests over institutional legitimacy. For doctoral researchers, the political economy of energy data is a promising and relatively unexplored field.

Institutional Reform and the Energy Transition

The energy transition poses fundamental challenges for energy institutions. Institutions designed for a world of centralised fossil fuel systems must adapt to a world of decentralised, digital, and decarbonised systems. Consumer-oriented institutions concerned with oil supply security must address new security challenges arising from critical mineral supply chains, cyber threats, and the variability of renewable generation. Producer-oriented institutions must address the prospect of declining demand. Regulators must design markets that accommodate variable renewable generation, storage, demand response, and distributed resources, and must balance the costs of transition against affordability for consumers. The capacity of institutions to adapt to these challenges will significantly influence the pace and fairness of the transition.

LEADING AUTHORITY

Federal Republic of Germany v Republic of Poland (OPAL), CJEU, 2021

Facts. In 2016, the European Commission approved a modification of the exemption from third-party access and tariff regulation previously granted to the OPAL pipeline, the German onshore extension of the Nord Stream pipeline, allowing Gazprom to use a greater share of OPAL's capacity. Poland challenged the decision before the General Court, arguing that it threatened Poland's security of gas supply by diverting transit flows from pipelines crossing Polish territory.

Holding. The General Court annulled the Commission's decision in 2019 for breach of the principle of energy solidarity. On appeal by Germany, the Court of Justice confirmed that energy solidarity under Article 194(1) TFEU is a general principle of EU law, which entails rights and obligations both for the EU and for the member

states and requires that the interests of all stakeholders liable to be affected be taken into account. The Commission had failed to examine the effects of its decision on the energy security of Poland and other member states.
Significance. The judgment transformed solidarity from a political aspiration into a justiciable legal principle that constrains EU energy decisions. It illustrates the role of courts in regional energy governance and the legal relevance of energy security considerations.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Designing an independent regulator (hypothetical). The fictional State of Morenia, a constituent state of a federal republic, has acquired legislative competence over its intrastate electricity market following a constitutional amendment. The Morenia State Assembly has asked you to prepare draft provisions for a Morenia Electricity Regulatory Commission Law addressing: (a) the Commission's statutory objectives, ranked in order of priority; (b) appointment, tenure, and removal of commissioners; (c) funding arrangements; (d) the relationship between the Commission and the State Commissioner for Energy, including the extent of permissible policy directions; (e) consultation, reason-giving, and publication obligations; (f) appeal and judicial review mechanisms; and (g) cooperation with the federal regulator during the transition of regulatory oversight. Accompany the draft provisions with a commentary of not more than 2,000 words explaining your choices by reference to comparative practice. *Morenia is fictitious.*

CHAPTER SUMMARY

- Global energy governance is a fragmented, polycentric regime complex, reflecting divergent producer and consumer interests, sovereignty sensitivities, path dependence, technological diversity, and shifting centres of economic gravity.
- The IEA, created in 1974, imposes binding emergency stockholding and response obligations on its OECD members and exercises broad analytical influence, but its limited membership constrains its global legitimacy.
- OPEC and the OPEC+ arrangements coordinate producer policies through politically binding but unenforceable production decisions, raising unresolved questions under competition and trade law.
- IRENA has near-universal membership and a promotional mandate without binding powers; the IAEA combines promotion with legally significant safeguards and safety functions.
- Regional institutions range from the supranational EU model to intergovernmental African power pools; the CJEU's recognition of energy solidarity as a binding principle illustrates the legal depth of EU integration.
- Independent regulators are justified by credible commitment, expertise, separation of roles, and insulation from capture, but must be accountable through

legislative oversight, judicial review, and procedural transparency; formal independence is often undermined in practice in developing countries.
- Reform proposals include expanding the IEA, creating a world energy organisation, strengthening the climate regime's energy role, modernising the ECT, and improving networked governance.

DOCTORAL RESEARCH QUESTIONS

1. Does the concept of a "regime complex" adequately explain the fragmentation of global energy governance, or does it underestimate the role of power asymmetries among states?
2. Assess the compatibility of OPEC+ production restrictions with Articles XI and XX(g) of the GATT. Why has no WTO member brought a challenge?
3. To what extent has the formal independence of energy regulators in sub-Saharan Africa translated into decisional independence in tariff-setting? Design a methodology to measure the gap.
4. How should the emergence of state-level electricity regulators in Nigeria be coordinated with federal regulation to avoid regulatory fragmentation in the national grid?
5. Could the energy solidarity principle articulated in *Germany v Poland* serve as a model for regional energy governance in ECOWAS?

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PART II

Petroleum, Gas, and Extractive Industries Law

The law governing the upstream petroleum industry, natural gas and LNG transactions, hard-rock mining, and the critical minerals on which the energy transition depends.

5 CHAPTER

International Petroleum Law and Upstream Resource Development

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the legal classification of petroleum resources and the stages of the upstream value chain.
2. Analyse the nature of exploration and production rights and the legal character of petroleum titles.
3. Compare licensing and concession systems with production sharing contracts and service contracts, and evaluate their respective allocation of risk and reward.
4. Analyse the structure and key provisions of joint operating agreements.
5. Evaluate the roles of national oil companies and state participation in upstream petroleum.
6. Explain the legal framework for unitisation and the development of transboundary reservoirs.
7. Analyse the legal regime governing decommissioning and abandonment liabilities.

Introduction

The upstream petroleum industry — the exploration for and production of crude oil and natural gas — is the historical core of international energy law. It was in the upstream sector that the foundational instruments of the field were developed: the concession, the licence, the production sharing contract, the joint operating agreement, and the unitisation agreement. It was disputes over upstream rights that generated the classic arbitral awards on stabilisation, internationalisation, and expropriation. And it is in the upstream sector that the tension between the sovereign interests of host states and the commercial interests of international oil companies is most acute.

The upstream sector has distinctive economic characteristics that shape its legal regime. Exploration is a high-risk activity: most exploration wells fail to discover commercial quantities of petroleum, and the costs of a single deep-water well can exceed one hundred million United States dollars. Development requires enormous capital investment, often billions of dollars, before any revenue is earned. Production continues for decades, during which prices, costs, technologies, and political conditions change unpredictably. And the resource is non-renewable and owned, in most jurisdictions, by the state. The legal instruments of the upstream sector are designed to allocate the risks and rewards of this activity between the state, as resource owner, and the investor, as provider of capital and technology.

This chapter examines the legal framework of upstream petroleum. It begins with the legal classification of petroleum resources, proceeds to the nature of exploration and production rights, and analyses the principal contractual and licensing systems. It then examines joint operating agreements, national oil companies, unitisation, and decommissioning. Throughout, Nigerian law — particularly the Petroleum Industry Act 2021 (PIA) — is used as a principal reference point, alongside the laws of the United Kingdom, Norway, and other leading jurisdictions.

5.1 Legal Classification of Petroleum Resources

What Is "Petroleum"?

The definition of "petroleum" in legislation and contracts determines the scope of rights granted and obligations imposed. Definitions vary, but most encompass naturally occurring hydrocarbons in liquid, gaseous, or solid form. Section 318 of the PIA defines "petroleum" as "crude oil, natural gas and condensates", and separately defines "crude oil" as "mineral oil, asphalt, ozokerite and all kinds of hydrocarbons and bitumen, whether in solid or in liquid form, in their natural state or obtained from oil shales, tar sands or any other mineral or rock, whether gaseous or not" and "natural gas" as "gas obtained from boreholes and wells and consisting primarily of hydrocarbons". Section 1(1) of the United Kingdom's Petroleum Act 1998 defines petroleum to include "any mineral oil or relative hydrocarbon and natural gas existing in its natural condition in strata", but expressly excludes coal and bituminous shales and other stratified deposits from which oil can be extracted by destructive distillation.

The precise definition can have significant legal consequences. Whether coal-bed methane, shale gas, oil shale, or helium falls within the definition of petroleum determines which licensing regime applies and who has the right to exploit the resource. The distinction between crude oil and natural gas matters because many

legal regimes treat them differently for fiscal purposes, ownership of associated and non-associated gas, and domestic supply obligations.

Associated and Non-Associated Gas

Natural gas may occur either in association with crude oil (associated gas), dissolved in the oil or as a gas cap above it, or in reservoirs that contain predominantly gas (non-associated gas). The distinction is legally important. Historically, many petroleum agreements granted rights to crude oil with only limited provision for associated gas, which was often flared because no market existed for it. Nigeria has long been among the world's largest gas flaring countries. The Associated Gas Re-Injection Act 1979 prohibited gas flaring without ministerial permission after 1 January 1984, but permission was routinely granted subject to payment of nominal fees. Section 104 of the PIA prohibits the flaring or venting of natural gas except in specified circumstances such as emergencies, or under an exemption granted by the NUPRC for facility start-up or for strategic operational reasons, and requires licensees and lessees to submit a natural gas flare elimination and monetisation plan. The Act further provides that payments of flare penalties shall not be cost-recoverable or tax-deductible. The Flare Gas (Prevention of Waste and Pollution) Regulations 2018 established the Nigerian Gas Flare Commercialisation Programme, under which flare gas is auctioned to third parties for commercial use.

Conventional and Unconventional Resources

A further classification distinguishes conventional resources — petroleum that can be produced from reservoirs by natural pressure, pumping, or conventional secondary recovery — from unconventional resources, such as shale oil and gas, tight gas, coal-bed methane, oil sands, and gas hydrates, which require specialised techniques such as horizontal drilling and hydraulic fracturing. The shale revolution in the United States, which transformed that country into the world's largest producer of both oil and natural gas, was made possible not only by technology but also by the American legal framework of private mineral ownership, which permitted rapid leasing and development without the need for state licensing. Elsewhere, unconventional resources have been subject to moratoria or bans on hydraulic fracturing, including in France (Law No 2011-835 of 13 July 2011), and, for a period, in the United Kingdom, where a moratorium imposed in 2019 was briefly lifted in 2022 and then reimposed.

Reserves Classification

The classification of petroleum reserves has legal and financial significance, since reserves underpin company valuations, borrowing capacity, and, in disputes, the assessment of damages. The Petroleum Resources Management System (PRMS), developed by the Society of Petroleum Engineers and other professional organ-

isations, classifies petroleum quantities according to their commerciality and the degree of uncertainty in recoverable volumes: reserves (proved, probable, and possible, commonly abbreviated as 1P, 2P, and 3P), contingent resources, and prospective resources. Securities regulators impose disclosure requirements based on reserves definitions; in the United States, the Securities and Exchange Commission's rules on oil and gas reserves disclosure apply to listed companies. The distinction between proved reserves and more speculative categories is frequently central to the valuation of expropriated petroleum assets in arbitration.

5.2 Exploration and Production Rights

The Nature of Petroleum Titles

In jurisdictions where petroleum is owned by the state, the right of an investor to explore for and produce petroleum derives from a grant by the state. The legal nature of that grant varies. It may be characterised as:

- an **administrative licence**, a permission granted by the state in the exercise of public power, which creates rights and obligations defined by statute and regulation and which may be varied or revoked in accordance with administrative law;
- a **contract**, a bilateral agreement between the state (or its agency) and the investor, creating reciprocal obligations enforceable in accordance with contract law;
- a **property right**, a real right in the petroleum or in the land, such as a lease or *profit à prendre*, enforceable against third parties; or
- a **hybrid** combining elements of each.

The characterisation has practical consequences. If the title is an administrative licence, the state may be able to vary its terms by legislation or regulation, subject to the protection afforded by administrative law and any applicable investment treaty. If it is a contract, unilateral variation will constitute a breach. If it is a property right, it may be mortgaged, assigned, and protected against third-party interference, and its taking may constitute expropriation.

In the United Kingdom, a petroleum licence granted under the Petroleum Act 1998 is a licence in the sense of a statutory permission, but it incorporates "model clauses" prescribed by regulation which take effect as terms of the licence and have a contractual character. The licence confers exclusive rights to "search and bore for, and get" petroleum within the licensed area. In *Shell UK Ltd v Enterprise Oil plc* and related litigation, English courts have treated licences as conferring valuable proprietary interests capable of assignment, subject to the consent of the regulator.

In Nigeria, petroleum titles under the PIA comprise the petroleum exploration licence (non-exclusive, for reconnaissance), the petroleum prospecting licence

(exclusive, for exploration and appraisal), and the petroleum mining lease (exclusive, for production). Under the PIA, a petroleum prospecting licence confers the exclusive right to drill exploration and appraisal wells and to carry out test production, while a petroleum mining lease confers the exclusive right to win, work, carry away, and dispose of petroleum. These titles are statutory instruments granted by the Minister on the recommendation of the NUPRC following a competitive bid round, and their terms are supplemented by the regulations made under the Act and by any associated contracts with NNPC Ltd.

Allocation of Rights: Competitive Bidding and Discretionary Award

States allocate petroleum rights through competitive bidding, discretionary award (negotiation), or first-come, first-served application systems. Competitive bidding, in which rights are awarded on the basis of bids comprising signature bonuses, work programme commitments, royalty or profit-share terms, or a combination of these, is now the dominant method in most producing states and is generally regarded as more transparent and efficient. The PIA requires that petroleum prospecting licences and petroleum mining leases be granted through a "fully transparent and competitive bid process" in accordance with regulations and guidelines issued by the NUPRC, subject to limited exceptions. Discretionary award was historically widespread in Nigeria and elsewhere and was associated with allegations of corruption, most notoriously in the award of Oil Prospecting Licence (OPL) 245, which generated criminal proceedings in Italy and Nigeria and civil litigation in England.

The design of bidding systems involves trade-offs. Signature bonuses generate immediate revenue for the state but may deter exploration in high-risk areas; work programme bidding encourages exploration activity but may lead to over-commitment; profit-share bidding defers state revenue but aligns state returns with project outcomes. Norway's licensing rounds, conducted through the Awards in Predefined Areas (APA) system for mature areas and numbered licensing rounds for frontier areas, award licences on the basis of discretionary criteria (technical competence, financial capacity, and the quality of work programmes) rather than monetary bids, reflecting a policy choice to capture rent through taxation rather than through bonuses.

Duration, Relinquishment, and Work Obligations

Petroleum titles are granted for defined periods. An exploration phase, typically lasting between three and ten years and often divided into sub-phases, is followed, upon discovery of a commercial field and approval of a development plan, by a production phase lasting twenty to thirty years or more. Under the PIA, a petroleum prospecting licence for onshore and shallow water acreage is granted for an initial exploration period of three years, which may be extended by a

further three years, with longer periods available for deep offshore and frontier acreage; a petroleum mining lease is granted for a term not exceeding twenty years, renewable in accordance with the Act.

Titles typically impose minimum work obligations — commitments to acquire seismic data and drill a specified number of wells within defined periods — backed by financial guarantees. Failure to perform the work programme may result in forfeiture of the guarantee and revocation of the title. Relinquishment provisions require the titleholder to surrender portions of the licensed area at specified intervals, ensuring that acreage that is not actively explored is returned to the state for re-allocation. The PIA provides for the relinquishment of fifty per cent of the area of a petroleum mining lease ten years after the grant, subject to specified conditions, a provision designed to address the problem of large, under-explored acreage held under legacy oil mining leases.

Assignment and Change of Control

Petroleum titles are valuable assets that are frequently transferred through farm-outs, sales, and corporate acquisitions. Because the identity, competence, and financial capacity of the titleholder are matters of public interest, most regimes require the consent of the state or regulator to any assignment of an interest in a title and, increasingly, to any change in control of the titleholder. The PIA prohibits the assignment, novation, or transfer of an interest in a licence or lease, or any change of control of the holder, without the prior consent of the Minister on the recommendation of the NUPRC. The divestment of onshore assets by international oil companies to Nigerian independents between 2021 and 2025 has generated significant regulatory scrutiny under these provisions, with the NUPRC requiring assurances concerning the purchasers' technical and financial capacity, their assumption of decommissioning and environmental liabilities, and the protection of host communities.

5.3 Petroleum Licensing and Concession Systems

Typology of Petroleum Arrangements

Petroleum arrangements between host states and investors are conventionally classified into two families: *concessionary* (or licensing) systems and *contractual* systems. Contractual systems are further divided into production sharing contracts and service contracts. The classification turns principally on the ownership of produced petroleum and the form of the state's financial return.

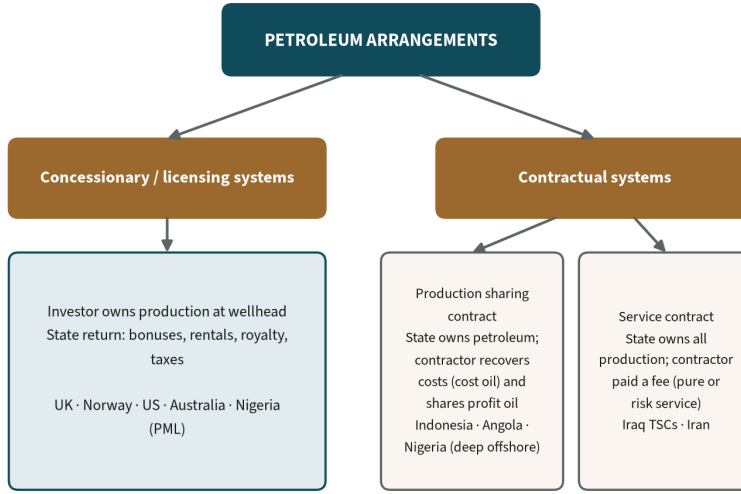


Figure 5.1 Principal families of petroleum arrangements, classified by ownership of production and the form of the state's return. *Source:* prepared for this edition.

Under a **concessionary or licensing system**, the investor acquires the exclusive right to explore for and produce petroleum within a defined area and owns the petroleum produced at the wellhead (or at another defined point), subject to the payment of royalties and taxes to the state. The state's return derives from fiscal instruments — bonuses, rentals, royalties, and taxes — rather than from a share of physical production. Modern licensing systems are used in the United Kingdom, Norway, the United States (for federal and state leases), Australia, Brazil (for its concession regime), and many other jurisdictions. Nigeria's historic oil mining lease system and the PIA's petroleum mining lease are licensing instruments, although in Nigeria they have been combined with joint ventures and production sharing contracts involving the national oil company.

Under a **production sharing contract (PSC)**, the state (usually acting through its national oil company) retains ownership of the petroleum, and the investor acts as a contractor that bears the exploration and development risk and costs. In return, the contractor is entitled to recover its costs from a portion of production ("cost oil") and to receive a share of the remaining production ("profit oil"), with the balance going to the state. Title to the contractor's share passes at a defined delivery point.

Under a **service contract**, the state retains ownership of all production, and the contractor is paid a fee for its services, either in cash or in kind. In a *pure* service contract, the fee is fixed and the state bears the risk; in a *risk* service contract, the contractor bears the exploration risk and is remunerated only if production

is achieved. Service contracts are used where constitutional or political considerations preclude any transfer of ownership of petroleum to foreign companies, notably in Iran (whose buyback contracts and the subsequent Iran Petroleum Contract reflect constitutional prohibitions on foreign ownership), Iraq (whose technical service contracts for major fields awarded from 2009 paid a fixed remuneration fee per barrel), and Mexico (prior to the 2013 reforms).

The Modern Licence

The modern petroleum licence differs fundamentally from the classical concession described in Chapter 1. Where the classical concession covered vast areas for very long periods with minimal state control, the modern licence covers a defined block for a limited exploration period, imposes work obligations and relinquishment requirements, subjects the licensee to comprehensive regulatory control over operations, and secures a substantial share of the resource rent for the state through a progressive fiscal regime. The modern licence is embedded in a statutory framework that regulates health and safety, environmental protection, resource management, decommissioning, and local content.

In the United Kingdom, for example, the North Sea Transition Authority (NSTA, formerly the Oil and Gas Authority) regulates licensing and resource management under the Petroleum Act 1998, the Energy Act 2016, and the model clauses; the Health and Safety Executive regulates safety under the Offshore Installations (Safety Case) Regulations 2015; the Offshore Petroleum Regulator for Environment and Decommissioning (OPRED), within the Department for Energy Security and Net Zero, regulates environmental matters and decommissioning; and HM Revenue and Customs administers a fiscal regime comprising ring-fence corporation tax, a supplementary charge, and the Energy Profits Levy introduced in 2022. Since 2021, the NSTA has been required to apply its "Strategy", which includes a central obligation on licensees and others to take the steps necessary to secure that the maximum value of economically recoverable petroleum is recovered from the strata beneath relevant UK waters, while also assisting the Secretary of State in meeting the net zero target.

Advantages and Disadvantages of Licensing

Licensing systems have several advantages. They are relatively simple to administer, since the state's return derives from fiscal instruments applied through the tax system. They provide investors with clear title to production, facilitating financing. They are compatible with general legislation applicable to all licensees, promoting transparency and equality of treatment. And they keep the state at arm's length from commercial operations, reducing the need for state technical capacity.

Their principal disadvantages, from the host state's perspective, are that the state has limited direct control over operations and limited access to information; that fiscal regimes based on royalties and taxes may fail to capture the full resource rent, especially when prices rise unexpectedly; and that the transfer of ownership of petroleum to foreign companies may be politically sensitive.

Local Content Obligations in Upstream Petroleum

Modern upstream regimes almost invariably impose local content obligations — requirements that licensees employ nationals, procure goods and services from domestic suppliers, use domestic facilities, and transfer technology and skills. Local content policies seek to convert petroleum extraction from an enclave activity into a driver of broader industrial development. They raise legal questions under trade law (where they discriminate against imported goods), investment law (where they impose performance requirements prohibited by some treaties), and domestic administrative law (where their enforcement is discretionary).

Nigeria's Nigerian Oil and Gas Industry Content Development Act 2010 (the NOGICD Act) is among the most comprehensive local content statutes in the world. It provides that Nigerian independent operators shall be given first consideration in the award of oil blocks, oil field licences, oil lifting licences, and all projects for which contracts are to be awarded in the Nigerian oil and gas industry, subject to the fulfilment of conditions specified by the Minister; that exclusive consideration shall be given to Nigerian indigenous service companies that demonstrate ownership of equipment, Nigerian personnel, and capacity to execute the work in bids for land and swamp operating areas; and that operators shall submit Nigerian content plans with every application for a licence, permit, or interest, demonstrating compliance with the minimum levels of Nigerian content set out in the Schedule to the Act for specified goods and services. The Act established the Nigerian Content Development and Monitoring Board to implement its provisions, and the Nigerian Content Development Fund, financed by a levy of one per cent of the value of every contract awarded in the upstream sector. Presidential directives issued in 2024 extended local content expectations to aspects of procurement and financing in the oil and gas industry.

Comparable statutes exist in Ghana (the Petroleum (Local Content and Local Participation) Regulations 2013, L.I. 2204), Angola, Brazil, Kazakhstan, and many other producer states. Empirical research suggests that local content policies are most effective when targets are realistic, linked to measurable capacity-building, and backed by investment in education, infrastructure, and access to finance; overly ambitious or poorly enforced targets may increase costs without delivering lasting capability. The design of local content provisions in licences and contracts — including the definitions of "local" goods, services, and companies,

the measurement methodology, the consequences of non-compliance, and the interaction with stabilisation clauses — is a significant area of legal practice.

5.4 Production Sharing Contracts

Origins

The production sharing contract was pioneered in Indonesia in the mid-1960s by the state oil company Permina (later Pertamina) under the leadership of Ibnu Sutowo, in response to political pressure to end the concession system and assert national control over petroleum. The Indonesian model, which provided for the state to retain ownership and management control while the foreign contractor bore the costs and risks and recovered them from production, was adopted by many other states in Asia, Africa, and Latin America. By the 1990s, the PSC had become the predominant form of petroleum arrangement in the developing world.

In Nigeria, PSCs were introduced in 1973 with Ashland Oil and became the primary vehicle for deep offshore development following the 1993 licensing round. The Deep Offshore and Inland Basin Production Sharing Contracts Act 1999 (as amended in 2019) established the fiscal framework for deep offshore PSCs, including royalty rates that decline with water depth. The 2019 amendment, which increased royalty rates and introduced a price-based royalty, was enacted after the federal government had lost significant revenue because the 1999 Act's provision for review of the royalty rates once oil prices exceeded USD 20 per barrel had never been implemented. The PIA now governs new upstream arrangements, while preserving the terms of existing PSCs subject to the conversion provisions discussed below.

Structure and Key Provisions

A typical PSC contains the following key provisions:

Parties and contract area. The PSC is concluded between the state or national oil company and the contractor (often a consortium of companies). It defines the contract area, usually by reference to geographical coordinates.

Term and phases. The PSC specifies an exploration period, divided into phases with associated work commitments, and a production period that commences upon a declaration of commerciality and approval of a development plan.

Work programme and budget. The contractor must submit annual work programmes and budgets for approval by a management committee on which the national oil company is represented.

Management committee. The management committee, composed of representatives of the national oil company and the contractor, supervises operations. The allocation of voting rights between the parties determines the degree of state control.

Cost recovery. The contractor is entitled to recover its exploration, development, and operating costs from a defined percentage of production — the "cost oil" or "cost recovery limit" — commonly between 50 and 80 per cent of production in any period. Costs that exceed the limit may usually be carried forward to future periods. Accounting procedures annexed to the PSC define which costs are recoverable and how they are to be allocated.

Profit oil split. Production remaining after royalty and cost oil — "profit oil" — is divided between the state and the contractor in accordance with an agreed formula. The split may be fixed or, more commonly, may vary according to a sliding scale based on cumulative production, daily production rates, the contractor's rate of return, or an "R-factor" (the ratio of the contractor's cumulative revenues to its cumulative costs).

Royalty and taxation. Many PSCs also require payment of a royalty and subject the contractor's share of profit oil to income tax. In some PSCs, the national oil company pays the contractor's income tax out of the state's share of profit oil (a "pay-on-behalf" arrangement), to enable the contractor to claim foreign tax credits in its home jurisdiction.

Title and lifting. Title to the contractor's share of production passes at the delivery point, and lifting arrangements govern the allocation of cargoes.

Ownership of assets. Title to equipment and facilities typically passes to the state either upon acquisition or upon full cost recovery, although the contractor retains the right to use them for operations.

Domestic supply obligations, local content, training, and stabilisation clauses are also common.

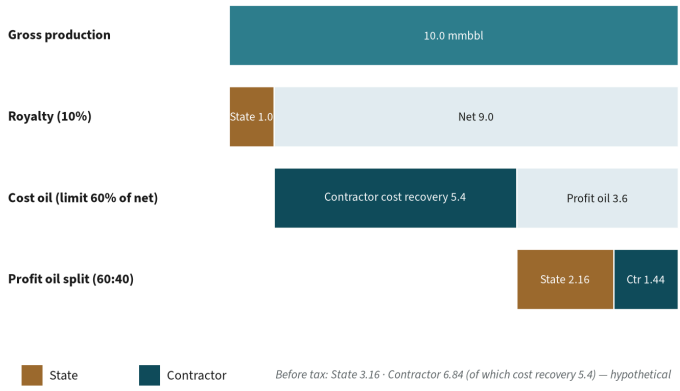


Figure 5.2 Illustrative production sharing waterfall. The percentages are hypothetical and are used solely to illustrate the mechanics of allocation. *Source:* prepared for this edition.

A Worked Illustration

The mechanics of a PSC may be illustrated with a simplified, hypothetical example. Suppose that in a given year a field produces 10 million barrels of oil, the PSC provides for a royalty of 10 per cent, a cost recovery limit of 60 per cent of production after royalty, and a profit oil split of 60:40 in favour of the state, and that the contractor has recoverable costs of 7 million barrels' worth for the year.

1. Royalty: 10 per cent of 10 million = 1 million barrels to the state, leaving 9 million barrels.
2. Cost oil: the cost recovery limit is 60 per cent of 9 million = 5.4 million barrels. The contractor recovers 5.4 million barrels; the unrecovered balance of 1.6 million barrels' worth is carried forward.
3. Profit oil: 9 million minus 5.4 million = 3.6 million barrels, of which the state receives 60 per cent (2.16 million) and the contractor 40 per cent (1.44 million).
4. Tax: if the contractor's profit oil is subject to income tax at, say, 30 per cent, a further portion of the contractor's share is payable to the state.

In this illustration, before tax the state receives 3.16 million barrels (31.6 per cent of production) and the contractor 6.84 million barrels, of which 5.4 million represent cost recovery. The "government take" — the share of the economic rent (revenues minus costs) accruing to the state — is a more meaningful measure than the share of gross production, and is the standard metric for comparing fiscal regimes. Chapter 16 examines fiscal design in more detail.

Advantages, Disadvantages, and Disputes

PSCs offer host states several advantages: formal ownership of petroleum is retained, satisfying political and constitutional requirements; the national oil company participates in management and gains access to information and expertise; and the state's share of production rises with profitability under progressive profit-sharing formulas. For investors, PSCs offer the right to recover costs from production and a share of production that can be booked as reserves under some accounting rules.

The disadvantages include administrative complexity, especially in auditing cost recovery; the incentive for contractors to "gold-plate" costs, since higher recoverable costs increase their share of production; and the potential for disputes over cost allocation, the interpretation of profit oil formulas, and the lifting of entitlements. The Nigerian experience includes significant disputes concerning the deep offshore PSCs. In the dispute concerning the Oil Mining Lease 133 (Erha) PSC, an arbitral tribunal in 2011 upheld claims by the contractors, Esso Exploration and Production Nigeria Ltd and Shell Nigeria Exploration and Production Co Ltd, against NNPC concerning the lifting of tax oil and royalty

oil. NNPC applied to the Federal High Court in Abuja, which set the award aside on the basis that the dispute concerned tax matters that were not arbitrable under Nigerian law, generating further appeals in Nigeria and parallel proceedings in the United States to enforce the award. The litigation illustrates the jurisdictional complexities that arise when fiscal provisions are embedded in contracts and the tension between the finality of arbitral awards and national courts' views on arbitrability.

The PIA and the Conversion of Existing Arrangements

The PIA introduced a framework for the voluntary conversion of existing oil prospecting licences and oil mining leases, including those underlying PSCs and joint ventures, to the new petroleum prospecting licences and petroleum mining leases. Conversion is incentivised by more favourable fiscal terms under the new regime, while holders that do not convert continue under their existing terms until expiry but cannot benefit from the new fiscal incentives and face restrictions on renewal. The Act also required the conversion of NNPC's joint venture interests into incorporated joint ventures within a specified period after its commencement. These transitional provisions illustrate a characteristic technique of petroleum law reform: the use of incentives rather than compulsion to migrate investors from legacy arrangements, thereby reducing the risk of investment treaty claims.

Stabilisation and Economic Equilibrium in PSCs

Because PSCs typically run for decades, contractors seek protection against changes in the legal and fiscal regime that would reduce the economic value of their entitlements. PSCs therefore frequently contain stabilisation provisions, which may take one of three principal forms. *Freezing clauses* provide that the law applicable to the contract is the law in force at the date of signature, and that subsequent legislation will not apply to the contractor to the extent that it is inconsistent with the contract. *Economic equilibrium clauses* permit the application of new laws but require the parties to restore the original economic balance of the contract, by adjusting fiscal terms or providing compensation, where a change in law materially affects the contractor's economic position. *Hybrid clauses* combine elements of both. Many modern PSCs prefer economic equilibrium clauses, because they preserve the state's legislative freedom while protecting the investor's economic expectations, and because freezing clauses have been criticised for insulating investors from environmental, labour, and human rights regulation.

The interaction between stabilisation and human rights has attracted particular attention. A study prepared for the United Nations Special Representative on Business and Human Rights in 2008 found that stabilisation clauses in many investment contracts in developing countries were drafted broadly enough to

exempt investors from new social and environmental legislation. The *Principles for Responsible Contracts*, endorsed by the Human Rights Council in 2011 as an addendum to the UN Guiding Principles, recommend that stabilisation clauses be drafted so as not to interfere with the state's good-faith efforts to implement laws, regulations, or policies in a non-discriminatory manner in order to meet its human rights obligations. Stabilisation clauses are examined further in Chapters 14 and 15.

Marginal Fields

Many mature producing provinces contain discovered but undeveloped fields that are too small to justify development by the original licensee. To promote their development by smaller companies, several jurisdictions have created special "marginal field" regimes. In Nigeria, the Petroleum (Amendment) Act 1996 introduced the marginal field programme, empowering the President to farm out marginal fields within an oil mining lease where they had been left unattended for not less than ten years from the date of first discovery. The programme facilitated the emergence of a cohort of Nigerian independent producers. The 2020–2021 marginal field bid round, conducted under the Marginal Field Operations (Fiscal Regime) Regulations and the Guidelines for Farm-out and Operation of Marginal Fields, awarded fifty-seven fields to Nigerian companies. The PIA continues to provide for the declaration and farm-out of fields that are not being developed, subject to the NUPRC's approval. Marginal field regimes illustrate how licensing law can be used as an instrument of industrial policy, but they also generate disputes with the original leaseholders concerning the identification of fields, access to infrastructure, and the terms of farm-out agreements.

Service Contracts in Practice

Service contracts, although less common than licences and PSCs, deserve attention because of their use in some of the world's largest oil provinces. The Iraqi technical service contracts awarded in 2009 and 2010 for super-giant fields such as Rumaila, West Qurna, and Majnoon paid contractors a fixed remuneration fee per barrel of incremental production above an agreed baseline, plus recovery of costs, with fees subject to an R-factor adjustment. Because contractors earned the same fee per barrel regardless of oil prices, the contracts transferred almost the entire price upside to the state, but they also left contractors exposed to production disruptions, export capacity constraints, and delays in cost recovery payments. Several contracts were subsequently renegotiated to lower production targets. The experience illustrates that the allocation of risk and reward in petroleum arrangements is determined not by the label attached to the contract but by the detailed economic and legal terms.

5.5 Joint Operating Agreements

Purpose of the Joint Operating Agreement

Petroleum titles are commonly held by several companies jointly, in order to spread risk, share capital costs, combine technical capabilities, and satisfy state participation requirements. The relationship among the co-venturers is governed by a joint operating agreement (JOA), a contract that establishes the rules for conducting joint operations, sharing costs and production, making decisions, and addressing default and withdrawal. The JOA does not usually create a partnership or a separate legal entity; most JOAs expressly provide that the parties' rights and obligations are several and not joint or collective, and that nothing in the agreement shall be construed as creating a partnership. The co-venturers hold undivided interests in the title as tenants in common (or the civil law equivalent), each entitled to take and separately dispose of its share of production.

The most widely used international model is the AIEN Model International Operating Agreement, first published in 1990 and most recently revised in 2012 (with subsequent updates). National models also exist, notably the UK Oil and Gas Authority (now NSTA) Master Deed and model JOAs developed under the industry's Standard Agreements initiative. In Nigeria, the historic NNPC joint ventures with international oil companies were governed by JOAs based in part on the AIEN model and in part on bespoke terms.

Key Provisions

Scope and participating interests. The JOA defines the joint property, the contract area, and each party's participating interest, which determines its share of costs, production, and voting power.

The operator. One party is designated as operator, responsible for conducting joint operations on behalf of all parties. The operator acts as an agent of the non-operators in some respects but is typically not entitled to profit from its role, recovering its costs through the joint account (including overhead charges) on a "no gain, no loss" basis. The JOA defines the operator's duties — commonly to conduct operations in a diligent, safe, and efficient manner in accordance with good oilfield practice — and limits its liability. A key feature of most international JOAs is that the operator is liable to the non-operators only for losses resulting from its "gross negligence or wilful misconduct", and in some forms only where such conduct is that of senior supervisory personnel. The interpretation of these terms has been considered in English cases, and their meaning depends heavily on the precise contractual wording.

Operating committee. Decisions on work programmes, budgets, and major operational matters are taken by an operating committee in which each party is represented. The JOA sets out voting thresholds, often requiring a majority

of participating interests together with a minimum number of parties ("pass marks"), and identifies matters requiring unanimity.

Work programmes, budgets, and authorisations for expenditure. The operator prepares annual work programmes and budgets for approval, and individual items of expenditure above specified thresholds require authorisations for expenditure (AFEs).

Sole risk and non-consent operations. Where a party wishes to undertake an operation that the operating committee has not approved, the JOA may permit it to proceed at its sole risk and cost. A party that elects not to participate in an approved operation ("non-consent") may be excluded from the benefits of that operation unless it pays a substantial premium — often several hundred per cent of the costs it avoided — to re-enter. These provisions balance the interests of parties wishing to proceed against those of parties wishing to limit risk.

Default. If a party fails to pay its share of costs ("cash calls"), the JOA provides remedies for the non-defaulting parties, which may include suspension of the defaulting party's rights to vote and to receive production, and ultimately forfeiture of its participating interest. Forfeiture provisions have been challenged as penalties. In *Cavendish Square Holding BV v Makdessi* and *ParkingEye Ltd v Beavis*, decided together in 2015, the United Kingdom Supreme Court reformulated the rule against penalties, holding that the test is whether the provision is a secondary obligation that imposes a detriment on the contract-breaker out of all proportion to any legitimate interest of the innocent party in the enforcement of the primary obligation.[1] In the oil and gas context, the legitimate interest of non-defaulting parties in ensuring the continuation of joint operations has generally been regarded as supporting the enforceability of forfeiture clauses, although each clause must be assessed on its terms.

Transfer and pre-emption. The JOA restricts transfers of participating interests, commonly requiring the consent of the other parties (not to be unreasonably withheld) and granting them pre-emption rights. Pre-emption provisions have been a frequent source of disputes, particularly regarding whether they are triggered by indirect transfers through changes of control of a party.

Withdrawal, decommissioning, and confidentiality. The JOA governs the right of a party to withdraw, the allocation of decommissioning costs (often addressed in a separate decommissioning security agreement), confidentiality, and dispute resolution.

[1] *Cavendish Square Holding BV v Makdessi; ParkingEye Ltd v Beavis* [2015] UKSC 67, [2016] AC 1172.

Joint Ventures with National Oil Companies

Where the state participates in upstream operations through its national oil company, the JOA governs the relationship between the national oil company and its private partners. In Nigeria, the joint venture arrangements between NNPC and international oil companies such as Shell, ExxonMobil, Chevron, TotalEnergies, and Eni historically required NNPC to fund its share of costs (typically 55 to 60 per cent) through cash calls. Chronic failure by NNPC to meet its cash call obligations, owing to budgetary constraints, led to underinvestment and the accumulation of arrears, prompting the development of alternative funding arrangements, including modified carry agreements, under which the private partners fund NNPC's share and recover it from production. The PIA's provision for the conversion of NNPC joint ventures into incorporated joint ventures, which could raise finance independently, was intended in part to address the cash call problem.

5.6 National Oil Companies and State Participation

Forms of State Participation

States participate in upstream petroleum in several ways:

1. **Full state operation**, in which the national oil company holds and operates all petroleum rights, as in Saudi Arabia (Saudi Aramco), Kuwait (Kuwait Petroleum Corporation), and Mexico (Pemex, historically).
2. **Majority or minority participation in joint ventures**, in which the national oil company holds an equity interest alongside private partners, either funding its share (paid participation) or being carried by the private partners through exploration (carried interest), as in Nigeria's joint ventures, Angola, and Ghana, where the Ghana National Petroleum Corporation holds a carried interest of not less than 15 per cent in petroleum agreements under the Petroleum (Exploration and Production) Act 2016 (Act 919).
3. **Contractual participation**, in which the national oil company is the contracting party under a PSC or service contract and participates in management through the management committee.
4. **Indirect participation through state-owned commercial companies or holding companies** that hold the state's direct financial interests, as in Norway's Petoro.

The Dual Role Problem

National oil companies frequently combine commercial and governmental functions: they may hold equity interests in joint ventures, act as the state's concessionaire or contracting party under PSCs, perform regulatory functions such as approving work programmes or monitoring compliance, market the state's share of production, and perform social or political functions such as fuel supply and

subsidy administration. This combination of roles creates conflicts of interest and impairs both commercial performance and regulatory effectiveness. A national oil company that is both a partner in a joint venture and the regulator of its partners cannot regulate impartially; one that is required to supply fuel at subsidised prices cannot operate on commercial principles.

The separation of the commercial and regulatory roles of national oil companies has been a central objective of petroleum sector reform, as discussed in Chapter 4 in the context of the Norwegian model and the PIA. The PIA's creation of NNPC Ltd as a commercial company without regulatory functions, and its vesting of regulatory functions in the NUPRC and NMDPRA, represents Nigeria's response to the dual role problem.

National Oil Companies Abroad

Many national oil companies have become international investors in their own right, operating outside their home states. Companies such as Saudi Aramco, Petronas, the China National Petroleum Corporation, the China National Offshore Oil Corporation, Equinor, and Petrobras hold significant upstream interests abroad. When national oil companies invest abroad, questions arise about their status as investors under investment treaties, their entitlement to sovereign immunity, and their susceptibility to the host state's competition and foreign investment screening laws. The ICSID Convention permits claims by "nationals of another Contracting State", and tribunals have held that state-owned companies may qualify as nationals unless they are acting as agents of the government or discharging an essentially governmental function — the test proposed by Aron Broches and applied in *Ceskoslovenska Obchodni Banka AS v Slovak Republic* (1999) and subsequent cases. In *Beijing Urban Construction Group Co Ltd v Republic of Yemen* (2017), the tribunal applied the Broches test to hold that a Chinese state-owned enterprise was a national for the purposes of the Convention.

5.7 Unitization and Cross-Border Reservoir Development

The Problem of Shared Reservoirs

Petroleum reservoirs do not respect licence boundaries or international borders. Where a reservoir straddles the boundary between two licence areas, or between two states, independent development by each owner would lead to the inefficiencies associated with the rule of capture: competitive drilling, wasteful duplication of infrastructure, loss of reservoir pressure, and reduced ultimate recovery. The solution is unitisation: the development of the reservoir as a single unit, with costs and production shared among the owners according to agreed "tract participation" percentages reflecting the proportion of the reservoir located within each area.

Domestic Unitisation

Within a single state, unitisation of a reservoir straddling licence boundaries may be achieved voluntarily or compelled by the regulator. Most petroleum regimes empower the regulator to require unitisation where it is necessary to secure the most effective recovery of petroleum. In the United Kingdom, the model clauses empower the Secretary of State (now acting through the NSTA) to require licensees to cooperate in the development of a field extending beyond the licensed area. In Nigeria, the PIA empowers the NUPRC to direct unitisation where a petroleum reservoir extends across licence boundaries and the holders cannot agree.

A unitisation and unit operating agreement (UUOA) establishes the unit, determines tract participations, appoints a unit operator, and provides for redeterminations of tract participations as geological knowledge improves. Redeterminations are a frequent source of disputes, because a change in tract participation can transfer very large values between the parties. UUOAs therefore contain detailed procedures for redetermination, often involving independent experts whose decisions are final and binding except for manifest error or fraud.

Transboundary Unitisation

Where a reservoir straddles an international boundary, unitisation requires cooperation between the states concerned. Customary international law does not clearly establish an obligation to unitise, but state practice, reflected in numerous bilateral agreements, demonstrates a strong preference for cooperation. Many maritime boundary agreements include a "mineral deposits clause", first appearing in the 1965 Agreement between the United Kingdom and Norway relating to the Delimitation of the Continental Shelf, which obliges the parties, where a single geological petroleum structure or field extends across the dividing line, to seek to reach agreement as to the manner in which the structure or field shall be most effectively exploited and the manner in which the proceeds shall be apportioned.

The Framework Agreement concerning Cross-Boundary Petroleum Co-operation between Norway and the United Kingdom of 2005 provides a comprehensive regime for the development of cross-boundary fields, including procedures for approval of unitisation agreements by both governments, apportionment of reserves, and the use of infrastructure on one side of the boundary to develop fields on the other. Similar arrangements exist in the Gulf of Mexico under the Agreement between the United States and Mexico concerning Transboundary Hydrocarbon Reservoirs in the Gulf of Mexico (2012).

Joint Development Zones

Where states cannot agree on a maritime boundary, they may establish a joint development zone (JDZ) in the disputed area, allowing petroleum activities to proceed without prejudice to their respective claims. The Nigeria–São Tomé and Príncipe Joint Development Zone, established by a treaty signed in Abuja in 2001, is administered by a Joint Ministerial Council and a Joint Development Authority, with revenues shared 60:40 in favour of Nigeria. Other examples include the Malaysia–Thailand Joint Development Area (1979) and the arrangements in the Timor Sea between Australia and Timor-Leste, which were superseded by the maritime boundary treaty of 2018 concluded following the first-ever compulsory conciliation under Annex V of UNCLOS. Joint development is examined further in Chapter 20.

5.8 Decommissioning and Abandonment Liabilities

The Decommissioning Challenge

At the end of their productive lives, petroleum installations must be decommissioned: wells must be plugged and abandoned, platforms and pipelines removed or made safe, and sites remediated. Decommissioning is technically complex and extremely expensive. Estimates of the total cost of decommissioning offshore infrastructure on the United Kingdom Continental Shelf alone run to tens of billions of pounds over the coming decades, and globally the liability runs to hundreds of billions of dollars. The energy transition is likely to accelerate decommissioning as fields become uneconomic earlier than anticipated.

Decommissioning raises four principal legal questions: what standard of decommissioning is required; who is liable to carry it out and pay for it; how the state can ensure that funds will be available when decommissioning becomes due; and how liability is affected by transfers of interests during the life of a field.

International Standards

UNCLOS Article 60(3) provides that "any installations or structures which are abandoned or disused shall be removed to ensure safety of navigation, taking into account any generally accepted international standards established in this regard by the competent international organization", and that "such removal shall also have due regard to fishing, the protection of the marine environment and the rights and duties of other States". The competent organisation is the International Maritime Organization, whose *Guidelines and Standards for the Removal of Offshore Installations and Structures on the Continental Shelf and in the Exclusive Economic Zone* (1989) require the removal of installations standing in less than 75 metres of water and weighing less than 4,000 tonnes in air (excluding the deck and superstructure), and of all installations placed after 1 January 1998 in waters of

less than 100 metres depth, while permitting partial removal of larger installations subject to specified conditions.

In the North-East Atlantic, OSPAR Decision 98/3 on the Disposal of Disused Offshore Installations, adopted in 1998 following the controversy over Shell's proposal to dispose of the Brent Spar storage buoy in deep water, prohibits the dumping and leaving wholly or partly in place of disused offshore installations, subject to derogations for the footings of large steel jackets weighing more than 10,000 tonnes and for concrete gravity-based installations, where the competent authority is satisfied that there are significant reasons why an alternative disposal option is preferable to reuse, recycling, or final disposal on land. The Brent Spar episode illustrates the role of public opinion and civil society in shaping decommissioning law.

The London Convention 1972 and its 1996 Protocol on the prevention of marine pollution by dumping also regulate the disposal of platforms at sea.

Domestic Regimes

In the United Kingdom, Part IV of the Petroleum Act 1998 establishes a detailed regime under which the Secretary of State may require persons responsible for offshore installations and pipelines to submit decommissioning programmes for approval. Section 29 notices may be served on a wide class of persons, including current and former licensees, owners of installations, and companies associated with them. The Energy Act 2008 extended the power to require security for decommissioning liabilities. Crucially, liability under Part IV can extend to former licensees: a company that has sold its interest may be called upon to carry out decommissioning if the purchaser defaults. This "residual liability" regime gives the state a powerful protection against insolvency of current operators but complicates transactions, since sellers must manage the risk of being called back.

In Nigeria, the PIA requires the holder of a petroleum mining lease to submit a decommissioning and abandonment plan and to establish a decommissioning and abandonment fund, to be maintained with a financial institution and managed in accordance with regulations issued by the NUPRC. Contributions to the fund are to be made over the life of the field, ensuring that adequate resources will be available. The PIA also provides that the obligations relating to decommissioning and abandonment are not extinguished by the transfer of an interest without the express agreement of the regulator, and the NUPRC's regulations on decommissioning and abandonment set out detailed requirements. The adequacy of decommissioning provisioning has been a central issue in the review of onshore divestments by international oil companies, given the age of much of the onshore infrastructure in the Niger Delta and the legacy of environmental contamination.

Liability Allocation in Contracts

Contracts allocate decommissioning liabilities among co-venturers and between buyers and sellers. JOAs typically provide that decommissioning costs are shared in proportion to participating interests at the time decommissioning is carried out, but require parties to provide security in advance — often through decommissioning security agreements (DSAs) — to protect against the default of any party. In asset sales, the seller typically seeks an indemnity from the buyer for decommissioning liabilities, supported by security such as letters of credit or parent company guarantees, because the seller may remain exposed to residual statutory liability. Disputes over the adequacy of security and the calculation of decommissioning cost estimates are frequent.

Decommissioning and the Energy Transition

The energy transition creates new opportunities and challenges for decommissioning. Infrastructure that would otherwise be removed may be repurposed for carbon dioxide transport and storage, hydrogen production or transport, or offshore wind support. Such repurposing requires legal frameworks that address the transfer of liability for legacy infrastructure, the long-term liability for stored carbon dioxide, and the interaction between petroleum, carbon storage, and renewable energy licensing regimes. The United Kingdom's Energy Act 2023 and the European Union's CCS Directive 2009/31/EC address some of these questions, examined in Chapter 12. At the same time, the risk that fields will be abandoned prematurely, by operators lacking the resources to decommission them properly, raises concerns about stranded liabilities that may ultimately fall on the state and taxpayers.

Onshore Abandonment and Legacy Pollution

Onshore, decommissioning intersects with the remediation of legacy pollution. In the Niger Delta, decades of oil spills, gas flaring, and inadequate decommissioning of wells and pipelines have left extensive environmental damage. The United Nations Environment Programme's *Environmental Assessment of Ogoniland* (2011) found widespread contamination of land, groundwater, and surface water and recommended a clean-up programme that it estimated would take up to thirty years. The Hydrocarbon Pollution Remediation Project (HYPREP) was established by the Federal Government in 2016 to implement the recommendations. The PIA's environmental remediation fund provisions, discussed in Chapter 17, are designed to ensure that operators provide for the remediation of environmental damage arising from their operations. The legal liability of parent companies for pollution caused by their subsidiaries in Nigeria has been addressed by English and Dutch courts, notably in *Okpabi v Royal Dutch Shell plc* (2021) and *Four Nigerian Farmers v Shell* (2021), considered in Chapter 21.

LEADING AUTHORITY

Ghana/Côte d'Ivoire Maritime Delimitation (ITLOS Special Chamber, 2017) — petroleum activities in disputed areas

Facts. Ghana had for decades licensed petroleum activities up to a line it regarded as the customary equidistance boundary with Côte d'Ivoire, and significant discoveries had been made, including the TEN fields. Côte d'Ivoire contested the boundary and claimed that Ghana's unilateral petroleum activities in the disputed area violated Côte d'Ivoire's sovereign rights and Article 83(3) of UNCLOS, which requires states, pending agreement on delimitation, to "make every effort to enter into provisional arrangements of a practical nature" and "not to jeopardize or hamper the reaching of the final agreement". In 2015 the Special Chamber had prescribed provisional measures requiring Ghana not to undertake new drilling in the disputed area.

Holding. The Special Chamber delimited the boundary using the equidistance/relevant circumstances methodology, producing a line close to that claimed by Ghana. It rejected Ghana's argument that a tacit agreement on the boundary had arisen from the parties' oil practice, but also held that Ghana had not violated Côte d'Ivoire's sovereign rights, because the activities had been carried out in an area that was ultimately attributed to Ghana, and that Ghana had not violated Article 83(3).

Significance. The decision is important for upstream petroleum practice in disputed maritime areas. It confirms that oil practice alone rarely establishes a tacit maritime boundary, and it illustrates the risks — including provisional measures halting new drilling — that investors face when operating in contested waters.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Allocating risk under a production sharing contract (hypothetical). The fictional Republic of Kandara is preparing a model PSC for its first deep-water licensing round. You are counsel to the Kandaran Ministry of Petroleum. Prepare (a) an annotated term sheet of the key fiscal provisions (royalty, cost recovery limit, profit oil split based on an R-factor, taxation, and state participation), explaining the policy rationale for each choice; (b) a short model calculation applying your terms to a hypothetical field producing 50,000 barrels per day at an assumed price of USD 70 per barrel, with stated assumptions; and (c) a note identifying the three provisions most likely to generate disputes, with drafting recommendations to reduce that risk. *Kandara, its institutions, and all figures are hypothetical and for teaching purposes only.*

CHAPTER SUMMARY

- Petroleum is defined by statute and contract; the distinctions between crude oil and gas, associated and non-associated gas, and conventional and unconven-

tional resources have significant legal and fiscal consequences, as illustrated by Nigeria's gas flaring provisions.

- Petroleum titles may be characterised as administrative licences, contracts, property rights, or hybrids; the characterisation affects the state's power to vary terms and the investor's protection.
- Competitive and transparent allocation of rights is increasingly required by law, as under the PIA, in reaction to the abuses associated with discretionary award.
- Petroleum arrangements fall into concessionary/licensing systems, production sharing contracts, and service contracts, distinguished by ownership of production and the form of the state's return; the economic substance matters more than the label.
- Joint operating agreements govern relations among co-venturers, including the operator's role and liability, operating committee decisions, sole risk, default, and transfers; NNPC cash call problems illustrate the difficulties of state participation.
- National oil companies' dual commercial and regulatory roles create conflicts of interest that reforms such as the PIA seek to resolve.
- Shared reservoirs require unitisation; transboundary fields are governed by mineral deposits clauses, framework agreements, and joint development zones.
- Decommissioning is governed by UNCLOS, IMO guidelines, regional instruments such as OSPAR Decision 98/3, and domestic regimes that increasingly require advance funding and impose residual liability.

DOCTORAL RESEARCH QUESTIONS

1. Is the conventional classification of petroleum arrangements into licences, PSCs, and service contracts analytically useful, given that fiscal terms can make any form economically equivalent to another?
2. Evaluate the PIA's incentive-based conversion regime as a technique for migrating legacy arrangements without triggering investment treaty claims.
3. Has state practice in maritime boundary agreements given rise to a customary obligation to cooperate in, or to unitise, the development of transboundary petroleum deposits?
4. How should residual decommissioning liability be designed to protect the state without deterring the transfer of mature assets to new operators better placed to extend field life?
5. Should international law impose obligations on states to ensure that fields are not abandoned without adequate remediation, and what would be the source of such obligations?

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6 CHAPTER

Natural Gas Law, LNG Transactions, and International Gas Markets

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the physical, economic, and commercial characteristics of natural gas that distinguish its legal regime from that of crude oil.
2. Analyse the regulation of gas exploration and production, including gas-specific licensing, domestic supply obligations, and flaring rules.
3. Evaluate the legal models for gas transportation and pipeline access, including negotiated and regulated third-party access and unbundling.
4. Compare the principal structures used to develop liquefied natural gas (LNG) projects and identify the legal risks associated with each.
5. Analyse the key provisions of long-term gas sales and purchase agreements, including take-or-pay, delivery, quality, and destination provisions.
6. Explain the law and practice of price review and the mechanisms by which gas contracts accommodate changing market conditions.
7. Evaluate the international legal framework governing cross-border gas pipelines and transit.
8. Assess the causes and resolution of gas supply disputes and the legal limits on regulatory intervention in gas markets.

Introduction

Natural gas occupies a distinctive and increasingly contested position in international energy law. For much of the twentieth century it was treated as an unwanted by-product of oil production, flared or reinjected because there was no economical way to bring it to market. Over the past five decades it has become a global commodity in its own right, supplying almost a quarter of world primary energy, fuelling a large share of electricity generation, providing heat for homes and industry, and serving as a feedstock for fertilisers and petrochemicals. Its lower carbon intensity relative to coal and oil has led many governments to describe

it as a "bridge" or "transition" fuel, while its methane emissions and long-lived infrastructure have led critics to warn that new gas investment risks locking in emissions incompatible with the goals of the Paris Agreement. The legal regime of natural gas must therefore accommodate both its role in energy security and its contested position in the energy transition.

The legal regime of natural gas differs from that of oil in ways that flow directly from the physical properties of the commodity. Oil is a dense liquid that can be stored easily and transported cheaply by tanker, rail, or pipeline to almost any market; this fungibility gave rise, from the 1980s onwards, to a global oil market with transparent spot and futures prices. Gas, by contrast, has a very low energy density at atmospheric pressure. It must be transported either through high-pressure pipelines, which are fixed, capital-intensive, and tie producer and consumer together for decades, or as liquefied natural gas, which requires cooling the gas to approximately minus 162 degrees Celsius, shipping it in specialised vessels, and regasifying it at the destination. Both modes of transport require very large, specific, and long-lived investments. As a consequence, gas markets historically developed as regional rather than global markets, organised around long-term contracts that allocated the risks of these investments between sellers and buyers, and subject to extensive regulation of the natural monopoly elements of the supply chain.

Three transformations over the past four decades have reshaped the legal regime of gas. The first is liberalisation: the introduction of competition in gas supply in North America from the 1980s, in the United Kingdom from the 1990s, and in the European Union from the late 1990s onwards, requiring the unbundling of transmission from supply and the creation of rights of third-party access to pipelines. The second is the growth of the LNG trade, which has expanded from a handful of point-to-point projects linking specific producers with Japanese and European buyers into a global market involving dozens of exporting and importing countries, with an increasing share of spot and short-term trade and greater price convergence among regions. The third is the politicisation of gas supply, illustrated most dramatically by the Russian Federation's curtailment of gas deliveries to Europe in 2021 and 2022, which precipitated an energy crisis, extraordinary regulatory interventions, and a fundamental restructuring of European gas supply.

This chapter examines the legal regime of natural gas from wellhead to burner tip. It begins with the legal and commercial characteristics of gas (section 6.1) and the regulation of gas exploration and production (section 6.2). It then examines gas transportation and pipeline access (section 6.3), the structures used to develop LNG projects (section 6.4), long-term gas sales and purchase agreements (section 6.5), and the mechanisms of price review, take-or-pay, and contractual

flexibility (section 6.6). It concludes with the legal framework for cross-border gas infrastructure and transit (section 6.7) and the law of gas supply disputes and regulatory intervention (section 6.8). The treatment of submarine pipelines under the law of the sea is reserved for Chapter 20, and the geopolitics of pipeline dependence for Chapter 25; this chapter focuses on the commercial and regulatory law of gas.

6.1 Legal and Commercial Characteristics of Natural Gas

Physical Properties and Their Legal Consequences

Natural gas is a mixture of hydrocarbon gases composed predominantly of methane, together with varying proportions of heavier hydrocarbons (ethane, propane, butane, and pentanes), and non-hydrocarbon components such as carbon dioxide, nitrogen, hydrogen sulphide, and water vapour. Before gas can be delivered to a transmission system, it must usually be processed to remove impurities and to extract natural gas liquids, which are valuable products in their own right. The composition of gas varies from field to field, and its energy content — its calorific or heating value — varies accordingly. Gas is therefore sold not by volume alone but by energy content, commonly measured in British thermal units (Btu), therms, gigajoules, or kilowatt hours, and gas contracts and network codes specify detailed quality specifications, including ranges for calorific value, the Wobbe index (a measure of the interchangeability of gases in burners), and maximum levels of contaminants.

These physical characteristics have several legal consequences. First, the specification of quality and the allocation of responsibility for off-specification gas are central issues in gas contracts and network codes. A seller that delivers gas outside the contractual specification may be in breach, and the buyer may be entitled to reject it, while a transmission system operator may refuse to accept gas that would compromise the safety or integrity of the network. Second, the measurement of gas — volume, pressure, temperature, and composition — requires agreed metering standards and procedures for resolving measurement disputes. Third, because gas from different sources is commingled in pipelines and storage, title to gas is usually a matter of contractual allocation rather than physical identification; the gas a buyer receives is not physically the same gas that its seller delivered. Gas law therefore relies heavily on concepts such as nominations, allocations, and balancing, which translate physical flows into contractual entitlements.

Natural Monopoly and Network Characteristics

High-pressure transmission pipelines and low-pressure distribution networks exhibit strong economies of scale and are commonly regarded as natural monopolies: it is usually more efficient for a single network to serve a given area than for competing networks to be duplicated. Natural monopoly provides the classic

justification for economic regulation: in the absence of competition, the owner of a network could charge monopoly prices or refuse access to competitors in upstream production or downstream supply. Gas law therefore distinguishes between the potentially competitive segments of the supply chain — production, wholesale trading, and retail supply — and the monopoly segments — transmission, distribution, and, in some cases, storage and LNG terminals — which are subject to regulation of tariffs, access, and investment.

Asset Specificity and the Long-Term Contract

The economics of gas infrastructure are characterised by high asset specificity. A pipeline linking a particular field to a particular market, or an LNG liquefaction plant built to process gas from a particular reservoir, has little value in any alternative use. Once the investment is made, its owner is exposed to opportunistic behaviour by its counterparties — the "hold-up" problem analysed in transaction cost economics by Oliver Williamson and others. A producer that has built a pipeline to a single buyer may find the buyer demanding lower prices once the pipeline is complete; a buyer that has built power stations relying on a single supplier may find the supplier demanding higher prices.

The long-term gas contract emerged as the principal legal device for managing this risk. By committing the buyer to purchase, or pay for, specified quantities of gas over periods of fifteen to twenty-five years, and the seller to deliver them, the long-term contract provides the revenue certainty necessary to finance the infrastructure, while price indexation and price review mechanisms allow the price to adjust to changing market conditions without reopening the fundamental bargain. The long-term contract is thus a form of private governance structure for a relationship in which neither party can easily exit. Its central provisions — take-or-pay, destination restrictions, price indexation, and price review — are examined in sections 6.5 and 6.6.

Regional Markets and Pricing Systems

Partly because of the costs of transport, gas markets developed historically as regional markets with distinct pricing systems. Four broad systems may be identified.

Gas-on-gas competition prevails in North America, the United Kingdom, and much of continental Europe, where liberalised markets with liquid trading hubs determine prices through the interaction of supply and demand. The principal hubs are Henry Hub in Louisiana, which serves as the delivery point for the New York Mercantile Exchange natural gas futures contract; the National Balancing Point (NBP) in Great Britain, a virtual trading point within the National Gas transmission system; and the Title Transfer Facility (TTF) in the Netherlands, which has become the benchmark for continental European gas prices.

Oil indexation was the traditional pricing method in continental Europe and remains significant in Asia. Under oil-indexed contracts, the gas price is linked by formula to the prices of competing fuels, typically crude oil or oil products. In Asian LNG contracts, the price has commonly been expressed as a percentage ("slope") of the Japan Customs-Cleared Crude price (the "JCC" or "Japanese Crude Cocktail") or, increasingly, Brent crude, sometimes with an "S-curve" that moderates the effect of very high or very low oil prices. The rationale for oil indexation, which originated in the pricing of Groningen gas in the Netherlands in the 1960s, was the "market value" or "replacement value" principle: gas should be priced at a level that allowed it to compete with the alternative fuel the buyer would otherwise use.

Regulated pricing prevails in many producing countries and some importing countries, where governments set gas prices administratively, often below the cost of supply, to support domestic industry and power generation or to protect consumers. Regulated domestic gas prices have been a significant obstacle to gas development in many countries, including Nigeria, where the historically low regulated domestic price discouraged investment in gas supply to the domestic market.

Spot and short-term LNG pricing has grown with the expansion of the LNG trade, and Asian spot LNG is now commonly priced by reference to the Japan Korea Marker (JKM), an assessed price published by a price reporting agency, while increasing quantities of LNG from the United States are sold on the basis of Henry Hub prices plus a fixed liquefaction fee.

The International Gas Union's periodic wholesale gas price surveys have documented a long-term shift from oil indexation towards gas-on-gas competition, particularly in Europe, although the extreme price volatility of hub prices during the 2021–2023 crisis led some buyers to reconsider the value of oil-linked contracts as a hedge.

Gas in the Energy Transition

The role of gas in the energy transition is legally significant because it shapes the policy frameworks within which gas investment takes place. The European Union's Taxonomy Regulation, through a Complementary Climate Delegated Act adopted in 2022, classifies certain gas-fired electricity generation and heat activities as transitional activities that may qualify as environmentally sustainable, subject to stringent emissions thresholds and conditions, including a commitment to switch to renewable or low-carbon gases by 2035. The inclusion of gas (and nuclear) in the taxonomy was controversial and was challenged before the General Court of the European Union by Austria and by environmental organisations; in September 2025 the General Court dismissed Austria's action. The

EU's Regulation on the reduction of methane emissions in the energy sector, adopted in 2024, imposes measurement, reporting, and verification obligations on oil, gas, and coal operators, restricts routine venting and flaring, and will progressively apply equivalent monitoring requirements to imported fossil fuels — a measure with significant implications for exporting states, including Nigeria. At the international level, the outcome of the first global stocktake under the Paris Agreement, adopted at COP28 in Dubai in 2023, called on parties to contribute to "transitioning away from fossil fuels in energy systems, in a just, orderly and equitable manner", while recognising that "transitional fuels can play a role in facilitating the energy transition while ensuring energy security".

For gas-producing developing countries, the transition raises profound questions of equity. African governments, including Nigeria, have argued that gas is essential to expanding energy access, industrialisation, and clean cooking, and that it would be unjust for countries that bear little historical responsibility for climate change to be denied financing for gas development. Nigeria declared 2021–2030 the "Decade of Gas", reflecting a strategy of using its large gas reserves — among the largest in Africa — to drive domestic industrialisation and power generation. The tension between this position and the climate policies of international lenders, which have restricted financing for upstream gas, is examined further in Chapters 18 and 23.

6.2 Gas Exploration and Production Regulation

The Historical Neglect of Gas in Petroleum Legislation

Petroleum legislation was designed historically around oil. Early concessions and petroleum statutes often made little distinction between oil and gas, or treated gas as an incidental by-product. As a result, petroleum regimes frequently failed to provide adequate incentives for gas development. Gas discovered in the course of oil exploration might be reinjected, flared, or left undeveloped because the fiscal terms designed for oil made gas development uneconomic, because there was no domestic market or infrastructure, or because the rights to gas were unclear. The development of gas-specific legal provisions — governing rights to non-associated gas, the fiscal treatment of gas, the obligations of producers to supply gas to the domestic market, and the restriction of flaring — has been a significant feature of petroleum law reform over the past three decades.

Rights to Gas Under Petroleum Titles

Petroleum titles usually confer rights to both oil and gas within the licence area, but they may differ in their treatment of the two. Some regimes provide for separate gas licences or for distinct terms applicable to non-associated gas discoveries, recognising that gas development requires longer periods for market development and a different fiscal structure. A common device is the extended "gas holding" or

"gas retention" period, which allows a licensee that has discovered non-associated gas but cannot yet develop it commercially to retain the discovery area for a defined period while it seeks a market, without being required to relinquish it. Without such provisions, licensees may be forced to relinquish gas discoveries before markets have developed.

Under Nigeria's Petroleum Industry Act 2021, the general framework for petroleum prospecting licences and petroleum mining leases applies to both oil and gas, but the Act contains specific provisions recognising the distinct characteristics of gas. Holders of petroleum prospecting licences that discover non-associated gas may be granted a period to establish its commerciality before conversion to a lease, and the Act's fiscal provisions treat gas production more favourably than oil, notably by subjecting gas to companies income tax rather than the hydrocarbon tax applicable to crude oil, and by providing reduced royalty rates for gas, with lower rates for gas delivered to the domestic market. These provisions reflect the policy of the Act to promote the development of gas for domestic use.

Domestic Gas Supply Obligations

Many gas-producing states impose domestic supply obligations requiring producers to make a proportion of their gas available for the domestic market, often at regulated prices. Such obligations reflect the policy of ensuring that national resources contribute to domestic energy supply and industrial development, rather than being exported exclusively. They may, however, deter investment if the regulated price is below the cost of supply or the export price.

In Nigeria, the Nigerian Gas Master Plan of 2008 and the National Domestic Gas Supply and Pricing Regulations 2008 introduced a domestic gas delivery obligation requiring producers to supply specified volumes of gas to the domestic market. The PIA continues this approach, empowering the Nigerian Midstream and Downstream Petroleum Regulatory Authority (NMDPRA) to determine the domestic gas delivery obligation and to set a domestic base price for gas supplied to the strategic sectors (notably power generation) until a willing-buyer, willing-seller market develops. In Western Australia, the Domestic Gas Reservation Policy, introduced in 2006, requires LNG export projects to make gas equivalent to fifteen per cent of their LNG production available to the domestic market, and the federal Australian Domestic Gas Security Mechanism, introduced in 2017, empowers the government to restrict LNG exports where there is a domestic shortfall. In 2022 and 2023, Australia introduced emergency price caps on wholesale gas and a mandatory code of conduct for the east coast gas market in response to the price surge triggered by the European energy crisis. Similar obligations exist in Indonesia, Egypt, Israel, and other producing countries.

Domestic supply obligations raise legal questions in international trade law, since they may function as export restrictions, and in investment law, since their introduction or tightening after an investment is made may affect the investor's legitimate expectations. They are discussed further in Chapter 13 in the context of export restrictions.

Gas Flaring Regulation

Gas flaring — the burning of associated gas at the wellsite — wastes a valuable resource, emits carbon dioxide and black carbon, and exposes nearby communities to air pollution, noise, heat, and light. The World Bank's Global Gas Flaring Tracker reports that over 140 billion cubic metres of gas are flared globally each year, with Russia, Iraq, Iran, the United States, Venezuela, Algeria, Libya, and Nigeria among the largest flaring countries. Venting — the direct release of unburned gas, predominantly methane — is even more damaging to the climate, since methane is a far more potent greenhouse gas than carbon dioxide over short time horizons.

Legal approaches to flaring include outright prohibitions with exceptions for safety and emergencies, flare fees or penalties, requirements for gas utilisation plans as a condition of development approval, and mechanisms to allocate flare gas to third parties that can commercialise it. Nigeria's experience with flaring regulation, examined in Chapter 5, illustrates both the difficulties of enforcement and the evolution of policy from prohibition to commercialisation through the Flare Gas (Prevention of Waste and Pollution) Regulations 2018 and the Nigerian Gas Flare Commercialisation Programme. The *Gbemre* case, also discussed in Chapter 5 and Chapter 21, established the human rights dimension of flaring in Nigerian law.

Internationally, the World Bank's Zero Routine Flaring by 2030 Initiative, launched in 2015, has been endorsed by governments, oil companies, and development institutions, which commit not to engage in routine flaring in new oil field developments and to end legacy routine flaring as soon as possible and no later than 2030. The Global Methane Pledge, launched at COP26 in 2021, commits participating countries to collective action to reduce global anthropogenic methane emissions by at least thirty per cent from 2020 levels by 2030. The Oil and Gas Methane Partnership 2.0 (OGMP 2.0), administered by the United Nations Environment Programme, has established a reporting framework for company methane emissions that has been incorporated by reference into the EU Methane Regulation. These instruments are largely voluntary, but they are increasingly being translated into binding domestic and regional law.

Gas Processing and Natural Gas Liquids

Gas processing — the separation of natural gas liquids, removal of impurities, and conditioning of gas to pipeline specification — occupies a regulatory borderland between upstream and midstream activity. In some regimes, processing facilities associated with a specific field are treated as part of upstream operations under the petroleum licence; in others, they require separate licences. The PIA treats gas processing as a midstream activity requiring a licence from the NMDPRA, distinct from the upstream licence regulated by the Nigerian Upstream Petroleum Regulatory Commission (NUPRC). The classification affects the applicable fiscal regime, the identity of the regulator, and the rights of third parties to access the facilities. Access to upstream infrastructure, including processing facilities, by third parties is examined in section 6.3.

Fiscal Treatment of Gas

Because gas development is typically less profitable than oil development, owing to higher transport costs, lower prices per unit of energy, and the need for market development, many petroleum fiscal regimes provide more favourable terms for gas. Common incentives include lower royalty rates, exemption from or reduced rates of resource rent taxes, accelerated depreciation for gas infrastructure, and tax holidays for gas utilisation projects. Nigeria's Associated Gas Framework Agreement of 1992 and the incentives for gas utilisation in the Companies Income Tax Act, now carried forward in modified form in the PIA, are examples. The design of gas fiscal incentives must balance the need to make gas development commercially viable against the risk of conferring windfall gains on projects that would have proceeded without them, and against the need to avoid distortions between oil and gas production that might encourage artificial classification of production. Fiscal regimes generally are examined in Chapter 16.

6.3 Gas Transportation and Pipeline Access

The Gas Supply Chain and Its Regulation

The physical supply chain for pipeline gas comprises upstream gathering pipelines, which collect gas from wells and convey it to processing facilities; high-pressure transmission pipelines, which carry processed gas over long distances; storage facilities, including depleted reservoirs, salt caverns, and aquifers, which balance seasonal and short-term fluctuations in demand; and low-pressure distribution networks, which deliver gas to end-users. LNG terminals, considered in section 6.4, perform an analogous function at the interface between maritime and pipeline transport. Each element raises distinct regulatory issues, but the central questions are the same: who may own and operate the infrastructure; on what terms must the owner grant access to third parties; how are tariffs determined; and how is new investment encouraged.

Models of Industry Structure

Historically, gas industries in most countries were organised as vertically integrated monopolies, whether state-owned (as in France, Italy, and much of the developing world) or privately owned but regulated (as in the United States). A single company, or a small number of regional companies, purchased gas from producers, transported it, and sold it to distributors or end-users. Liberalisation sought to introduce competition in the potentially competitive segments of the chain by separating them from the monopoly networks and requiring network owners to provide access to competing suppliers on non-discriminatory terms.

The United States led this process. Under the Natural Gas Act of 1938, interstate pipelines were regulated by the Federal Power Commission (now the Federal Energy Regulatory Commission, FERC) as merchants that bought and resold gas. Following the partial decontrol of wellhead prices under the Natural Gas Policy Act of 1978, FERC Order No 436 (1985) encouraged pipelines to offer open-access transportation services, and Order No 636 (1992) required interstate pipelines to unbundle their sales and transportation services, transforming them into pure transporters and creating a competitive wholesale gas market. The United Kingdom followed with the privatisation of British Gas in 1986, the Gas Act 1995, and the separation of transportation from supply, producing one of the most liquid gas markets in the world. The European Union adopted a sequence of liberalisation directives, culminating in the Third Energy Package of 2009, comprising Directive 2009/73/EC concerning common rules for the internal market in natural gas and Regulation (EC) No 715/2009 on conditions for access to the natural gas transmission networks. In 2024, the EU adopted the Hydrogen and Decarbonised Gas Market Package — Directive (EU) 2024/1788 and Regulation (EU) 2024/1789 — which recasts the gas market rules and extends them to the emerging hydrogen sector.

Third-Party Access

Third-party access (TPA) is the right of a person other than the owner of a pipeline or other facility to use its capacity to transport gas. TPA may be negotiated or regulated.

Under **negotiated TPA**, the owner and the access seeker negotiate the terms of access commercially, subject to a general obligation to negotiate in good faith and not to discriminate, and to the oversight of a regulator or competition authority in the event of a dispute. Negotiated TPA is common for upstream pipelines and processing facilities, where the circumstances of each access request are highly specific. In the United Kingdom, for example, the Energy Act 2011 and the Infrastructure Code of Practice govern access to upstream oil and gas infra-

structure on the UK Continental Shelf, and the North Sea Transition Authority may determine the terms of access in default of agreement.

Under **regulated TPA**, the terms of access, including tariffs, are published in advance and approved or set by an independent regulator. Regulated TPA is the norm for transmission and distribution networks in the European Union, where Directive 2009/73/EC (now the recast Directive) requires member states to ensure a system of third-party access to transmission and distribution systems and LNG facilities based on published tariffs, applicable to all eligible customers and applied objectively and without discrimination. Access to storage may be either negotiated or regulated, at the option of the member state.

Access may be refused on limited grounds. The EU directive permits a natural gas undertaking to refuse access on the basis of lack of capacity, or where access would prevent it from carrying out public service obligations, or on the basis of serious economic and financial difficulties with take-or-pay contracts. Refusals must be duly substantiated.

Unbundling

Third-party access is unlikely to be effective if the network owner is also a supplier of gas in competition with the access seekers, because it will have both the incentive and the means to discriminate against them — for example, by underinvesting in capacity, manipulating information, or delaying connections. Liberalisation has therefore been accompanied by requirements for the *unbundling* of network activities from supply and production activities. Unbundling may take several forms, of increasing stringency:

1. **Accounting unbundling**, requiring separate accounts for network and supply activities, to prevent cross-subsidisation.
2. **Functional or legal unbundling**, requiring network activities to be carried out by a separate legal entity with independent management, although it may remain within a vertically integrated group.
3. **Independent system operator (ISO)**, under which the integrated group retains ownership of the network but its operation is entrusted to an independent operator.
4. **Independent transmission operator (ITO)**, under which the network remains within the integrated group but is subject to detailed rules ensuring its independence in management and decision-making.
5. **Ownership unbundling**, requiring the separation of ownership of transmission networks from production and supply interests.

The Third Energy Package required member states to choose among ownership unbundling, the ISO model, and the ITO model for transmission system operators, and the choice was the subject of intense political controversy, with France

and Germany defending the ITO model to protect their integrated incumbents. The "third country clause" in Article 11 of the Directive required certification of transmission system operators controlled by persons from third countries, permitting refusal of certification where the grant would put at risk the security of energy supply of the member state or the Union — a provision directed in large part at Gazprom.

Tariff Regulation

The regulation of transportation tariffs seeks to allow network owners to recover their efficiently incurred costs, including a reasonable return on capital, while preventing monopoly pricing. The principal methodologies are *cost-of-service* or *rate-of-return* regulation, which sets tariffs to recover a defined revenue requirement based on operating costs, depreciation, and a return on the regulated asset base; and *incentive regulation*, such as price caps or revenue caps of the "RPI–X" type pioneered in the United Kingdom, under which tariffs are set for multi-year periods and the network owner retains the benefits of efficiency improvements. Tariff structures may be *distance-related* (point-to-point), *postage stamp* (uniform irrespective of distance), or *entry–exit*, under which users book capacity separately at entry and exit points and gas can flow freely within the system between them. The EU Network Code on Harmonised Transmission Tariff Structures for Gas (Regulation (EU) 2017/460) requires entry–exit systems and sets out detailed rules for reference price methodologies.

Pipeline Regulation in Nigeria

Nigeria's gas transportation infrastructure has historically been owned and operated by the Nigerian Gas Company, a subsidiary of NNPC, now succeeded within NNPC Ltd's corporate structure. The PIA establishes a framework for midstream gas regulation by the NMDPRA, including licensing of gas transportation pipelines, gas distribution systems, gas processing facilities, and gas storage; the establishment of a Gas Transportation Network Code governing access to the transportation network; open access to pipelines and facilities on a non-discriminatory basis, subject to available capacity; and the regulation of tariffs. The Act requires the separation of gas transportation network operation from gas wholesale and supply activities, reflecting the European unbundling model. Major infrastructure projects, including the Ajaokuta–Kaduna–Kano (AKK) pipeline and the Obiafu–Obrikom–Oben (OB3) pipeline, are intended to connect gas resources in the south with demand centres in the north, but have faced delays associated with financing, security, and construction challenges.

6.4 Liquefied Natural Gas Project Structures

The LNG Value Chain

The LNG value chain comprises upstream production of feed gas; liquefaction, in which the gas is treated and cooled to approximately minus 162 degrees Celsius, reducing its volume by a factor of around six hundred; shipping in specialised LNG carriers; and regasification at import terminals, either onshore or on floating storage and regasification units (FSRUs). Each stage requires very large capital investments: a large liquefaction plant may cost tens of billions of United States dollars, a modern LNG carrier more than two hundred million dollars, and a regasification terminal several hundred million dollars or more. The capital intensity of the chain, together with its specificity and long life, means that LNG projects have historically required long-term contracts covering most of their output before financing can be secured.

Physical chain (top) and principal legal instruments (bottom)

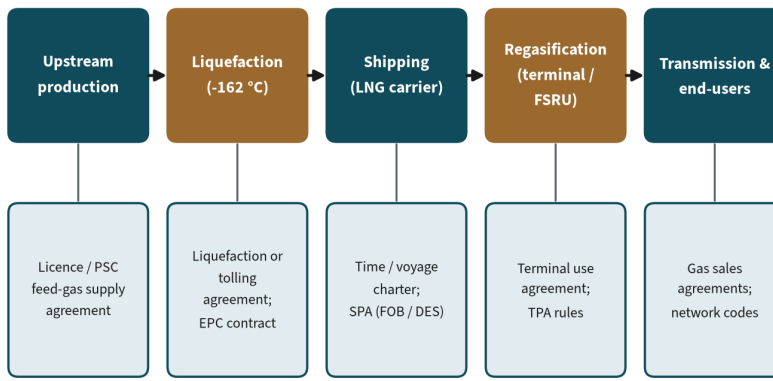


Figure 6.1 The LNG value chain and the principal contracts that link its stages. *Source:* prepared for this edition.

Principal Project Structures

LNG export projects have been organised according to several structures, distinguished by the relationship between ownership of the upstream gas, the liquefaction facilities, and the LNG product.

The integrated structure. Under the integrated structure, the same parties (in the same proportions) own the upstream gas reserves and the liquefaction plant, and sell LNG directly to buyers. The upstream and liquefaction activities are treated as a single project for fiscal and commercial purposes. The structure aligns the interests of upstream and midstream investors and avoids the need for

a transfer price between them, but it requires all investors to participate in both segments. Many projects in Qatar, Australia, and elsewhere have used integrated structures.

The merchant structure. Under the merchant structure, the liquefaction company is owned separately from the upstream producers (or by them in different proportions). It purchases feed gas from the producers under gas supply agreements, liquefies it, and sells LNG to buyers on its own account, taking the margin between its purchase and sales prices. Nigeria LNG Limited (NLNG), owned by NNPC (49 per cent), Shell (25.6 per cent), TotalEnergies (15 per cent), and Eni (10.4 per cent), operates on a broadly merchant basis, purchasing feed gas from upstream joint ventures and selling LNG under long-term contracts to buyers in Europe and elsewhere. The merchant structure requires a negotiated feed gas price, which allocates value between upstream and midstream, and it may raise fiscal issues where the liquefaction company is subject to a different tax regime from the upstream producers.

The tolling structure. Under the tolling structure, the liquefaction company does not take title to the gas. It provides liquefaction services to the gas owners (or to LNG offtakers) in return for a tolling fee, which is typically fixed and payable regardless of whether capacity is used, providing a utility-like return to the plant owners. The gas owners retain title through the liquefaction process and sell the LNG themselves. The tolling model was used for later trains of Atlantic LNG in Trinidad and Tobago and for several of the export projects developed in the United States from the mid-2010s, such as Freeport LNG, whose capacity is contracted to offtakers under liquefaction tolling agreements. Other United States projects, such as Cheniere's Sabine Pass, sell LNG under sale and purchase agreements priced at a percentage of Henry Hub plus a fixed fee, which replicate the economics of tolling. Under the United States model, buyers enter into long-term "liquefaction services" or sale and purchase agreements under which they pay a fixed liquefaction fee for capacity and purchase feed gas at Henry Hub prices plus a percentage, giving them flexibility to lift cargoes or cancel them (paying the fixed fee) and to resell LNG anywhere in the world.

Hybrid structures. Many projects combine elements of these structures, and the choice depends on the fiscal regime, the identity and objectives of the investors, the availability of financing, and the requirements of the host state.

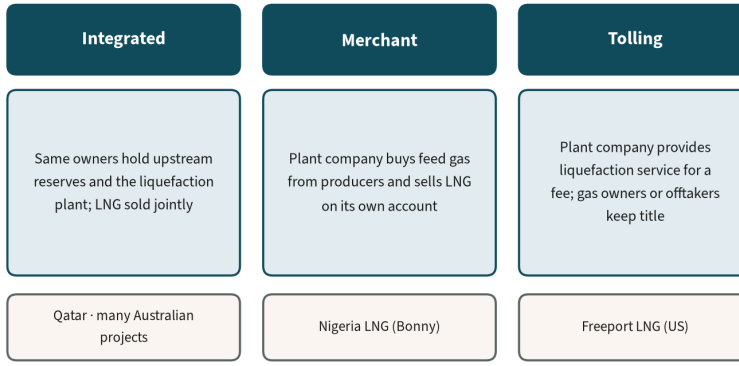


Figure 6.2 Integrated, merchant, and tolling structures for LNG export projects. *Source:* prepared for this edition.

The Host State Framework

LNG projects typically require a dedicated legal framework, often in the form of a project-specific statute or host government agreement, providing fiscal terms, guarantees against expropriation, stability of the legal regime, rights to export LNG, and rights to retain export proceeds offshore. The Nigeria LNG (Fiscal Incentives, Guarantees and Assurances) Act 1990 is an example: it conferred on NLNG a package of fiscal incentives, guarantees, and assurances, including exemption from certain taxes for an initial period, a guarantee against nationalisation, the right to maintain offshore accounts, and a commitment that the Act could not be amended without the consent of the company's shareholders. The NLNG Act illustrates both the attractions and the controversies of project-specific legislation. When the Nigerian Maritime Administration and Safety Agency (NIMASA) sought to impose levies on NLNG's shipping activities, and when the Federal Government sought to subject NLNG to contributions to the Niger Delta Development Commission, the company invoked the protections of the Act, generating litigation concerning the extent to which a statute can bind future legislatures. The question whether a legislature may validly restrict its own future legislative competence by statute raises fundamental constitutional issues; in common law systems influenced by the doctrine of parliamentary sovereignty, such entrenchment is generally ineffective as a matter of domestic law, although it may still give rise to liability under contract or investment treaties.

Shipping

LNG shipping arrangements depend on the delivery terms of the sales contract. Under a *free on board* (FOB) contract, the buyer takes delivery at the loading port and is responsible for shipping; under a *delivered ex-ship* (DES) or *cost, insurance and freight* (CIF) contract, the seller is responsible for shipping and delivers the LNG at the discharge port. LNG carriers are chartered under long-term time charters, voyage charters, or, increasingly, short-term charters for spot cargoes. The legal issues include the compatibility of vessels with terminals (ship–shore compatibility), the allocation of boil-off gas (LNG that evaporates during the voyage), demurrage, and the application of maritime conventions.

Regasification and FSRUs

The legal structure of regasification terminals varies. Some terminals are owned by the buyers that use them; others are owned by independent operators that offer capacity to multiple users under terminal use agreements, subject to regulated or negotiated third-party access. In the European Union, LNG terminals are subject to regulated third-party access, but major new infrastructure may obtain an exemption under Article 36 of Directive 2009/73/EC (now carried forward in the recast) where the investment enhances competition and security of supply and would not take place without the exemption. Many European terminals, including the Adriatic LNG terminal in Italy and the Gate terminal in Rotterdam, have obtained exemptions.

Floating storage and regasification units have dramatically reduced the time and cost required to establish LNG import capacity, enabling countries such as Egypt, Pakistan, Bangladesh, and, following the 2022 crisis, Germany to begin importing LNG within months rather than years. Germany's LNG Acceleration Act (LNG-Beschleunigungsgesetz) of May 2022 expedited permitting for FSRUs and onshore terminals by, among other things, temporarily exempting projects from certain environmental impact assessment requirements, illustrating the use of emergency legislation to accelerate energy infrastructure.

LNG Export Authorisation

Some states require specific authorisation for the export of LNG. In the United States, section 3 of the Natural Gas Act requires authorisation by the Department of Energy for exports of natural gas. Exports to countries with which the United States has a free trade agreement requiring national treatment for trade in natural gas are deemed to be consistent with the public interest and must be granted without modification or delay; exports to non-FTA countries must be authorised unless the Department finds that they would not be consistent with the public interest. In January 2024, the Department announced a temporary pause on pending decisions on non-FTA exports while it updated the economic and envi-

ronmental analyses underlying its public interest determinations; the pause was lifted in January 2025 by the incoming administration. The episode illustrates the regulatory risk facing LNG projects, whose commercial viability depends on export authorisation, and the growing incorporation of climate considerations into export licensing.

6.5 Long-Term Gas Sales and Purchase Agreements

Function and Forms

The gas sales and purchase agreement (GSPA, or SPA in the LNG context) is the central commercial instrument of the gas industry. It allocates between seller and buyer the volume risk (the risk that the buyer will not need, or the seller will not be able to supply, the contracted quantities), the price risk (the risk that market prices will move against one party), and the operational risks of delivery. Long-term GSPAs of fifteen to twenty-five years' duration underpin the financing of most major pipeline and LNG export projects. Medium-term contracts (two to ten years), short-term contracts (up to two years), and spot transactions have grown in importance with the development of liquid hubs and the LNG spot market, and are commonly concluded under master agreements that set out general terms, with individual transactions confirmed by short confirmation notices.

Several standard forms are used in the industry. For trading on European hubs, the European Federation of Energy Traders (EFET) General Agreement Concerning the Delivery and Acceptance of Natural Gas is widely used; in North America, the North American Energy Standards Board (NAESB) Base Contract for Sale and Purchase of Natural Gas performs a similar function. For LNG, the Association of International Energy Negotiators (AIEN) has published a Model Master LNG Sale and Purchase Agreement, and the International Group of Liquefied Natural Gas Importers (GIIGNL) has published a Master LNG Sale and Purchase Agreement for spot and short-term trades. Long-term SPAs, by contrast, are typically bespoke, although they follow well-established structures. Appendix D contains a model international gas sales agreement with annotations.

Contract Quantity Provisions

The quantity provisions of a long-term GSPA typically define:

- the **Annual Contract Quantity (ACQ)**, the quantity of gas the seller undertakes to make available and the buyer undertakes to take or pay for in each contract year;
- the **Daily Contract Quantity (DCQ)**, usually the ACQ divided by the number of days in the year, and the **Maximum Daily Quantity**, often 110 or 120 per cent of the DCQ, which defines the buyer's daily flexibility;

- **downward quantity tolerance** or **downward flexibility**, permitting the buyer to reduce its annual nominations by a specified percentage without incurring take-or-pay liability;
- adjustments to the ACQ for gas not taken because of the seller's failure to deliver, force majeure, or the buyer's rejection of off-specification gas; and
- the **build-up period** at the start of the contract and the **plateau period** during which the ACQ remains constant.

LNG SPAs specify quantities in terms of the Annual Contract Quantity expressed in tonnes or million British thermal units, and contain an **Annual Delivery Programme** procedure under which the parties agree before each contract year on the schedule of cargoes, subject to adjustments for operational reasons. The seller may have rights to reduce quantities for scheduled maintenance of the liquefaction facilities, and the buyer may have rights to divert or cancel cargoes, depending on the terms of the contract.

Delivery Point and Transfer of Title and Risk

The GSPA defines the delivery point at which title and risk pass from seller to buyer. In pipeline contracts, the delivery point may be at the outlet of the seller's processing facilities, at a border crossing, at a trading hub, or at the buyer's facilities. In LNG contracts, the delivery point depends on whether the contract is FOB (title and risk pass at the loading port, usually as the LNG passes the flange connecting the loading arms to the vessel's manifold) or DES (title and risk pass at the discharge port). Under a CIF contract, risk passes at the loading port but the seller arranges and pays for shipping and insurance. The choice affects which party controls the vessel and, consequently, the ability to divert cargoes.

Quality and Measurement

GSPAs specify the quality of gas to be delivered and the consequences of off-specification deliveries. The buyer may be entitled to reject off-specification gas, or to accept it subject to a price reduction or compensation for costs incurred in treating it. The allocation of risk for off-specification gas is particularly important where the gas is to be used in sensitive equipment, such as gas turbines, or injected into networks with strict specifications. Measurement provisions define the methods, standards, and equipment to be used for measuring quantity and quality, the procedures for calibration and testing, and the resolution of measurement disputes, often by reference to an independent surveyor in the LNG context.

Destination Clauses and Diversion Rights

Traditional long-term gas contracts frequently contained **destination clauses** (or territorial restriction clauses) prohibiting the buyer from reselling gas outside a defined territory, and **profit-sharing mechanisms** requiring the buyer to share with the seller any profit from resale to a different destination. Sellers defended

these provisions as necessary to prevent arbitrage between markets in which they sold at different prices, and to protect their pricing strategies. Buyers and competition authorities criticised them as restrictions on competition and on the development of integrated markets.

In the European Union, destination clauses in contracts with EU buyers have been treated as restrictions of competition contrary to Article 101 of the Treaty on the Functioning of the European Union (TFEU). From the early 2000s, the European Commission pursued investigations concerning the contracts of suppliers including Gazprom, Sonatrach, and Nigeria LNG, resulting in the removal of destination clauses from their contracts with European buyers. In its 2018 decision accepting commitments from Gazprom in Case AT.39816 (*Upstream Gas Supplies in Central and Eastern Europe*), the Commission addressed concerns that Gazprom had pursued an overall strategy of partitioning Central and Eastern European gas markets, including through territorial restrictions, and Gazprom committed to remove contractual barriers to the cross-border resale of gas and to facilitate gas flows to isolated markets. In 2018 the Commission also opened an investigation into whether Qatar Petroleum's long-term LNG supply agreements with European importers contained territorial restrictions.

In Asia, the Japan Fair Trade Commission published a report in 2017 concluding that destination restrictions in FOB LNG contracts were likely to violate Japan's Antimonopoly Act, and that profit-sharing mechanisms and certain restrictions in DES contracts were also problematic. The report has encouraged Japanese buyers to negotiate greater flexibility in new contracts, contributing, together with the growth of United States LNG supplied on destination-free FOB terms, to the increasing liquidity and flexibility of the global LNG market.

Force Majeure and Seller's Failure to Deliver

GSPAs contain detailed force majeure provisions excusing performance where a party is prevented from performing by events beyond its reasonable control. In the gas context, force majeure clauses often enumerate specific events, such as failures of facilities, governmental action, and natural disasters, and may treat certain events — such as failures of the seller's upstream supply or of the buyer's downstream market — as either within or outside the clause depending on the parties' negotiated allocation of risk. Force majeure in energy contracts is examined in Chapter 15; here it is sufficient to note that the treatment of upstream and transportation failures has been a recurring source of disputes in gas contracts, and that the 2022 crisis generated significant litigation concerning Gazprom's declaration of force majeure in relation to deliveries to European buyers.

GSPAs also provide remedies for the seller's failure to deliver contracted quantities otherwise than as a result of force majeure. These commonly include

a price discount on shortfall quantities, sometimes expressed as a percentage of the contract price, and may be the buyer's exclusive remedy for shortfalls up to a specified level. The adequacy of such remedies was tested during the 2022 crisis, when replacement gas cost many times the contract price; buyers argued that the contractual shortfall remedies did not apply to deliberate non-performance, and sought full damages in arbitration.

6.6 Price Review, Take-or-Pay, and Contractual Flexibility

Take-or-Pay

The **take-or-pay** clause is the defining feature of the long-term gas contract. It obliges the buyer to pay for a minimum quantity of gas in each contract year (the "take-or-pay quantity", often 80 to 90 per cent of the ACQ, after allowing for downward flexibility) whether or not it actually takes delivery of that quantity. If the buyer takes less than the minimum, it must pay for the shortfall, but it typically acquires a right to take the paid-for but untaken gas ("make-up gas") in subsequent years, usually after it has taken the ACQ for that year, and sometimes at the price applicable at the time of payment. Conversely, if the buyer takes more than the ACQ in a year, it may accumulate "carry-forward" credits that reduce its minimum take obligation in future years.

The economic function of take-or-pay is to allocate volume risk to the buyer, providing the seller with predictable revenues sufficient to service the debt raised to finance the production and transportation facilities. In return, the buyer obtains security of supply over the long term. Lenders to gas projects typically require a substantial proportion of the project's output to be committed under take-or-pay contracts with creditworthy buyers before they will provide financing.

The enforceability of take-or-pay clauses has been challenged on the ground that they are penalties. Under English law, the rule against penalties applies only to clauses that operate upon a breach of contract; a clause that provides for payment as a primary obligation in the event of a contractual contingency, rather than a breach, is not subject to the rule. In *M&J Polymers Ltd v Imerys Minerals Ltd*, the High Court considered a take-or-pay clause in a supply contract and held that, although the rule against penalties was potentially applicable on the particular construction of that clause, the clause was commercially justifiable and not a penalty.[1] Following the restatement of the penalty rule in *Cavendish Square Holding BV v Makdessi* (discussed in Chapter 5), the predominant view is that a properly drafted take-or-pay clause, which creates a primary obligation to pay for gas made available, does not engage the penalty rule at all, and that even if it did, it would usually be justified by the seller's legitimate interest in recovering its

[1] *M&J Polymers Ltd v Imerys Minerals Ltd* [2008] EWHC 344 (Comm), [2008] 1 Lloyd's Rep 541.

investment. In the United States, courts have generally upheld take-or-pay clauses, treating them as alternative performance obligations, although extensive litigation followed the collapse of gas prices in the 1980s, when pipeline companies that had contracted to take gas at high prices sought to escape their take-or-pay obligations.

Price Indexation

The price provisions of a long-term GSPA typically combine a base price with an indexation formula that adjusts the price periodically to reflect changes in the value of gas. In oil-indexed contracts, the formula links the gas price to the prices of crude oil, gas oil, or fuel oil, often averaged over a preceding period (for example, six or nine months) and applied with a time lag, smoothing the effect of short-term volatility. In hub-indexed contracts, the price is linked to a hub price, such as TTF or NBP, perhaps with a fixed discount or premium. Hybrid formulas combine oil and hub indexation.

Price Review

Because long-term contracts run for decades, and because the indexation formula may cease to reflect the market value of gas as markets evolve, most long-term GSPAs contain a **price review** (or price reopener) clause permitting either party to request a revision of the price formula at specified intervals (commonly every three or five years) and, in some cases, at any time upon the occurrence of exceptional changes in market circumstances ("wildcard" reviews). Price review clauses typically provide that if, since the last price review, there has been a significant change in the economic circumstances in the buyer's market beyond the control of the parties, and the contract price no longer reflects the value of gas in that market, the parties shall negotiate in good faith a revision of the price formula. If they fail to agree within a specified period, either party may refer the matter to arbitration or to an expert.

Price review clauses raise distinctive legal issues. The *trigger* question is whether the conditions for a revision are satisfied — whether there has been a sufficiently significant change in circumstances of the specified kind. The *revision* question is what the revised price should be — what price reflects the value of gas in the buyer's market, often measured by reference to the prices paid by comparable buyers under comparable contracts or by reference to the "netback" value of gas from the price of competing fuels. The *scope* question is what the tribunal is empowered to change — whether it may alter only the base price, or also the indexation, the weighting of indices, or the introduction of hub indexation.

The wave of price review arbitrations that followed the decoupling of European hub prices from oil-indexed contract prices after 2009 — when the development of hub trading and the arrival of LNG intended for the United States market led to hub prices falling well below oil-indexed prices — trans-

formed the European gas market. European buyers, committed to purchasing large quantities at oil-indexed prices while their customers could buy cheaper gas at the hubs, invoked price review clauses against their suppliers, including Gazprom, Statoil (now Equinor), and Sonatrach. Many disputes were settled through renegotiation, often introducing a significant element of hub indexation; others were decided by arbitral tribunals, which in a number of reported cases accepted that the emergence of hub prices was a relevant change in circumstances and introduced hub indexation into the contract formula. Because most price review awards are confidential, the jurisprudence is known primarily through practitioners' commentaries and the limited number of awards that have entered the public domain through court proceedings.

One such case is *Atlantic LNG Co of Trinidad and Tobago v Gas Natural Aprovisionamientos SDG SA*, in which a price review award in favour of Gas Natural concerning LNG supplied from Trinidad was challenged before the United States District Court for the Southern District of New York. The court confirmed the award in 2008, rejecting the argument that the tribunal had exceeded its powers in determining the revised price.[2] The case illustrates the deference shown by courts in leading arbitral seats to tribunals' interpretation of price review clauses.

Hardship and Price Review Compared

Price review clauses perform a function similar to hardship clauses in general commercial contracts and to doctrines such as *imprévision* in French administrative law, *Wegfall der Geschäftsgrundlage* (now codified in § 313 of the German Civil Code), and Article 6.2.2 of the UNIDROIT Principles of International Commercial Contracts. They differ, however, in that they are designed to operate routinely and periodically, to keep the price aligned with the market, rather than to address exceptional events that fundamentally alter the equilibrium of the contract. A price review tribunal is not concerned with whether performance has become excessively onerous for one party, but with whether the contract price reflects the market value of gas. Chapter 15 examines hardship clauses in energy contracts generally.

Contractual Flexibility in a Changing Market

The trend in gas contracting over the past two decades has been towards greater flexibility: shorter contract durations, smaller volumes, greater downward quantity tolerance, removal of destination restrictions, hub indexation, and rights to divert and cancel cargoes. In the LNG market, portfolio players — international oil companies and trading houses that aggregate supply from multiple projects and sell to multiple buyers — have intermediated between sellers seeking long-term commitments and buyers seeking flexibility.

[2] *Atlantic LNG Co of Trinidad and Tobago v Gas Natural Aprovisionamientos SDG SA*, (SDNY 2008).

The 2021–2023 crisis complicated this trend. The extreme volatility of hub prices and the scarcity of uncommitted LNG led some buyers, particularly in Asia and in Europe, to sign long-term contracts again, including twenty-year contracts with Qatari and United States suppliers, while European buyers sought to balance supply security against the risk that gas demand would fall over the life of the contract as a result of decarbonisation policies. The resulting contracts frequently include provisions addressing the energy transition, such as rights to reduce volumes in response to changes in climate policy, commitments to reduce methane emissions across the supply chain, and provisions for the future supply of low-carbon gases. The capacity of contract law to accommodate the energy transition is examined further in Chapter 15.

6.7 Cross-Border Gas Infrastructure and Transit

The Distinctive Legal Problem of Cross-Border Pipelines

A cross-border gas pipeline links a producing state, a consuming state, and frequently one or more transit states through a single, fixed, and very expensive physical asset. Once built, the pipeline cannot be relocated, and each state through which it passes acquires a degree of leverage over the others: the producing state can withhold supply, the consuming state can refuse to purchase or delay payment, and the transit state can interrupt flows, demand higher tariffs, or divert gas for its own use. Unlike LNG, which can be redirected to alternative markets, pipeline gas offers no easy exit. The legal framework for cross-border pipelines must therefore address not only the ordinary questions of project finance and regulation but also the political risks that arise from the interdependence of sovereign states.

The legal architecture of cross-border pipelines typically operates at three levels: an **intergovernmental agreement** (IGA) among the states concerned, establishing the principles governing the project, the rights and obligations of the states, and the framework for dispute resolution; **host government agreements** (HGAs) between each state and the project company, setting out the detailed legal and fiscal regime applicable in each territory; and the commercial contracts among the project participants, shippers, and buyers. This structure was developed and refined in the Baku–Tbilisi–Ceyhan (BTC) oil pipeline project, completed in 2006, and has been applied to gas pipelines including the South Caucasus Pipeline, the Trans-Anatolian Natural Gas Pipeline (TANAP), and the Trans Adriatic Pipeline (TAP), which together form the Southern Gas Corridor bringing gas from Azerbaijan to Europe.

International Law of Transit

General international law does not confer a comprehensive right of transit for energy across the territory of another state. Several treaty regimes, however, address transit.

Article V of the General Agreement on Tariffs and Trade 1994 provides that there shall be "freedom of transit through the territory of each contracting party, via the routes most convenient for international transit, for traffic in transit to or from the territory of other contracting parties", and that such traffic shall not be subject to unnecessary delays or restrictions and shall be exempt from customs duties and transit duties, other than charges commensurate with the administrative expenses entailed by transit or with the cost of services rendered. Whether Article V applies to fixed infrastructure such as pipelines, as distinct from goods moving by conventional means of transport, has been debated. The WTO panel in *Russia — Measures Concerning Traffic in Transit* (2019), which concerned Russian restrictions on road and rail transit from Ukraine, is significant chiefly for its interpretation of the national security exception in Article XXI, but it confirmed the justiciability of Article XXI defences, a matter of considerable importance for energy trade (discussed in Chapters 13 and 25).

Article 7 of the Energy Charter Treaty provides that each contracting party shall take the necessary measures to facilitate the transit of energy materials and products consistent with the principle of freedom of transit and without distinction as to the origin, destination, or ownership of such energy materials and products or discrimination as to pricing on the basis of such distinctions, and without imposing any unreasonable delays, restrictions, or charges. Article 7(6) provides that a contracting party through whose area energy materials and products transit shall not, in the event of a dispute over any matter arising from that transit, interrupt or reduce the existing flow prior to the conclusion of the conciliation procedure set out in Article 7(7), except where specifically provided in a contract or other agreement or permitted in accordance with the conclusion of the conciliator. Article 7 does not, however, oblige a contracting party to permit the construction or modification of energy transport facilities or to allow new or additional transit through existing facilities that it demonstrates would endanger the security or efficiency of its energy systems.

Negotiations on a more detailed **Energy Charter Transit Protocol** began in 2000 but were suspended after Russia, which had signed but never ratified the ECT and which terminated its provisional application in 2009, objected to key provisions, including the treatment of existing long-term transit contracts and the "right of first refusal" for holders of expiring transit capacity. The failure of the Transit Protocol, and Russia's withdrawal from provisional application of the ECT, left European gas transit governed largely by bilateral arrangements

and by EU law. The modernisation of the ECT adopted in December 2024, and the withdrawal of the EU and many of its member states from the Treaty, are discussed in Chapters 4 and 14.

Article 125 of the United Nations Convention on the Law of the Sea confers on land-locked states a right of access to and from the sea and freedom of transit through the territory of transit states by all means of transport, the terms and modalities of which are to be agreed through bilateral, subregional, or regional agreements. Submarine pipelines on the continental shelf and in the exclusive economic zone are governed by Articles 58 and 79 of UNCLOS, examined in Chapter 20.

Transit Tariffs and Transit Disputes

Transit states are compensated for the use of their territory through transit fees or tariffs, which may be set by reference to the costs of providing the service, the value of the gas transported, or the bargaining power of the parties. Transit tariffs have been a recurrent source of disputes. The gas disputes between Russia and Ukraine in January 2006 and January 2009 — the latter resulting in a complete cessation of gas flows through Ukraine to south-eastern Europe for almost two weeks in mid-winter — arose from linked disputes over the price of gas supplied to Ukraine, the tariff for transit of Russian gas to Europe, and Ukraine's accumulated debts. They exposed the vulnerability of European consumers to disputes between producer and transit states and prompted the European Union to adopt measures to strengthen security of supply, including the requirement of reverse-flow capacity on interconnectors, and to support the diversification of supply routes. The 2009 crisis was resolved by a set of ten-year contracts between Gazprom and Naftogaz, the disputes under which are examined in the casebox at the end of this chapter. Gas transit through Ukraine to the EU ended on 1 January 2025, when the transit agreement concluded in 2019 expired and was not renewed.

Regional Gas Pipeline Frameworks in Africa

Africa offers several examples of regional and transnational gas pipeline frameworks. The **West African Gas Pipeline** (WAGP), which transports Nigerian gas to Benin, Togo, and Ghana, is governed by the Treaty on the West African Gas Pipeline Project signed by the four states in 2003 and by an International Project Agreement between the states and the West African Gas Pipeline Company. The Treaty established the West African Gas Pipeline Authority as a regional regulator, with powers over tariffs and access, and provided for a harmonised legal and fiscal regime for the pipeline across the four states. The WAGP has suffered from supply interruptions, attributable in part to supply shortfalls in Nigeria and to damage to the pipeline, and from payment arrears by downstream buyers, illustrating the

challenges of regional gas integration where domestic supply and payment discipline are weak.

Other initiatives include the proposed **Nigeria–Morocco Gas Pipeline** (also referred to as the African Atlantic Gas Pipeline), intended to transport Nigerian gas along the West African coast to Morocco and potentially to Europe, for which intergovernmental memoranda have been signed with numerous ECOWAS states and Mauritania; and the **Trans-Saharan Gas Pipeline**, intended to link Nigeria with Algeria via Niger, for which the three governments signed a memorandum of understanding in 2022. These projects face formidable challenges of financing, security, and coordination, but they illustrate the aspiration to use gas infrastructure as an instrument of regional integration. The ECOWAS Energy Protocol of 2003, modelled on the Energy Charter Treaty, contains provisions on investment protection and transit, and the African Continental Free Trade Area (AfCFTA) may in time provide a broader framework for cross-border energy trade (examined in Chapter 13).

Cross-Border LNG: The Greater Tortue Ahmeyim Project

Cross-border cooperation is not confined to pipelines. The Greater Tortue Ahmeyim (GTA) gas field straddles the maritime boundary between Mauritania and Senegal. In 2018, the two governments concluded an Inter-Governmental Cooperation Agreement providing for the joint development of the field as a single unit, with resources and revenues shared equally in the initial phase, subject to future redeterminations, and for the liquefaction of the gas on a floating LNG vessel located at a hub near the maritime boundary. The project, operated by BP with Kosmos Energy and the national oil companies of both states, illustrates the application of the unitisation principles examined in Chapter 5 to a transboundary LNG development, and the use of a bilateral treaty framework to provide the legal stability required for financing in states with no prior history of petroleum production. First LNG cargoes from the project were exported in 2025.

Elsewhere in Africa, the Mozambique LNG project in the Rovuma Basin illustrates the vulnerability of LNG projects to security risks: TotalEnergies declared force majeure in April 2021 following insurgent attacks on the town of Palma near the project site, suspending construction for an extended period, while the Coral South floating LNG project, operated by Eni, began production in 2022. The force majeure declaration, and the subsequent review of the human rights situation around the project commissioned by the operator, raise issues examined in Chapters 15 and 21.

Host Government Agreements and Human Rights

Host government agreements for cross-border pipelines have attracted criticism for conferring extensive stabilisation protections on project companies, poten-

tially restricting the ability of host states to adopt new environmental, labour, and human rights legislation. In response to such criticism concerning the BTC pipeline, the project participants executed a Human Rights Undertaking in 2003 clarifying that the stabilisation provisions would not be invoked to challenge host state regulation in the fields of health, safety, environment, and human rights that was consistent with international standards. The episode contributed to the development of the *Principles for Responsible Contracts* discussed in Chapter 5 and has influenced the drafting of later agreements.

6.8 Gas Supply Disputes and Regulatory Intervention

Typology of Gas Disputes

Gas disputes may be classified into several categories: price review disputes between sellers and buyers under long-term contracts (examined in section 6.6); disputes over volumes, including take-or-pay claims and claims for failure to deliver; disputes over quality and measurement; transit disputes; disputes between investors and host states arising from regulatory measures, such as changes in fiscal terms, domestic supply obligations, or export restrictions; and disputes arising from emergency interventions in gas markets. Many of these disputes are resolved through international commercial arbitration under the rules of the ICC, the LCIA, or the Arbitration Institute of the Stockholm Chamber of Commerce (SCC), or through investment treaty arbitration.

Supply Failure and Diversion: The Egyptian Disputes

The supply failures in Egypt following the political upheavals of 2011 generated a series of significant awards. Egypt had committed to supply gas for export, both by pipeline to Israel and Jordan through the Arab Gas Pipeline and the East Mediterranean Gas (EMG) pipeline, and to LNG export plants at Damietta and Idku. As domestic demand outstripped production and the pipeline to Israel was repeatedly attacked in the Sinai, Egypt diverted gas to the domestic market and terminated the EMG supply contract in 2012.

The resulting claims included an ICC arbitration brought by the Israel Electric Corporation against the Egyptian General Petroleum Corporation (EGPC) and the Egyptian Natural Gas Holding Company (EGAS), in which the tribunal in 2015 awarded the Israel Electric Corporation approximately USD 1.76 billion; and the ICSID case *Ampal-American Israel Corp v Egypt*, in which the tribunal found that Egypt had breached the fair and equitable treatment standard by failing to protect the pipeline and by the manner in which the supply contract was terminated. In *Unión Fenosa Gas SA v Arab Republic of Egypt* (2018), an ICSID tribunal held that Egypt had breached the fair and equitable treatment standard of the Spain–Egypt bilateral investment treaty by failing to supply gas to the Damietta LNG plant, in which the claimant held an interest, and awarded damages of

approximately USD 2 billion. The tribunal rejected Egypt's defence of necessity, holding that Egypt had contributed to the situation of necessity through its domestic gas pricing policies, which had encouraged domestic consumption. These disputes were eventually resolved through settlements, including arrangements that facilitated the resumption of LNG exports from Damietta and the reversal of the EMG pipeline to carry Israeli gas to Egypt.

The Egyptian disputes illustrate the tension between export commitments and domestic supply obligations, the limits of the necessity defence in investment law (examined in Chapter 14), and the way in which regional gas flows can reverse dramatically as production patterns change.

The European Gas Crisis of 2021–2023

The most significant gas supply crisis of the modern era began in 2021, when Gazprom reduced deliveries to European buyers to the minimum contractual levels and ceased selling additional volumes on the spot market, while European storage levels fell. Following the Russian invasion of Ukraine in February 2022, Russia required European buyers to pay for gas in roubles through a mechanism established by Presidential Decree No 172 of 31 March 2022, and suspended supplies to buyers that refused to comply, including buyers in Poland, Bulgaria, Finland, Denmark, and the Netherlands. Gazprom progressively reduced flows through the Nord Stream pipeline, citing technical problems with turbines, before halting them entirely in August 2022. In September 2022, explosions damaged both Nord Stream pipelines. European hub prices reached unprecedented levels, peaking in August 2022 at more than ten times their historical average.

The crisis generated extensive commercial arbitration between European buyers and Gazprom. In 2024, the German importer Uniper announced that an arbitral tribunal had awarded it damages of more than EUR 13 billion for Gazprom's failure to deliver contracted gas, and that it had been granted the right to terminate its long-term contracts. Other European buyers obtained similar awards. Gazprom responded by obtaining anti-arbitration and anti-suit injunctions from Russian courts under Articles 248.1 and 248.2 of the Russian Arbitrazh Procedure Code, introduced in 2020, which assert exclusive jurisdiction of Russian courts over disputes involving sanctioned Russian persons. The enforcement of awards against Gazprom's assets outside Russia raises complex questions examined in Chapter 26.

Regulatory Intervention: The European Emergency Measures

The European Union responded to the crisis with an unprecedented series of emergency measures, many adopted under Article 122 TFEU, which permits the Council to adopt measures appropriate to the economic situation "in a spirit of solidarity between Member States", in particular if severe difficulties arise in the

supply of certain products, notably in the area of energy. The principal measures included:

- **Regulation (EU) 2022/1032** on gas storage, amending the Security of Gas Supply Regulation (EU) 2017/1938, which required member states to fill their underground storage facilities to 80 per cent of capacity by 1 November 2022 and 90 per cent by 1 November in subsequent years;
- **Council Regulation (EU) 2022/1369** on coordinated demand-reduction measures, which established a voluntary target to reduce gas consumption by 15 per cent compared with the average of the previous five years, with the possibility of a mandatory target in a "Union alert";
- **Council Regulation (EU) 2022/2576** on enhancing solidarity, which established the AggregateEU mechanism for joint purchasing of gas and default solidarity rules;
- **Council Regulation (EU) 2022/2578** establishing a market correction mechanism, which capped prices for TTF front-month derivatives in defined circumstances; and
- the **REPowerEU** plan of May 2022, which set the objective of ending EU dependence on Russian fossil fuels, and subsequent legislative proposals in 2025 to phase out remaining imports of Russian gas.

At national level, member states nationalised or recapitalised gas importers facing insolvency as a result of the need to purchase replacement gas at spot prices to meet their supply obligations — notably Germany's stabilisation of Uniper, in which the federal government acquired a stake of approximately 99 per cent — placed Russian-owned subsidiaries under trusteeship, as Germany did with Gazprom Germania (later renamed SEFE), and introduced price caps, windfall taxes, and subsidies for consumers.

The Principle of Energy Solidarity

The crisis gave new prominence to the principle of energy solidarity in Article 194(1) TFEU, which provides that EU energy policy shall aim "in a spirit of solidarity between Member States" to ensure the functioning of the energy market, ensure security of energy supply, promote energy efficiency and renewable energy, and promote the interconnection of networks. In *Germany v Poland* (Case C-848/19 P), decided in July 2021, the Court of Justice confirmed the judgment of the General Court in the OPAL case (examined as a casebox in Chapter 4), holding that the principle of energy solidarity is a legally binding principle that entails rights and obligations for both the EU and its member states, and that EU institutions must assess, in adopting measures in the energy field, whether there is a risk to the energy interests of member states. The principle has been invoked in support of the emergency measures adopted in 2022 and 2023.

Limits on Regulatory Intervention

Regulatory interventions in gas markets, however justified by emergency, raise legal questions concerning their compatibility with property rights, contract rights, investment treaties, and the rule of law. Windfall taxes and price caps have been challenged before national courts and the Court of Justice of the European Union, and some interventions have given rise to investment treaty claims. The legal limits of emergency intervention, including the doctrine of necessity in international law and the treatment of emergency measures under investment treaties, are examined in Chapters 14 and 25. The general lesson of the crisis for gas law is that the commercial framework of long-term contracts and liberalised markets operates within a political and security environment that can override it, and that the law must provide mechanisms both for decisive intervention in emergencies and for the protection of legitimate expectations and the restoration of normal market functioning.

Gas Supply Security Regulation Beyond Europe

The European experience is the most fully developed, but security of gas supply regulation exists in many jurisdictions. In Japan and Korea, which depend almost entirely on imported LNG, governments have encouraged buyers to maintain diversified portfolios of long-term contracts and have established strategic buffer LNG arrangements; Japan introduced a "strategic buffer LNG" mechanism in 2023 under which the state-backed Japan Organization for Metals and Energy Security (JOGMEC) may procure LNG for use in emergencies. China has imposed gas storage obligations on suppliers and local governments following winter shortages in 2017 and 2018, when an aggressive coal-to-gas switching policy outpaced supply. In the United States, the Federal Energy Regulatory Commission and the North American Electric Reliability Corporation have examined the interdependence of gas and electricity systems following Winter Storm Uri in Texas in February 2021, during which freezing of gas production and processing facilities contributed to the failure of the Texas electricity system and the loss of life. These experiences demonstrate that gas security is a function not only of supply sources but also of the physical resilience of infrastructure and the coordination of gas and power markets.

In Nigeria, the principal security of supply challenges are domestic: inadequate gas supply to power stations, attributable to pipeline vandalism, insufficient processing and transportation capacity, payment arrears in the electricity value chain, and the historic preference of producers for export markets that pay in foreign currency. The PIA's domestic gas delivery obligation, the Gas Transportation Network Code, and the initiatives of the Decade of Gas are intended to address these challenges, but their success depends on resolving the liquidity crisis in the

electricity sector examined in Chapter 9, on the security of pipeline infrastructure, and on the gradual movement of domestic gas prices towards cost-reflective levels.

LEADING AUTHORITY

The Gazprom–Naftogaz Arbitrations (SCC, 2017–2018)

Background. Following the January 2009 gas crisis, Gazprom and Naftogaz of Ukraine concluded two ten-year contracts: a supply contract, under which Gazprom sold gas to Naftogaz for consumption in Ukraine, and a transit contract, under which Naftogaz transported Russian gas across Ukraine to European buyers. The supply contract contained a take-or-pay obligation for large annual quantities at an oil-indexed price that, following the fall in oil prices and the deterioration of relations after 2014, Naftogaz regarded as far above the market value of gas. In 2014 the parties commenced arbitration under the Rules of the Arbitration Institute of the Stockholm Chamber of Commerce, with claims by Gazprom for unpaid gas and take-or-pay amounts exceeding USD 30 billion, and claims by Naftogaz for a revision of the supply contract price, the removal of the take-or-pay obligation and the prohibition on re-export, and compensation for Gazprom's failure to deliver contracted transit volumes.

Awards. In the supply arbitration, the tribunal's final award of December 2017 substantially reduced Gazprom's take-or-pay claim, revised the contract price to reflect the market in continental Europe, and invalidated the prohibition on re-export, while ordering Naftogaz to pay for gas delivered but unpaid. In the transit arbitration, the tribunal's final award of February 2018 held that Gazprom had failed to deliver the agreed transit volumes and awarded Naftogaz damages of approximately USD 4.6 billion. After netting off the amounts awarded in the two proceedings, Gazprom owed Naftogaz approximately USD 2.56 billion. Gazprom's challenges before the Swedish courts were unsuccessful, and Naftogaz commenced enforcement proceedings against Gazprom's assets in several European jurisdictions. In December 2019, alongside the conclusion of a new five-year transit arrangement, Gazprom paid Naftogaz approximately USD 2.9 billion in settlement of the awards, and the parties withdrew pending claims.

Significance. The arbitrations are among the largest commercial arbitrations in history. They demonstrate the willingness of arbitral tribunals to revise price and volume provisions in long-term gas contracts where market circumstances have fundamentally changed and the contract so permits, and to sever export restrictions incompatible with competition law. They also illustrate the interaction between commercial arbitration and geopolitics: the enforcement of the awards and their eventual settlement were intertwined with intergovernmental negotiations over the future of gas transit through Ukraine, mediated by the European Commission.

PRACTICE NOTE

Drafting checklist: key protections for a buyer in a long-term LNG SPA

1. Downward quantity tolerance sufficient to absorb foreseeable demand fluctuations, including those resulting from decarbonisation policies.
2. Destination flexibility (FOB delivery or DES with diversion rights) and the absence of profit-sharing on resale, subject to applicable competition law.
3. A price review clause with a clear trigger, a defined reference market, express authority for the tribunal to change the indexation, and a time limit for determination.
4. Shortfall remedies that are not exclusive in cases of wilful or deliberate non-delivery, and a cap on the seller's ability to invoke upstream supply failures as force majeure.
5. Quality specifications compatible with the buyer's terminal and downstream network, with clear allocation of the costs of treating off-specification LNG.
6. Methane intensity reporting and reduction commitments across the supply chain, aligned with applicable import regulations such as the EU Methane Regulation.
7. Termination rights for prolonged force majeure and for sanctions that render performance unlawful, and a neutral seat of arbitration in a New York Convention state.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Simulated price review arbitration (hypothetical). *All parties, facts, and figures in this exercise are fictitious and created solely for teaching purposes.* Coastal Gas Exporters Ltd, a fictional liquefaction company in the fictional Republic of Ardenia, supplies 2 million tonnes per annum of LNG to Northern Utilities AG, a fictional utility in the fictional state of Valdora, under a twenty-year DES SPA concluded in 2015. The price is 14 per cent of the Brent crude oil price (three-month average) plus USD 0.50 per MMBtu. Clause 12 permits either party, every four years, to request a price review if "there has been a significant change in circumstances in the Buyer's Market, beyond the control of the Parties, as a result of which the Contract Price no longer reflects the value of LNG in the Buyer's Market", and provides that in default of agreement the tribunal "shall determine a revised Contract Price that reflects the value of LNG in the Buyer's Market". In 2025, Northern Utilities requests a price review, arguing that the Valdoran market is now priced predominantly by reference to TTF and that hub prices have been consistently below the contract price. Coastal Gas Exporters argues that the shift to hub pricing predated the 2021 review and that the 2021–2023 crisis showed that oil-indexed prices provide valuable security.

Divide into teams representing the claimant, the respondent, and the tribunal. (a) Prepare written submissions of not more than 2,000 words on the trigger question and the revision question, identifying the evidence each party would need (for example, comparable contract data and expert evidence on market value). (b) Conduct a thirty-minute oral hearing. (c) As the tribunal, draft the dispositive section of a

reasoned award, addressing whether the tribunal has authority to introduce hub indexation into the price formula.

CHAPTER SUMMARY

- The physical properties of natural gas — low energy density, the need for fixed transport infrastructure, and variable composition — explain its distinctive legal regime: long-term contracts, regulation of network monopolies, and detailed provisions on quality, measurement, and balancing.
- Gas markets have developed as regional markets with distinct pricing systems — gas-on-gas competition at hubs such as Henry Hub, NBP, and TTF; oil indexation; regulated pricing; and spot LNG — although LNG trade has promoted convergence.
- Gas-specific upstream regulation addresses rights to non-associated gas, domestic supply obligations, fiscal incentives, and the restriction of flaring and venting, increasingly through binding methane regulation.
- Liberalisation has required third-party access to pipelines and LNG terminals, unbundling of network and supply activities, and regulation of tariffs, models that Nigeria's PIA has adopted for the midstream gas sector.
- LNG projects are structured as integrated, merchant, tolling, or hybrid projects, within host-state frameworks that frequently include project-specific statutes or agreements.
- Long-term gas sales agreements allocate volume risk through take-or-pay clauses and price risk through indexation and price review clauses; competition law has curtailed destination restrictions.
- Cross-border pipelines are governed by intergovernmental and host government agreements and by transit regimes in GATT Article V and ECT Article 7; African initiatives such as the WAGP illustrate the promise and difficulties of regional integration.
- Gas supply disputes — from the Egyptian supply failures to the European crisis of 2021–2023 — reveal the interaction between commercial contracts, investment protection, emergency regulation, and geopolitics.

DOCTORAL RESEARCH QUESTIONS

1. Has the take-or-pay clause outlived its economic rationale in liberalised gas markets, or does the experience of the 2021–2023 crisis demonstrate its continuing value for security of supply?
2. To what extent do price review tribunals act as market regulators, and what are the implications of the confidentiality of their awards for the development of a coherent body of gas law?

3. Evaluate the adequacy of GATT Article V and ECT Article 7 as frameworks for energy transit in the light of the Russia–Ukraine transit experience and the failure of the Energy Charter Transit Protocol.
4. How should the law reconcile domestic gas supply obligations in producing developing countries with export commitments and investment protection, in the light of the Egyptian experience?
5. Does the principle of energy solidarity in Article 194 TFEU, as interpreted in *Germany v Poland*, offer a model for regional energy integration in West Africa?
6. Can the extraterritorial reach of import-side methane regulation, such as the EU Methane Regulation, be reconciled with the principle of common but differentiated responsibilities and with WTO law?

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